

HACCP Plan: Produce

Hazard Analysis and Critical Control Point Compliance Record

STATUS: FINAL APPROVAL RECORD

FoodSafe Compass

SITE COMPLIANCE EVIDENCE

Product Description & Intended Use

PRODUCT NAME

sliced cabbage

PRODUCT TYPE

Produce

DESCRIPTION

Fresh, raw, pre-cut *Brassica oleracea* var. *capitata*, presented as uniformly sliced or shredded leaves. The product maintains its natural cellular structure and turgidity. Typical pH range is 5.5-6.7. Water activity (aw) is high, generally >0.98. Preservation relies primarily on refrigeration and hygienic processing, with potential use of modified atmosphere packaging (MAP) to extend shelf life by reducing respiration rates and inhibiting microbial growth.

INGREDIENTS

Cabbage (*Brassica oleracea* var. *capitata*). No declared allergens inherent to the product. Potential for cross-contamination with other produce or processing aids should be assessed if applicable.

PACKAGING

Primary packaging consists of food-grade, sealed polyethylene (PE) bags or rigid polypropylene (PP) clam-shell containers, designed to maintain product integrity and atmosphere. Secondary packaging typically involves corrugated cardboard cartons or reusable plastic containers (RPCs) for bulk transport and protection.

SHELF LIFE

Under optimal storage conditions, the expected shelf life is 7-14 days from the date of packaging. This duration is contingent upon strict adherence to temperature controls and packaging integrity.

INTENDED USE

This product is intended for culinary use, requiring thorough washing prior to consumption. It can be consumed raw (e.g., salads, coleslaw) or cooked (e.g., stir-fries, soups, stews). It is not considered a 'ready-to-eat' product without further preparation.

INTENDED CONSUMERS

The product is intended for consumption by the general public. However, due to its raw nature, vulnerable populations including immunocompromised individuals, the elderly, pregnant women, and young children should ensure proper washing and, if consumed raw, be aware of potential microbiological risks associated with fresh produce.

STORAGE & DISTRIBUTION

The product must be stored and distributed under continuous refrigeration at a temperature range of 0-4°C (32-40°F) from the point of harvest through to the final consumer. Temperature excursions above 4°C must be minimized to prevent accelerated spoilage and potential microbial proliferation. Transport vehicles and storage facilities must be clean and free from contaminants.

LABELING INSTRUCTIONS

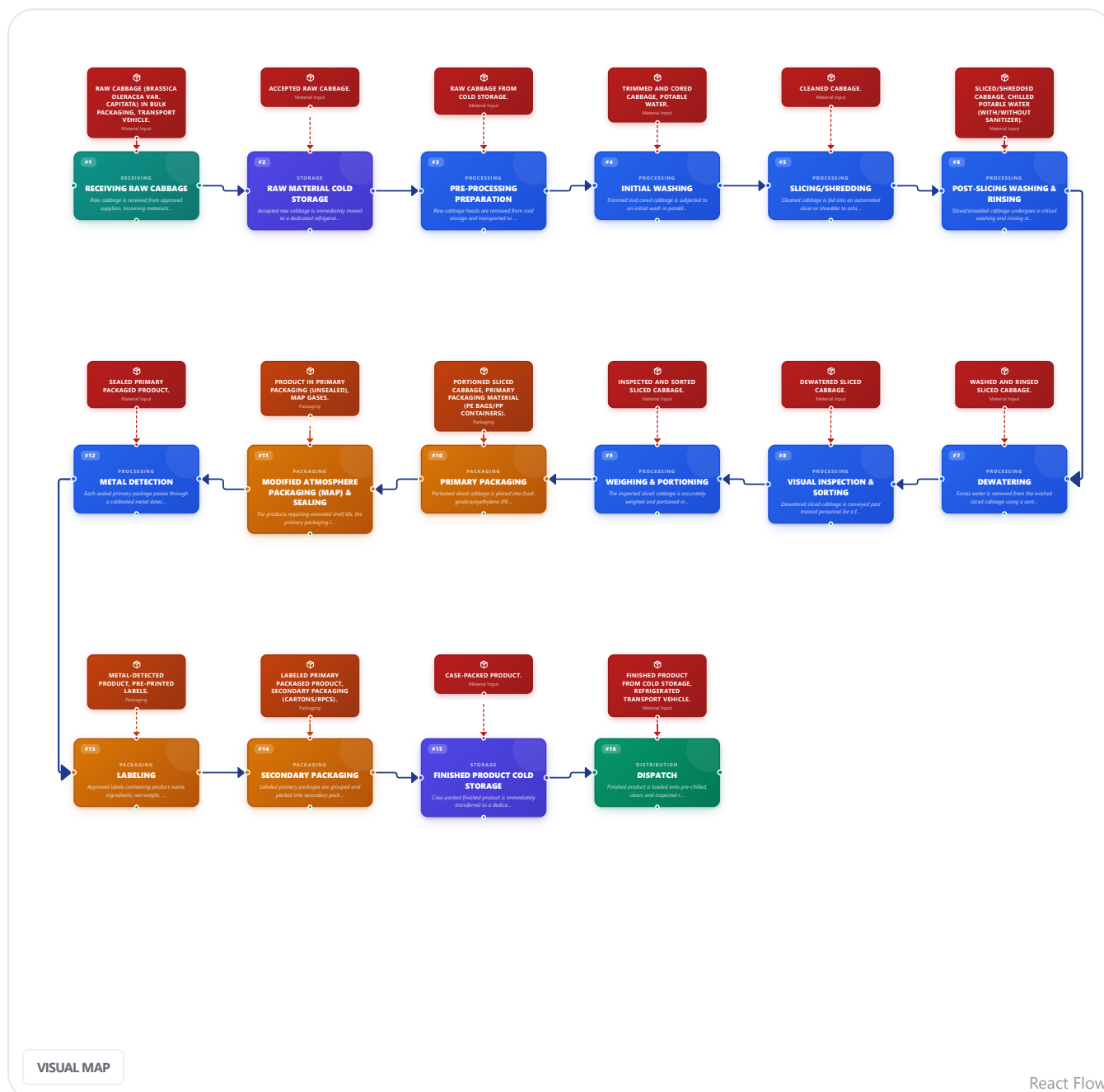
Labeling shall include: 'INGREDIENTS: Cabbage'. Allergen declaration: 'Contains: None' (unless cross-contamination risks necessitate a 'May Contain' statement). Storage instructions: 'KEEP REFRIGERATED', 'WASH BEFORE USE', 'PERISHABLE – KEEP REFRIGERATED AT 0-4°C (32-40°F)', and 'BEST BY' or 'USE BY' date.

Process Flow

1. **RECEIVING RAW MATERIALS:** Raw cabbage heads are received from approved suppliers. Incoming materials are inspected for quality, temperature, physical damage, foreign material, and pest infestation. Documentation (e.g., COA, delivery notes) is verified.
2. **RAW MATERIAL COLD STORAGE:** Accepted raw cabbage heads are immediately moved to a dedicated refrigerated storage area to maintain freshness and inhibit microbial growth. Storage conditions (temperature, humidity) are monitored.
3. **PREPARATION & TRIMMING:** Raw cabbage heads are brought to the preparation area. Outer leaves are removed, the core is extracted, and any damaged, discolored, or diseased portions are trimmed away manually or mechanically.
4. **INITIAL WASHING:** Trimmed cabbage portions are subjected to an initial wash in potable water to remove surface dirt, soil, and any remaining debris. This may involve a spray wash or immersion.
5. **SLICING/SHREDDING:** The initially cleaned cabbage portions are fed into a commercial slicer or shredder to achieve the desired uniform slice or shred size and shape. Equipment is cleaned and sanitized regularly.
6. **FINAL WASHING/FLUMING:** Sliced/shredded cabbage undergoes a thorough final wash in chilled, potable water, often in a flume system. This step removes cellular sap, further cleans the product, and may incorporate a food-grade sanitizer to reduce microbial load.
7. **DEWATERING/DRYING:** Excess water is removed from the washed sliced cabbage using centrifugal dryers or air knives. Effective dewatering is crucial for product quality, shelf life, and packaging integrity.
8. **INSPECTION & SORTING:** Dewatered sliced cabbage is visually inspected on a conveyor belt for any remaining foreign material, discolored pieces, or other defects. Non-conforming product is manually removed.
9. **WEIGHING & PORTIONING:** The inspected sliced cabbage is accurately weighed and portioned into specified batch sizes or individual package weights, typically using multi-head weighers or manual scales.
10. **PACKAGING:** Portioned sliced cabbage is transferred into pre-formed bags or packaging film. For extended shelf life, Modified Atmosphere Packaging (MAP) may be used, where the air inside the package is replaced with a specific gas mixture (e.g., N₂, CO₂). Packages are sealed.
11. **METAL DETECTION:** Each sealed package of sliced cabbage passes through a calibrated metal detector to identify and reject any products containing metallic foreign objects. Rejected products are quarantined for investigation.
12. **LABELING & CODING:** Approved labels containing product information (e.g., ingredients, net weight, nutritional facts, storage instructions) are applied. Batch codes, production dates, and best-before dates are printed or applied to each package.

13. FINISHED PRODUCT COLD STORAGE: Labeled and coded finished product is immediately transferred to a dedicated refrigerated finished goods storage area. Temperature and humidity are strictly controlled and monitored to maintain product quality and safety until dispatch.

14. DISPATCH: Finished product is retrieved from cold storage and loaded onto pre-chilled, refrigerated transport vehicles. Loading procedures ensure product temperature is maintained and proper stacking prevents damage. Dispatch documentation is completed.



STEP TYPES

- Receiving
- Storage
- Processing
- Mixing/Combining

- Packaging
- Distribution
- Other

INPUTS / OUTPUTS

FLAGS & HACCP

-  Material Input
-  Packaging Input
-  Waste / Output

-  Outsourced Process
-  Rework / Recycle
-  Critical Control Point



Hazard Analysis Summary

RECEIVING RAW CABBAGE

B

Hazard: Pathogenic bacteria (e.g., Salmonella, E. coli O157:H7, Listeria monocytogenes), viruses (e.g., Norovirus), or parasites (e.g., Cyclospora cayetanensis) on raw cabbage.

Control Measures: Approved supplier program (supplier qualification, audits, COA for microbial testing if applicable), visual inspection for quality/damage/pests, temperature verification upon receipt (0-4°C), rejection of non-conforming product.

RLTO: YES

Result: CP

RECEIVING RAW CABBAGE

C

Hazard: Pesticide residues, heavy metals, sanitizers (from transport vehicle), or allergens (cross-contamination from other produce in transport).

Control Measures: Approved supplier program (requiring adherence to MRLs, COA for pesticide residues/heavy metals if high risk), visual inspection of transport vehicle for cleanliness, segregation of produce, rejection of non-conforming product.

RLTO: YES

Result: CP

RECEIVING RAW CABBAGE

P

Hazard: Foreign material (e.g., stones, soil, insects, plant debris, metal/plastic from transport/packaging).

Control Measures: Approved supplier program, visual inspection upon receipt, rejection of non-conforming product.

RLTO: YES

Result: CP

RAW MATERIAL COLD STORAGE

B

Hazard: Proliferation of pathogenic bacteria (e.g., Listeria monocytogenes, psychrotrophic pathogens) due to temperature abuse or cross-contamination.

Control Measures: Maintain storage temperature at 0-4°C, continuous temperature monitoring, regular cleaning and sanitization of storage area, segregation of raw materials, FIFO stock rotation.

RLTO: YES

Result: CP

RAW MATERIAL COLD STORAGE

C

Hazard: Cross-contamination with cleaning chemicals or allergens from other stored products.

Control Measures: Segregation of chemicals from food products, proper labeling and storage of chemicals, strict allergen control program (if other allergens are stored), proper cleaning procedures.

RLTO: YES

Result: CP

RAW MATERIAL COLD STORAGE

P

Hazard: Foreign material (e.g., pests, packaging debris, facility-related debris).

Control Measures: Pest control program, good manufacturing practices (GMPs), regular inspection of storage area, proper handling of packaging materials.

RLTO: YES

Result: CP

PRE-PROCESSING PREPARATION

B

Hazard: Pathogen transfer from outer leaves/core to inner leaves, cross-contamination from unsanitized equipment/surfaces/personnel, or proliferation if product is held at ambient temperature for too long.

Control Measures: Good Hygienic Practices (GHPs) for personnel (handwashing, gloves), regular cleaning and sanitization of equipment and surfaces, visual inspection of cabbage, removal of outer leaves and core, strict time/temperature control during preparation (minimize time out of refrigeration).

RLTO: YES

Result: CP

PRE-PROCESSING PREPARATION

C

Hazard: Cross-contamination from cleaning chemicals, lubricants, or allergens from other products processed on shared lines.

Control Measures: Strict cleaning and sanitization procedures, proper use and storage of lubricants, allergen control program (if applicable), visual inspection of equipment.

RLTO: YES

Result: CP

PRE-PROCESSING PREPARATION**P**

Hazard: Foreign material (e.g., plant debris, insects, plastic/metal from equipment, personal effects).

Control Measures: Visual inspection of cabbage, GMPs for personnel (no jewelry, hairnets), regular maintenance of equipment, foreign material control program.

RLTO: YES**Result: CP****INITIAL WASHING****B**

Hazard: Pathogenic bacteria (e.g., Salmonella spp., Listeria monocytogenes, E. coli O157:H7, Shigella spp., Cyclospora cayetanensis, Norovirus) from soil, water, animal feces, human handling, or cross-contamination during harvest/transport.

Control Measures: Supplier approval program (GAP/GMP verification), visual inspection upon receipt, proper storage conditions (temperature control), subsequent washing steps in the process.

RLTO: YES**Result: CP****INITIAL WASHING****C**

Hazard: Pesticide residues, heavy metals (e.g., lead, cadmium), nitrates, sanitizers/cleaning chemicals (from field equipment).

Control Measures: Supplier approval program (GAP verification, COA for pesticide residues/heavy metals if high risk), adherence to MRLs (Maximum Residue Limits), visual inspection.

RLTO: YES**Result: CP****INITIAL WASHING****P**

Hazard: Foreign material (e.g., stones, soil, insects, plant debris, metal fragments, plastic pieces, glass) from field, harvest, or transport.

Control Measures: Supplier approval program, visual inspection upon receipt, subsequent washing steps, proper handling.

RLTO: YES**Result: CP****SLICING/SHREDDING****B**

Hazard: Pathogenic bacteria (e.g., E. coli, Salmonella, Listeria) or protozoa (e.g., Giardia, Cryptosporidium) if water is not potable or becomes contaminated.

Control Measures: Use of municipal potable water supply or approved private well water, regular water testing (microbiological, chemical), backflow prevention, proper maintenance of water lines.

RLTO: YES**Result: CP****SLICING/SHREDDING****C**

Hazard: Excessive levels of chlorine, heavy metals, industrial contaminants, cleaning chemicals from water lines.

Control Measures: Regular water testing (chemical parameters), adherence to municipal water standards, proper maintenance of water treatment systems.

RLTO: YES**Result: CP****SLICING/SHREDDING****P**

Hazard: Sediment, rust particles, pipe scale.

Control Measures: N/A

RLTO: NO**Result: CP****POST-SLICING WASHING & RINSING****B**

Hazard: Pathogenic bacteria (e.g., E. coli, Salmonella, Listeria) or protozoa if water is not potable or if sanitizer is ineffective/not used correctly.

Control Measures: Use of municipal potable water supply or approved private well water, regular water testing, backflow prevention, proper maintenance of water lines, monitoring of sanitizer concentration and contact time (if used), monitoring of water temperature.

RLTO: YES**Result: CP**

POST-SLICING WASHING & RINSING

C

Hazard: Excessive levels of sanitizer (e.g., chlorine, peracetic acid), heavy metals, industrial contaminants, cleaning chemicals from water lines.

Control Measures: Regular water testing (chemical parameters), adherence to municipal water standards, proper maintenance of water treatment systems, monitoring of sanitizer concentration.

RLTO: YES

Result: CP

POST-SLICING WASHING & RINSING

P

Hazard: Sediment, rust particles, pipe scale.

Control Measures: N/A

RLTO: NO

Result: CP

DEWATERING

B

Hazard: Pathogenic bacteria (e.g., Salmonella spp., E. coli O157:H7, Listeria monocytogenes, Shigella spp.)

Control Measures: Supplier approval program (GAP/GMP verification), Certificate of Analysis (COA) for microbiological parameters (if risk-based), visual inspection upon receiving, proper handling and storage to prevent cross-contamination and temperature abuse. Subsequent washing steps in the process are critical.

RLTO: YES

Result: CP

DEWATERING

C

Hazard: Pesticide residues, heavy metals, environmental contaminants (e.g., nitrates)

Control Measures: Supplier approval program (GAP verification), COA for pesticide residues/heavy metals (if risk-based), adherence to MRLs (Maximum Residue Limits), visual inspection for signs of contamination.

RLTO: YES

Result: CP

DEWATERING

P

Hazard: Foreign material (e.g., stones, soil, insects, plant debris, metal fragments from harvesting equipment)

Control Measures: Supplier approval program, visual inspection upon receiving, proper handling, subsequent washing and sorting steps in the process.

RLTO: YES

Result: CP

VISUAL INSPECTION & SORTING

B

Hazard: Proliferation of pathogenic bacteria (e.g., Salmonella spp., E. coli O157:H7, Listeria monocytogenes) due to temperature abuse or cross-contamination from equipment.

Control Measures: Maintain product temperature below 4°C (40°F) during dewatering. Implement effective sanitation program (SSOPs) for dewatering equipment, including pre-operational and operational checks. Minimize processing time.

RLTO: YES

Result: CP

VISUAL INSPECTION & SORTING

C

Hazard: Contamination from lubricants, cleaning chemicals, or sanitizers.

Control Measures: Use food-grade lubricants. Implement effective sanitation program (SSOPs) with proper rinsing procedures. Train personnel on chemical handling and storage.

RLTO: YES

Result: CP

VISUAL INSPECTION & SORTING

P

Hazard: Foreign material (e.g., metal fragments from equipment wear, plastic pieces, loose parts from machinery)

Control Measures: Implement a preventive maintenance program for dewatering equipment. Regular inspection of equipment for wear and tear. Use of screens or filters where appropriate.

RLTO: YES

Result: CP

WEIGHING & PORTIONING

B

Hazard: Cross-contamination from personnel (e.g., unwashed hands, soiled gloves) or unsanitized surfaces.

Control Measures: Implement strict Good Manufacturing Practices (GMPs) for personnel hygiene (handwashing, gloves, clean uniforms). Train personnel on proper handling techniques. Maintain clean and sanitized sorting surfaces.

RLTO: YES

Result: CP

WEIGHING & PORTIONING

C

Hazard: Contamination from cleaning chemicals, sanitizers, or personal care products (e.g., hand lotions, perfumes).

Control Measures: Implement effective sanitation program (SSOPs) for sorting tables and tools. Enforce GMPs regarding personal care products for personnel.

RLTO: YES

Result: CP

WEIGHING & PORTIONING

P

Hazard: Foreign material (e.g., hair, jewelry, personal items from personnel; plastic/metal from equipment; plant debris missed in previous steps)

Control Measures: Implement strict GMPs for personnel (hairnets, beard nets, no jewelry, no loose items). Regular inspection of sorting lines and tools. Training for personnel on identifying and removing foreign material.

RLTO: YES

Result: CP

PRIMARY PACKAGING

B

Hazard: Pathogenic bacteria (e.g., Salmonella spp., E. coli O157:H7, Listeria monocytogenes), viruses (e.g., Norovirus), parasites (e.g., Cyclospora cayetanensis)

Control Measures: Supplier approval program (GAP/GMP verification), incoming material specifications (microbiological criteria), proper washing and sanitizing of raw cabbage prior to slicing (pre-processing step), maintaining cold chain. Consumer washing instruction on label.

RLTO: YES

Result: CP

PRIMARY PACKAGING

C

Hazard: Pesticide residues, heavy metals (e.g., lead, cadmium), environmental contaminants, sanitizers (if used in pre-processing and not rinsed)

Control Measures: Supplier approval program (GAP verification, pesticide residue testing, heavy metal testing), incoming material specifications, proper rinsing procedures for sanitizers (if applicable).

RLTO: YES

Result: CP

PRIMARY PACKAGING

P

Hazard: Stones, dirt, insects, plant debris, foreign objects from field (e.g., plastic pieces, glass shards)

Control Measures: Supplier approval program, visual inspection of incoming cabbage, washing/trimming/sorting steps (pre-processing), metal detection (later in process).

RLTO: YES

Result: CP

MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING

B

Hazard: Introduction or proliferation of pathogenic bacteria (e.g., Listeria monocytogenes, Salmonella spp.) due to unsanitary packaging environment, contaminated packaging materials, or temperature abuse during packaging.

Control Measures: Good Manufacturing Practices (GMPs) including sanitation of packaging lines, employee hygiene, packaging material specifications (food-grade, clean), proper storage of packaging materials, maintaining cold chain during packaging, environmental monitoring.

RLTO: YES

Result: CP

MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING

C

Hazard: Chemical contamination from packaging materials (e.g., migration of non-food grade chemicals), cleaning chemicals (if not properly rinsed/sanitized), or lubricants.

Control Measures: Packaging material specifications (food-grade, FDA compliant), proper storage of chemicals, documented cleaning and sanitizing procedures, use of food-grade lubricants.

RLTO: YES

Result: CP

MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING

P

Hazard: Foreign material from packaging materials (e.g., plastic fragments from damaged bags/containers), equipment (e.g., metal shards, plastic pieces), or environment (e.g., dust, hair).

Control Measures: Visual inspection of packaging materials, equipment maintenance program, GMPs (employee hygiene, foreign material control program), metal detection (later step).

RLTO: **YES**Result: **CP****METAL DETECTION**

B

Hazard: Growth of anaerobic or facultative anaerobic pathogens (e.g., *Clostridium botulinum*, *Listeria monocytogenes*) due to improper gas mixture, seal integrity failure, or temperature abuse.

Control Measures: Critical limits for gas mixture composition (e.g., minimum CO2 level), seal integrity checks (visual, destructive testing), temperature control during and after packaging, validation studies for MAP shelf life and safety.

RLTO: **YES**Result: **CP****METAL DETECTION**

C

Hazard: Contamination from MAP gases (e.g., non-food grade gases, impurities).

Control Measures: Specifications for MAP gases (food-grade, purity certificates from supplier), supplier approval program.

RLTO: **YES**Result: **CP****METAL DETECTION**

P

Hazard: None reasonably likely to occur at this step.

Control Measures: N/A

RLTO: **NO**Result: **CP****LABELING**

B

Hazard: Microbial contamination (e.g., environmental pathogens like *Listeria monocytogenes*, *Salmonella*) transferred from labels, equipment surfaces, or personnel to the exterior of the primary package.

Control Measures: Good Manufacturing Practices (GMPs), personnel hygiene program, equipment sanitation procedures, hygienic storage and handling of labels.

RLTO: **NO**Result: **CP****LABELING**

C

Hazard: Undeclared allergens due to mislabeling (e.g., applying a label from an allergen-containing product to sliced cabbage) or incorrect product information (e.g., missing 'WASH BEFORE USE' instruction).

Control Measures: Label control program (e.g., line clearance, label reconciliation, visual verification of correct labels for product run), allergen management program, personnel training, regular label audits.

RLTO: **YES**Result: **CP****LABELING**

P

Hazard: Foreign material (e.g., label fragments, plastic pieces from label rolls, staples, dust, hair) introduced during the labeling process.

Control Measures: Good Manufacturing Practices (GMPs), equipment maintenance, visual inspection, foreign material control program, personnel training.

RLTO: **YES**Result: **CP****SECONDARY PACKAGING**

B

Hazard: Microbial contamination (e.g., environmental pathogens) from secondary packaging materials (cartons, RPCs), equipment, or personnel transferred to the exterior of the primary package.

Control Measures: Good Manufacturing Practices (GMPs), personnel hygiene program, sanitation of equipment, supplier control program for secondary packaging materials, cleaning and sanitation program for Reusable Plastic Containers (RPCs).

RLTO: **NO**Result: **CP**

SECONDARY PACKAGING

C

Hazard: Chemical contamination from secondary packaging materials (e.g., inks, adhesives, cleaning chemical residues on RPCs) or cross-contamination from other products if RPCs are reused without proper sanitation.

Control Measures: Supplier specifications for secondary packaging materials, robust cleaning and sanitation program for RPCs, visual inspection of RPCs prior to use, chemical control program.

RLTO: NO

Result: CP

SECONDARY PACKAGING

P

Hazard: Foreign material (e.g., cardboard dust/fibers, plastic fragments from RPCs, staples, tape, hair, pests) from secondary packaging materials or the packing process.

Control Measures: Good Manufacturing Practices (GMPs), equipment maintenance, visual inspection, foreign material control program, supplier control for packaging materials, pest control program.

RLTO: YES

Result: CP

FINISHED PRODUCT COLD STORAGE

B

Hazard: Pathogen growth (e.g., *Listeria monocytogenes*, psychrotrophic spoilage organisms) due to temperature abuse during storage.

Control Measures: Continuous temperature monitoring and recording of cold storage units, alarm systems for temperature excursions, preventative maintenance program for refrigeration equipment, adherence to FIFO (First-In, First-Out) stock rotation, rapid cooling of product upon entry to storage.

RLTO: YES

Result: CP

FINISHED PRODUCT COLD STORAGE

C

Hazard: Chemical contamination from cleaning agents, refrigerants, or cross-contamination from other stored products (e.g., non-food chemicals stored nearby).

Control Measures: Segregation of chemicals and non-food items from food products, proper labeling and storage of all chemicals, chemical control program, spill procedures, Good Manufacturing Practices (GMPs).

RLTO: NO

Result: CP

FINISHED PRODUCT COLD STORAGE

P

Hazard: Foreign material (e.g., pests, dust, debris from facility, packaging damage leading to product exposure) from the storage environment.

Control Measures: Robust pest control program, regular facility maintenance, cleaning and sanitation program for storage areas, visual inspection of packaging for integrity, Good Manufacturing Practices (GMPs).

RLTO: YES

Result: CP

DISPATCH

B

Hazard: Pathogenic bacterial growth (e.g., *Listeria monocytogenes*, *Salmonella* spp., pathogenic *E. coli*) due to temperature abuse during loading or transit.

Control Measures: Verify product temperature prior to loading (0-4°C). Ensure transport vehicles are pre-chilled to 0-4°C and clean prior to loading. Minimize loading time to prevent temperature excursions. Monitor and record temperature during transit to ensure continuous cold chain integrity (0-4°C). Implement corrective actions for temperature deviations (e.g., reject load, re-cool, divert for cooking only). Driver training on cold chain management.

RLTO: YES

Result: CP

DISPATCH

C

Hazard: Contamination with cleaning chemicals, lubricants, or fuel from the transport vehicle.

Control Measures: Inspect transport vehicles for cleanliness, absence of odors, and structural integrity (no leaks) prior to loading. Ensure cleaning chemicals used in vehicles are food-grade or properly rinsed. Driver training on proper vehicle maintenance and loading procedures.

RLTO: **YES**

Result: **CP**

DISPATCH

P

Hazard: Foreign material contamination (e.g., glass, metal, plastic, wood, insects, debris) from the transport vehicle or loading environment.

Control Measures: Inspect transport vehicles for cleanliness and absence of foreign material prior to loading. Ensure loading dock area is clean and free of debris. Inspect product packaging for integrity prior to loading. Driver training on proper loading and handling procedures.

RLTO: **YES**

Result: **CP**

CCP Determination (Codex 2023)

STEP & HAZARD	RISK ANALYSIS	CODEX Q1-Q4	RESULT
Step 1: RECEIVING RAW CABBAGE B: Pathogenic bacteria (e.g., Salmonella, E. coli O157:H7, Listeria monocytogenes), viruses (e.g., Norovirus), or parasites (e.g., Cyclospora cayetanensis) on raw cabbage.	Score: 12 (High)	Q1:yes Q2:no Q3:yes Q4:no	CCP
Step 1: RECEIVING RAW CABBAGE C: Pesticide residues, heavy metals, sanitizers (from transport vehicle), or allergens (cross-contamination from other produce in transport).	Score: 6 (Moderate)	Q1:yes Q2:no Q3:yes Q4:no	CCP
Step 1: RECEIVING RAW CABBAGE P: Foreign material (e.g., stones, soil, insects, plant debris, metal/plastic from transport/packaging).	Score: 9 (High)	Q1:yes Q2:no Q3:yes Q4:yes	CP/OPRP
Step 2: RAW MATERIAL COLD STORAGE B: Proliferation of pathogenic bacteria (e.g., Listeria monocytogenes, psychrotrophic pathogens) due to temperature abuse or cross-contamination.	Score: 8 (Moderate)	Q1:yes Q2:yes Q3:skip Q4:skip	CCP
Step 2: RAW MATERIAL COLD STORAGE C: Cross-contamination with cleaning chemicals or allergens from other stored products.	Score: 4 (Low)	Q1:yes Q2:no Q3:yes Q4:yes	CP/OPRP
Step 2: RAW MATERIAL COLD STORAGE P: Foreign material (e.g., pests, packaging debris, facility-related debris).	Score: 1 (Low)	Q1:skip Q2:skip Q3:skip Q4:skip	CP/OPRP
Step 3: PRE-PROCESSING PREPARATION B: Pathogen transfer from outer leaves/core to inner leaves, cross-contamination from unsanitized equipment/surfaces/personnel, or proliferation if product is held at ambient temperature for too long.	Score: 9 (High)	Q1:yes Q2:no Q3:yes Q4:no	CCP
Step 3: PRE-PROCESSING PREPARATION C: Cross-contamination from cleaning chemicals, lubricants, or allergens from other products processed on shared lines.	Score: 6 (Moderate)	Q1:yes Q2:no Q3:yes Q4:no	CCP
Step 3: PRE-PROCESSING PREPARATION P: Foreign material (e.g., plant debris, insects, plastic/metal from equipment, personal effects).	Score: 6 (Moderate)	Q1:yes Q2:no Q3:yes Q4:no	CCP
Step 4: INITIAL WASHING B: Pathogenic bacteria (e.g., Salmonella spp., Listeria monocytogenes, E. coli O157:H7, Shigella spp., Cyclospora cayetanensis, Norovirus) from soil, water, animal feces, human handling, or cross-contamination during harvest/transport.	Score: 12 (High)	Q1:yes Q2:yes Q3:skip Q4:skip	CP/OPRP
Step 4: INITIAL WASHING C: Pesticide residues, heavy metals (e.g., lead, cadmium), nitrates, sanitizers/cleaning chemicals (from field equipment).	Score: 6 (Moderate)	Q1:yes Q2:no Q3:yes Q4:no	CP/OPRP
Step 4: INITIAL WASHING P: Foreign material (e.g., stones, soil, insects, plant debris, metal fragments, plastic pieces, glass) from field, harvest, or transport.	Score: 12 (High)	Q1:yes Q2:no Q3:yes Q4:no	CP/OPRP
Step 5: SLICING/SHREDDING B: Pathogenic bacteria (e.g., E. coli, Salmonella, Listeria) or protozoa (e.g., Giardia, Cryptosporidium) if water is not potable or becomes contaminated.	Score: 12 (High)	Q1:yes Q2:no Q3:yes Q4:no	CCP

STEP & HAZARD	RISK ANALYSIS	CODEX Q1-Q4	RESULT
Step 5: SLICING/SHREDDING C: Excessive levels of chlorine, heavy metals, industrial contaminants, cleaning chemicals from water lines.	Score: 6 (Moderate)	Q1:yes Q2:no Q3:yes Q4:no	CCP
Step 6: POST-SLICING WASHING & RINSING B: Pathogenic bacteria (e.g., E. coli, Salmonella, Listeria) or protozoa if water is not potable or if sanitizer is ineffective/not used correctly.	Score: 12 (High)	Q1:yes Q2:yes Q3:skip Q4:skip	CCP
Step 6: POST-SLICING WASHING & RINSING C: Excessive levels of sanitizer (e.g., chlorine, peracetic acid), heavy metals, industrial contaminants, cleaning chemicals from water lines.	Score: 9 (High)	Q1:no Q2:skip Q3:skip Q4:skip	CP/OPRP
Step 7: DEWATERING B: Pathogenic bacteria (e.g., Salmonella spp., E. coli O157:H7, Listeria monocytogenes, Shigella spp.)	Score: 6 (Moderate)	Q1:no Q2:skip Q3:skip Q4:skip	CP/OPRP
Step 7: DEWATERING C: Pesticide residues, heavy metals, environmental contaminants (e.g., nitrates)	Score: 1 (Low)	Q1:skip Q2:skip Q3:skip Q4:skip	CP/OPRP
Step 7: DEWATERING P: Foreign material (e.g., stones, soil, insects, plant debris, metal fragments from harvesting equipment)	Score: 9 (High)	Q1:no Q2:skip Q3:skip Q4:skip	CP/OPRP
Step 8: VISUAL INSPECTION & SORTING B: Proliferation of pathogenic bacteria (e.g., Salmonella spp., E. coli O157:H7, Listeria monocytogenes) due to temperature abuse or cross-contamination from equipment.	Score: 12 (High)	Q1:yes Q2:yes Q3:skip Q4:skip	CCP
Step 8: VISUAL INSPECTION & SORTING C: Contamination from lubricants, cleaning chemicals, or sanitizers.	Score: 6 (Moderate)	Q1:yes Q2:yes Q3:skip Q4:skip	CCP
Step 8: VISUAL INSPECTION & SORTING P: Foreign material (e.g., metal fragments from equipment wear, plastic pieces, loose parts from machinery)	Score: 9 (High)	Q1:yes Q2:no Q3:yes Q4:no	CCP
Step 9: WEIGHING & PORTIONING B: Cross-contamination from personnel (e.g., unwashed hands, soiled gloves) or unsanitized surfaces.	Score: 12 (High)	Q1:yes Q2:no Q3:yes Q4:yes	CP/OPRP
Step 9: WEIGHING & PORTIONING C: Contamination from cleaning chemicals, sanitizers, or personal care products (e.g., hand lotions, perfumes).	Score: 6 (Moderate)	Q1:yes Q2:no Q3:yes Q4:yes	CP/OPRP
Step 9: WEIGHING & PORTIONING P: Foreign material (e.g., hair, jewelry, personal items from personnel; plastic/metal from equipment; plant debris missed in previous steps)	Score: 9 (High)	Q1:yes Q2:no Q3:yes Q4:no	CCP
Step 10: PRIMARY PACKAGING B: Pathogenic bacteria (e.g., Salmonella spp., E. coli O157:H7, Listeria monocytogenes), viruses (e.g., Norovirus), parasites (e.g., Cyclospora cayetanensis)	Score: 12 (High)	Q1:yes Q2:no Q3:yes Q4:no	CCP
Step 10: PRIMARY PACKAGING C: Pesticide residues, heavy metals (e.g., lead, cadmium), environmental contaminants, sanitizers (if used in pre-processing and not rinsed)	Score: 6 (Moderate)	Q1:yes Q2:no Q3:no Q4:skip	CP/OPRP

STEP & HAZARD	RISK ANALYSIS	CODEX Q1-Q4	RESULT
Step 10: PRIMARY PACKAGING P: Stones, dirt, insects, plant debris, foreign objects from field (e.g., plastic pieces, glass shards)	Score: 6 (Moderate)	Q1:no Q2:skip Q3:skip Q4:skip	CP/OPRP
Step 11: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING B: Introduction or proliferation of pathogenic bacteria (e.g., <i>Listeria monocytogenes</i> , <i>Salmonella</i> spp.) due to unsanitary packaging environment, contaminated packaging materials, or temperature abuse during packaging.	Score: 12 (High)	Q1:yes Q2:yes Q3:skip Q4:skip	CCP
Step 11: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING C: Chemical contamination from packaging materials (e.g., migration of non-food grade chemicals), cleaning chemicals (if not properly rinsed/sanitized), or lubricants.	Score: 8 (Moderate)	Q1:yes Q2:yes Q3:skip Q4:skip	CCP
Step 11: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING P: Foreign material from packaging materials (e.g., plastic fragments from damaged bags/containers), equipment (e.g., metal shards, plastic pieces), or environment (e.g., dust, hair).	Score: 9 (High)	Q1:yes Q2:no Q3:yes Q4:yes	CP/OPRP
Step 12: METAL DETECTION B: Growth of anaerobic or facultative anaerobic pathogens (e.g., <i>Clostridium botulinum</i> , <i>Listeria monocytogenes</i>) due to improper gas mixture, seal integrity failure, or temperature abuse.	Score: 12 (High)	Q1:yes Q2:yes Q3:skip Q4:skip	CCP
Step 12: METAL DETECTION C: Contamination from MAP gases (e.g., non-food grade gases, impurities).	Score: 6 (Moderate)	Q1:yes Q2:yes Q3:skip Q4:skip	CCP
Step 13: LABELING C: Undeclared allergens due to mislabeling (e.g., applying a label from an allergen-containing product to sliced cabbage) or incorrect product information (e.g., missing 'WASH BEFORE USE' instruction).	Score: 12 (High)	Q1:yes Q2:no Q3:yes Q4:no	CCP
Step 13: LABELING P: Foreign material (e.g., label fragments, plastic pieces from label rolls, staples, dust, hair) introduced during the labeling process.	Score: 4 (Low)	Q1:yes Q2:no Q3:yes Q4:yes	CP/OPRP
Step 14: SECONDARY PACKAGING P: Foreign material (e.g., cardboard dust/fibers, plastic fragments from RPCs, staples, tape, hair, pests) from secondary packaging materials or the packing process.	Score: 6 (Moderate)	Q1:yes Q2:no Q3:yes Q4:yes	CP/OPRP
Step 15: FINISHED PRODUCT COLD STORAGE B: Pathogen growth (e.g., <i>Listeria monocytogenes</i> , psychrotrophic spoilage organisms) due to temperature abuse during storage.	Score: 12 (High)	Q1:yes Q2:no Q3:yes Q4:no	CCP
Step 15: FINISHED PRODUCT COLD STORAGE P: Foreign material (e.g., pests, dust, debris from facility, packaging damage leading to product exposure) from the storage environment.	Score: 6 (Moderate)	Q1:yes Q2:yes Q3:skip Q4:skip	CP/OPRP
Step 16: DISPATCH B: Pathogenic bacterial growth (e.g., <i>Listeria monocytogenes</i> , <i>Salmonella</i> spp., pathogenic <i>E. coli</i>) due to temperature abuse during loading or transit.	Score: 12 (High)	Q1:yes Q2:no Q3:yes Q4:no	CCP
Step 16: DISPATCH C: Contamination with cleaning chemicals, lubricants, or fuel from the transport vehicle.	Score: 8 (Moderate)	Q1:yes Q2:no Q3:yes Q4:no	CCP

STEP & HAZARD	RISK ANALYSIS	CODEX Q1-Q4	RESULT
Step 16: DISPATCH P: Foreign material contamination (e.g., glass, metal, plastic, wood, insects, debris) from the transport vehicle or loading environment.	Score: 6 (Moderate)	Q1:yes Q2:no Q3:yes Q4:yes	CP/OPRP

Step 5: Critical Limits at Each CCP

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 1 Step 1: RECEIVING RAW CABBAGE Hazard: Pathogenic bacteria (e.g., Salmonella, E. coli O157:H7, Listeria monocytogenes), viruses (e.g., Norovirus), or parasites (e.g., Cyclospora cayetanensis) on raw cabbage.	PARAMETER: SUPPLIER STATUS Raw cabbage must be sourced from an approved supplier.	An approved supplier program ensures that the supplier has controls in place to manage chemical hazards, such as adherence to maximum residue limits (MRLs) for pesticides, monitoring for heavy metals in growing environments, and proper sanitation practices for transport vehicles. This minimizes the risk of chemical contamination that cannot be removed by subsequent processing.	FSMA Produce Safety Rule (21 CFR Part 112)
	PARAMETER: CERTIFICATE OF ANALYSIS (COA) FOR CHEMICAL RESIDUES COA must be provided with each lot, confirming pesticide residues and heavy metals are below established regulatory limits.	A Certificate of Analysis (COA) from the supplier provides documented evidence that the raw cabbage meets regulatory standards for chemical contaminants. This is a direct verification that the product is safe from harmful levels of pesticides and heavy metals, which cannot be removed by subsequent processing steps.	FDA Food, Drug, and Cosmetic Act (21 U.S.C. § 342 - Adulterated Food)
	PARAMETER: VISUAL INSPECTION OF TRANSPORT VEHICLE AND PRODUCT FOR CHEMICAL CONTAMINATION/ALLERGENS Transport vehicle must be clean, free from chemical residues/odors, and raw cabbage must be segregated from known allergens or other incompatible materials during transport.	Visual inspection of the transport vehicle ensures it is clean and free from residues of sanitizers or other chemicals that could contaminate the product. Proper segregation during transport prevents cross-contamination with undeclared allergens or other chemical substances from other goods.	FDA Food Code 2022 (Section 3-302.11)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 2 Step 1: RECEIVING RAW CABBAGE Hazard: Pathogenic bacteria (e.g., Salmonella, E. coli O157:H7, Listeria monocytogenes), viruses (e.g., Norovirus), or parasites (e.g., Cyclospora cayetanensis) on raw cabbage.	PARAMETER: SUPPLIER STATUS Raw cabbage must be sourced from an approved supplier.	An approved supplier program ensures that the raw material originates from a source that has demonstrated adherence to food safety standards (e.g., GAP, GMPs), thereby minimizing the risk of initial contamination with pathogens. This proactive control prevents the introduction of hazards that cannot be eliminated later in the process.	FSMA Produce Safety Rule (21 CFR Part 112)
	PARAMETER: PRODUCT TEMPERATURE UPON RECEIPT Internal temperature of raw cabbage must be <= 4°C (40°F).	Maintaining raw produce at refrigeration temperatures (0-4°C) significantly slows the growth of most pathogenic bacteria (e.g., Listeria monocytogenes, Salmonella, E. coli O157:H7) and spoilage microorganisms. This prevents the proliferation of any pathogens that may be present on the produce during transit and storage prior to processing.	FDA Food Code 2022 (Section 3-202.11)
	PARAMETER: VISUAL INSPECTION FOR INTEGRITY AND SANITATION Raw cabbage must be free from visible signs of spoilage, pest infestation, foreign material, or unsanitary conditions (e.g., excessive dirt, mold, damaged packaging).	Visual inspection upon receipt helps identify potential contamination or conditions that could lead to pathogen growth or introduction. Damaged produce, signs of pest activity, or unsanitary transport conditions can compromise food safety and indicate a failure in the supplier's controls.	FSMA Produce Safety Rule (21 CFR Part 112.111)
CCP 3 Step 1: RECEIVING RAW CABBAGE Hazard: Pesticide residues, heavy metals, sanitizers (from transport vehicle), or allergens (cross-contamination from other produce in transport).	PARAMETER: STORAGE TEMPERATURE <= 4°C (40°F)	Maintaining raw material storage temperatures at or below 4°C (40°F) is critical to inhibit the growth of psychrotrophic pathogens like Listeria monocytogenes and other spoilage microorganisms. This temperature range significantly slows microbial proliferation, ensuring the raw cabbage remains safe before processing. It aligns with established food safety guidelines for refrigerated storage of perishable produce.	FDA Food Code 2022, Chapter 3-501.16 (Potentially Hazardous Food, Hot and Cold Holding)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 4 Step 1: RECEIVING RAW CABBAGE <i>Hazard: Pesticide residues, heavy metals, sanitizers (from transport vehicle), or allergens (cross-contamination from other produce in transport).</i>	PARAMETER: SUPPLIER STATUS Raw cabbage must be sourced from an approved supplier.	<i>An approved supplier program ensures that the supplier has controls in place to manage chemical hazards, such as adherence to maximum residue limits (MRLs) for pesticides, monitoring for heavy metals in growing environments, and proper sanitation practices for transport vehicles. This minimizes the risk of chemical contamination that cannot be removed by subsequent processing.</i>	FSMA Produce Safety Rule (21 CFR Part 112)
	PARAMETER: CERTIFICATE OF ANALYSIS (COA) FOR CHEMICAL RESIDUES COA must be provided with each lot, confirming pesticide residues and heavy metals are below established regulatory limits.	<i>A Certificate of Analysis (COA) from the supplier provides documented evidence that the raw cabbage meets regulatory standards for chemical contaminants. This is a direct verification that the product is safe from harmful levels of pesticides and heavy metals, which cannot be removed by subsequent processing steps.</i>	FDA Food, Drug, and Cosmetic Act (21 U.S.C. § 342 - Adulterated Food)
	PARAMETER: VISUAL INSPECTION OF TRANSPORT VEHICLE AND PRODUCT FOR CHEMICAL CONTAMINATION/ALLERGENS Transport vehicle must be clean, free from chemical residues/odors, and raw cabbage must be segregated from known allergens or other incompatible materials during transport.	<i>Visual inspection of the transport vehicle ensures it is clean and free from residues of sanitizers or other chemicals that could contaminate the product. Proper segregation during transport prevents cross-contamination with undeclared allergens or other chemical substances from other goods.</i>	FDA Food Code 2022 (Section 3-302.11)
CCP 5 Step 2: RAW MATERIAL COLD STORAGE <i>Hazard: Proliferation of pathogenic bacteria (e.g., Listeria monocytogenes, psychrotrophic pathogens) due to temperature abuse or cross-contamination.</i>	PARAMETER: PRODUCT HOLDING TIME/TEMPERATURE Cumulative time product is held above 4°C (40°F) must not exceed 4 hours.	<i>Limiting the cumulative time that raw cabbage is held at temperatures above 4°C (40°F) during pre-processing prevents the rapid proliferation of pathogenic bacteria. This critical limit minimizes the opportunity for microbial growth to unsafe levels, especially for pathogens that can multiply quickly at ambient temperatures. The 4-hour limit is a widely accepted standard for time/temperature control for safety foods.</i>	FDA Food Code 2022, Chapter 3-501.19 (Time as a Public Health Control)
	PARAMETER: FOOD CONTACT SURFACE SANITATION Sanitizer concentration (e.g., free chlorine) meets manufacturer's specifications and regulatory requirements (e.g., 50-200 ppm free chlorine) with a minimum contact time of 1 minute.	<i>Effective sanitation of food contact surfaces (e.g., knives, cutting boards, conveyors) is crucial to prevent cross-contamination of pathogens from equipment to the product. The specified sanitizer concentration and contact time ensure that microbial loads on these surfaces are reduced to acceptable levels, thereby preventing the transfer of bacteria to the sliced cabbage. This directly addresses the hazard of pathogen transfer.</i>	21 CFR 178.1010 (Sanitizing Solutions), FDA Food Code 2022, Chapter 4-501.114 (Manual and Mechanical Warewashing Equipment, Chemical Sanitization)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 6 Step 2: RAW MATERIAL COLD STORAGE <i>Hazard: Proliferation of pathogenic bacteria (e.g., Listeria monocytogenes, psychrotrophic pathogens) due to temperature abuse or cross-contamination.</i>	PARAMETER: STORAGE TEMPERATURE <= 4°C (40°F)	Maintaining raw material storage temperatures at or below 4°C (40°F) is critical to inhibit the growth of psychrotrophic pathogens like <i>Listeria monocytogenes</i> and other spoilage microorganisms. This temperature range significantly slows microbial proliferation, ensuring the raw cabbage remains safe before processing. It aligns with established food safety guidelines for refrigerated storage of perishable produce.	FDA Food Code 2022, Chapter 3-501.16 (Potentially Hazardous Food, Hot and Cold Holding)
CCP 7 Step 3: PRE-PROCESSING PREPARATION <i>Hazard: Pathogen transfer from outer leaves/core to inner leaves, cross-contamination from unsanitized equipment/surfaces/personnel, or proliferation if product is held at ambient temperature for too long.</i>	PARAMETER: ABSENCE OF VISIBLE RESIDUE No visible residue of cleaning chemicals, lubricants, or previous product material on food contact surfaces.	Visual inspection is a fundamental control to ensure the physical removal of gross contamination. The absence of visible residues indicates effective cleaning, which is a prerequisite for effective sanitation. This prevents direct chemical or allergen transfer to the product.	21 CFR Part 117 - Current Good Manufacturing Practice, Hazard Analysis, and Risk-Based Preventive Controls for Human Food
	PARAMETER: ATP SWAB RESULTS < 200 RLU (Relative Light Units) on food contact surfaces.	ATP testing quantitatively measures organic residue, indicating the effectiveness of cleaning procedures. Achieving a low RLU value confirms that surfaces are microbiologically clean. This significantly reduces the risk of chemical or allergen cross-contamination by ensuring thorough cleaning before sanitation.	Industry Best Practices (e.g., GFSI recognized schemes like BRCS, SQF)
	PARAMETER: ALLERGEN TEST RESULTS Negative (< Limit of Detection) for specified allergens on food contact surfaces after cleaning.	When shared lines process allergenic products, specific allergen testing is crucial to validate cleaning effectiveness and prevent cross-contact. Negative results confirm the absence of allergenic proteins to a detectable level. This protects sensitive consumers from potential allergic reactions.	FDA Food Allergen Labeling and Consumer Protection Act (FALCPA)
CCP 8 Step 3: PRE-PROCESSING PREPARATION <i>Hazard: Pathogen transfer from outer leaves/core to inner leaves, cross-contamination from unsanitized equipment/surfaces/personnel, or proliferation if product is held at ambient temperature for too long.</i>	PARAMETER: PRODUCT HOLDING TIME/TEMPERATURE Cumulative time product is held above 4°C (40°F) must not exceed 4 hours.	Limiting the cumulative time that raw cabbage is held at temperatures above 4°C (40°F) during pre-processing prevents the rapid proliferation of pathogenic bacteria. This critical limit minimizes the opportunity for microbial growth to unsafe levels, especially for pathogens that can multiply quickly at ambient temperatures. The 4-hour limit is a widely accepted standard for time/temperature control for safety foods.	FDA Food Code 2022, Chapter 3-501.19 (Time as a Public Health Control)
	PARAMETER: FOOD CONTACT SURFACE SANITATION Sanitizer concentration (e.g., free chlorine) meets manufacturer's specifications and regulatory requirements (e.g., 50-200 ppm free chlorine) with a minimum contact time of 1 minute.	Effective sanitation of food contact surfaces (e.g., knives, cutting boards, conveyors) is crucial to prevent cross-contamination of pathogens from equipment to the product. The specified sanitizer concentration and contact time ensure that microbial loads on these surfaces are reduced to acceptable levels, thereby preventing the transfer of bacteria to the sliced cabbage. This directly addresses the hazard of pathogen transfer.	21 CFR 178.1010 (Sanitizing Solutions), FDA Food Code 2022, Chapter 4-501.114 (Manual and Mechanical Warewashing Equipment, Chemical Sanitization)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 9 Step 3: PRE-PROCESSING PREPARATION <i>Hazard: Cross-contamination from cleaning chemicals, lubricants, or allergens from other products processed on shared lines.</i>	PARAMETER: ABSENCE OF VISIBLE FOREIGN MATERIAL No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.	<i>Visual inspection by trained personnel is a primary control measure to identify and remove gross physical contaminants. This proactive step prevents the incorporation of foreign materials into the final product. Such contaminants could pose choking hazards or other physical injuries to consumers.</i>	21 CFR Part 117 - Current Good Manufacturing Practice, Hazard Analysis, and Risk-Based Preventive Controls for Human Food
	PARAMETER: INTEGRITY OF SCREENS/SIEVES All screens and sieves are intact, free from damage, and of appropriate mesh size (e.g., < 5 mm opening) to prevent passage of specified foreign materials.	<i>Screens and sieves act as physical barriers designed to remove specific sizes of foreign material. Maintaining their integrity and using the correct mesh size ensures their effectiveness. This prevents contaminants like insects or large plant debris from proceeding further in the process.</i>	Industry Best Practices (e.g., GFSI recognized schemes like BRCGS, SQF)
	PARAMETER: METAL DETECTOR SENSITIVITY Metal detector set to detect ferrous (e.g., 1.5 mm), non-ferrous (e.g., 2.0 mm), and stainless steel (e.g., 2.5 mm) contaminants.	<i>Metal detectors are highly effective in identifying and rejecting metallic foreign objects that could pose a physical hazard. Setting appropriate sensitivity levels ensures that common types of metal contaminants are detected. This protects consumers from potential injury.</i>	Industry Best Practices (e.g., GFSI recognized schemes like BRCGS, SQF)
CCP 10 Step 3: PRE-PROCESSING PREPARATION <i>Hazard: Cross-contamination from cleaning chemicals, lubricants, or allergens from other products processed on shared lines.</i>	PARAMETER: ABSENCE OF VISIBLE RESIDUE No visible residue of cleaning chemicals, lubricants, or previous product material on food contact surfaces.	<i>Visual inspection is a fundamental control to ensure the physical removal of gross contamination. The absence of visible residues indicates effective cleaning, which is a prerequisite for effective sanitation. This prevents direct chemical or allergen transfer to the product.</i>	21 CFR Part 117 - Current Good Manufacturing Practice, Hazard Analysis, and Risk-Based Preventive Controls for Human Food
	PARAMETER: ATP SWAB RESULTS < 200 RLU (Relative Light Units) on food contact surfaces.	<i>ATP testing quantitatively measures organic residue, indicating the effectiveness of cleaning procedures. Achieving a low RLU value confirms that surfaces are microbiologically clean. This significantly reduces the risk of chemical or allergen cross-contamination by ensuring thorough cleaning before sanitation.</i>	Industry Best Practices (e.g., GFSI recognized schemes like BRCGS, SQF)
	PARAMETER: ALLERGEN TEST RESULTS Negative (< Limit of Detection) for specified allergens on food contact surfaces after cleaning.	<i>When shared lines process allergenic products, specific allergen testing is crucial to validate cleaning effectiveness and prevent cross-contact. Negative results confirm the absence of allergenic proteins to a detectable level. This protects sensitive consumers from potential allergic reactions.</i>	FDA Food Allergen Labeling and Consumer Protection Act (FALCPA)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 11 Step 3: PRE-PROCESSING PREPARATION <i>Hazard: Foreign material (e.g., plant debris, insects, plastic/metal from equipment, personal effects).</i>	PARAMETER: MICROBIOLOGICAL POTABILITY OF PROCESS WATER E. coli absent in 100 mL; Total Coliforms absent in 100 mL	<i>Potable water standards are designed to ensure water is safe for human consumption and processing, preventing the introduction of microbial pathogens. E. coli is a primary indicator of fecal contamination, signaling the potential presence of other enteric pathogens like Salmonella or Giardia. Total coliforms indicate general water quality and potential for contamination.</i>	U.S. Environmental Protection Agency (EPA) National Primary Drinking Water Regulations (NPDWRs)
CCP 12 Step 3: PRE-PROCESSING PREPARATION <i>Hazard: Foreign material (e.g., plant debris, insects, plastic/metal from equipment, personal effects).</i>	PARAMETER: ABSENCE OF VISIBLE FOREIGN MATERIAL No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.	<i>Visual inspection by trained personnel is a primary control measure to identify and remove gross physical contaminants. This proactive step prevents the incorporation of foreign materials into the final product. Such contaminants could pose choking hazards or other physical injuries to consumers.</i>	21 CFR Part 117 - Current Good Manufacturing Practice, Hazard Analysis, and Risk-Based Preventive Controls for Human Food
	PARAMETER: INTEGRITY OF SCREENS/SIEVES All screens and sieves are intact, free from damage, and of appropriate mesh size (e.g., < 5 mm opening) to prevent passage of specified foreign materials.	<i>Screens and sieves act as physical barriers designed to remove specific sizes of foreign material. Maintaining their integrity and using the correct mesh size ensures their effectiveness. This prevents contaminants like insects or large plant debris from proceeding further in the process.</i>	Industry Best Practices (e.g., GFSI recognized schemes like BRCGS, SQF)
	PARAMETER: METAL DETECTOR SENSITIVITY Metal detector set to detect ferrous (e.g., 1.5 mm), non-ferrous (e.g., 2.0 mm), and stainless steel (e.g., 2.5 mm) contaminants.	<i>Metal detectors are highly effective in identifying and rejecting metallic foreign objects that could pose a physical hazard. Setting appropriate sensitivity levels ensures that common types of metal contaminants are detected. This protects consumers from potential injury.</i>	Industry Best Practices (e.g., GFSI recognized schemes like BRCGS, SQF)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 13 Step 5: SLICING/SHREDDING <i>Hazard: Pathogenic bacteria (e.g., E. coli, Salmonella, Listeria) or protozoa (e.g., Giardia, Cryptosporidium) if water is not potable or becomes contaminated.</i>	PARAMETER: FREE CHLORINE RESIDUAL IN PROCESS WATER ≤ 4 mg/L (ppm)	<i>The EPA sets a Maximum Contaminant Level (MCL) for chlorine and chloramines in drinking water to prevent adverse health effects from long-term exposure and to avoid undesirable sensory impacts on food products. Levels above this can also indicate issues with water treatment or contamination, posing a chemical hazard to the product.</i>	U.S. Environmental Protection Agency (EPA) National Primary Drinking Water Regulations (NPDWRs) for Disinfectants and Disinfection Byproducts
	PARAMETER: HEAVY METALS AND INDUSTRIAL CONTAMINANTS IN PROCESS WATER Levels of regulated heavy metals (e.g., Lead, Arsenic, Mercury) and industrial contaminants (e.g., Benzene, Vinyl Chloride) must be below their respective Maximum Contaminant Levels (MCLs)	<i>MCLs are legally enforceable standards that apply to public water systems and are set to protect public health by limiting the levels of contaminants in drinking water. These contaminants are known to cause various adverse health effects and their presence in process water would contaminate the food product.</i>	U.S. Environmental Protection Agency (EPA) National Primary Drinking Water Regulations (NPDWRs) for Inorganic and Organic Chemicals
	PARAMETER: ABSENCE OF CLEANING CHEMICALS IN PROCESS WATER Undetectable levels of cleaning chemicals (e.g., detergents, sanitizers not intentionally added for water treatment)	<i>Cleaning chemicals are not food-grade and can pose significant chemical hazards if present in process water, leading to product contamination and potential consumer illness. Their presence indicates inadequate rinsing of water lines or cross-contamination, which violates fundamental food safety principles.</i>	Generally Recognized as Good Manufacturing Practices (GMPs) and Food Safety Principles (e.g., 21 CFR Part 117)
CCP 14 Step 5: SLICING/SHREDDING <i>Hazard: Pathogenic bacteria (e.g., E. coli, Salmonella, Listeria) or protozoa (e.g., Giardia, Cryptosporidium) if water is not potable or becomes contaminated.</i>	PARAMETER: MICROBIOLOGICAL POTABILITY OF PROCESS WATER E. coli absent in 100 mL; Total Coliforms absent in 100 mL	<i>Potable water standards are designed to ensure water is safe for human consumption and processing, preventing the introduction of microbial pathogens. E. coli is a primary indicator of fecal contamination, signaling the potential presence of other enteric pathogens like Salmonella or Giardia. Total coliforms indicate general water quality and potential for contamination.</i>	U.S. Environmental Protection Agency (EPA) National Primary Drinking Water Regulations (NPDWRs)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 15 Step 5: SLICING/SHREDDING <i>Hazard: Excessive levels of chlorine, heavy metals, industrial contaminants, cleaning chemicals from water lines.</i>	PARAMETER: SANITIZER CONCENTRATION (E.G., FREE CHLORINE) 50-200 ppm free chlorine (or other approved sanitizer at manufacturer's recommended effective concentration)	<i>Maintaining the sanitizer concentration within the specified range ensures effective reduction of pathogenic bacteria and protozoa on the surface of the cabbage. This range is scientifically proven to be effective against common foodborne pathogens while remaining safe for food contact and product quality.</i>	FDA Guide to Minimize Microbial Food Safety Hazards for Fresh Fruits and Vegetables (2008), EPA regulations for food-contact sanitizers.
	PARAMETER: WATER PH 6.0-7.5 (for chlorine-based sanitizers)	<i>The efficacy of chlorine-based sanitizers is highly dependent on pH. Within this range, hypochlorous acid (HOCl), the most effective antimicrobial form of chlorine, predominates, ensuring optimal pathogen reduction. Deviations outside this range significantly reduce sanitizer effectiveness.</i>	FDA Guide to Minimize Microbial Food Safety Hazards for Fresh Fruits and Vegetables (2008).
	PARAMETER: WATER POTABILITY Meets EPA National Primary Drinking Water Regulations	<i>All water used in direct contact with food, including washing and rinsing steps, must be potable to prevent the introduction of microbial, chemical, or physical hazards. Using non-potable water would negate the purpose of this CCP and introduce significant food safety risks.</i>	EPA National Primary Drinking Water Regulations, FDA Food Code 2022.
CCP 16 Step 5: SLICING/SHREDDING <i>Hazard: Excessive levels of chlorine, heavy metals, industrial contaminants, cleaning chemicals from water lines.</i>	PARAMETER: FREE CHLORINE RESIDUAL IN PROCESS WATER ≤ 4 mg/L (ppm)	<i>The EPA sets a Maximum Contaminant Level (MCL) for chlorine and chloramines in drinking water to prevent adverse health effects from long-term exposure and to avoid undesirable sensory impacts on food products. Levels above this can also indicate issues with water treatment or contamination, posing a chemical hazard to the product.</i>	U.S. Environmental Protection Agency (EPA) National Primary Drinking Water Regulations (NPDWRs) for Disinfectants and Disinfection Byproducts
	PARAMETER: HEAVY METALS AND INDUSTRIAL CONTAMINANTS IN PROCESS WATER Levels of regulated heavy metals (e.g., Lead, Arsenic, Mercury) and industrial contaminants (e.g., Benzene, Vinyl Chloride) must be below their respective Maximum Contaminant Levels (MCLs)	<i>MCLs are legally enforceable standards that apply to public water systems and are set to protect public health by limiting the levels of contaminants in drinking water. These contaminants are known to cause various adverse health effects and their presence in process water would contaminate the food product.</i>	U.S. Environmental Protection Agency (EPA) National Primary Drinking Water Regulations (NPDWRs) for Inorganic and Organic Chemicals
	PARAMETER: ABSENCE OF CLEANING CHEMICALS IN PROCESS WATER Undetectable levels of cleaning chemicals (e.g., detergents, sanitizers not intentionally added for water treatment)	<i>Cleaning chemicals are not food-grade and can pose significant chemical hazards if present in process water, leading to product contamination and potential consumer illness. Their presence indicates inadequate rinsing of water lines or cross-contamination, which violates fundamental food safety principles.</i>	Generally Recognized as Good Manufacturing Practices (GMPs) and Food Safety Principles (e.g., 21 CFR Part 117)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 17 Step 6: POST-SLICING WASHING & RINSING <i>Hazard: Pathogenic bacteria (e.g., E. coli, Salmonella, Listeria) or protozoa if water is not potable or if sanitizer is ineffective/not used correctly.</i>	PARAMETER: PRODUCT TEMPERATURE Product internal temperature <= 4°C (40°F)	<i>Maintaining the product at or below 4°C (40°F) is critical to inhibit the growth and proliferation of pathogenic bacteria such as Salmonella spp., E. coli O157:H7, and Listeria monocytogenes. This temperature range is a recognized safe holding temperature for refrigerated produce, preventing rapid microbial multiplication.</i>	FDA Food Code 2022, USDA FSIS regulations for refrigerated foods.
	PARAMETER: EQUIPMENT FOOD CONTACT SURFACE CLEANLINESS ATP swab readings < 100 Relative Light Units (RLU) on all food contact surfaces post-sanitation, AND/OR no detectable Listeria monocytogenes (or other relevant pathogens) on environmental swabs from food contact surfaces.	<i>Effective cleaning and sanitation of equipment food contact surfaces are essential to prevent cross-contamination with pathogenic bacteria. ATP testing provides a rapid indicator of organic residue, while microbiological swabbing directly verifies the absence of specific pathogens, ensuring the equipment does not contribute to microbial hazards.</i>	FDA Current Good Manufacturing Practices (21 CFR Part 117), industry best practices for sanitation verification.
CCP 18 Step 6: POST-SLICING WASHING & RINSING <i>Hazard: Pathogenic bacteria (e.g., E. coli, Salmonella, Listeria) or protozoa if water is not potable or if sanitizer is ineffective/not used correctly.</i>	PARAMETER: SANITIZER CONCENTRATION (E.G., FREE CHLORINE) 50-200 ppm free chlorine (or other approved sanitizer at manufacturer's recommended effective concentration)	<i>Maintaining the sanitizer concentration within the specified range ensures effective reduction of pathogenic bacteria and protozoa on the surface of the cabbage. This range is scientifically proven to be effective against common foodborne pathogens while remaining safe for food contact and product quality.</i>	FDA Guide to Minimize Microbial Food Safety Hazards for Fresh Fruits and Vegetables (2008), EPA regulations for food-contact sanitizers.
	PARAMETER: WATER PH 6.0-7.5 (for chlorine-based sanitizers)	<i>The efficacy of chlorine-based sanitizers is highly dependent on pH. Within this range, hypochlorous acid (HOCl), the most effective antimicrobial form of chlorine, predominates, ensuring optimal pathogen reduction. Deviations outside this range significantly reduce sanitizer effectiveness.</i>	FDA Guide to Minimize Microbial Food Safety Hazards for Fresh Fruits and Vegetables (2008).
	PARAMETER: WATER POTABILITY Meets EPA National Primary Drinking Water Regulations	<i>All water used in direct contact with food, including washing and rinsing steps, must be potable to prevent the introduction of microbial, chemical, or physical hazards. Using non-potable water would negate the purpose of this CCP and introduce significant food safety risks.</i>	EPA National Primary Drinking Water Regulations, FDA Food Code 2022.

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 19 Step 8: VISUAL INSPECTION & SORTING <i>Hazard: Proliferation of pathogenic bacteria (e.g., Salmonella spp., E. coli O157:H7, Listeria monocytogenes) due to temperature abuse or cross-contamination from equipment.</i>	PARAMETER: LUBRICANT TYPE Only NSF H1 or H3 certified food-grade lubricants used on equipment with potential food contact.	<i>Using NSF H1 (incidental food contact) or H3 (direct food contact) certified lubricants ensures that any accidental contact with the product does not introduce harmful chemical contaminants. Non-food grade lubricants can contain toxic compounds that pose a chemical hazard if ingested.</i>	NSF International Standards for Nonfood Compounds
	PARAMETER: CHEMICAL RESIDUE ON FOOD CONTACT SURFACES No detectable chemical residues (e.g., cleaning agents, sanitizers) on food contact surfaces after rinsing, verified by visual inspection and/or appropriate testing (e.g., pH strips, ATP swabs).	<i>Effective rinsing is critical to remove cleaning and sanitizing agents, which can be harmful if ingested. Detectable residues indicate inadequate rinsing, posing a chemical hazard to the consumer. Visual inspection and/or chemical tests confirm the absence of these contaminants.</i>	21 CFR 117.35(a) - Current Good Manufacturing Practice in Manufacturing, Packing, or Holding Human Food (Sanitation)
CCP 20 Step 8: VISUAL INSPECTION & SORTING <i>Hazard: Proliferation of pathogenic bacteria (e.g., Salmonella spp., E. coli O157:H7, Listeria monocytogenes) due to temperature abuse or cross-contamination from equipment.</i>	PARAMETER: PRODUCT TEMPERATURE Product internal temperature $\leq 4^{\circ}\text{C}$ (40°F)	<i>Maintaining the product at or below 4°C (40°F) is critical to inhibit the growth and proliferation of pathogenic bacteria such as Salmonella spp., E. coli O157:H7, and Listeria monocytogenes. This temperature range is a recognized safe holding temperature for refrigerated produce, preventing rapid microbial multiplication.</i>	FDA Food Code 2022, USDA FSIS regulations for refrigerated foods.
	PARAMETER: EQUIPMENT FOOD CONTACT SURFACE CLEANLINESS ATP swab readings < 100 Relative Light Units (RLU) on all food contact surfaces post-sanitation, AND/OR no detectable Listeria monocytogenes (or other relevant pathogens) on environmental swabs from food contact surfaces.	<i>Effective cleaning and sanitation of equipment food contact surfaces are essential to prevent cross-contamination with pathogenic bacteria. ATP testing provides a rapid indicator of organic residue, while microbiological swabbing directly verifies the absence of specific pathogens, ensuring the equipment does not contribute to microbial hazards.</i>	FDA Current Good Manufacturing Practices (21 CFR Part 117), industry best practices for sanitation verification.
CCP 21 Step 8: VISUAL INSPECTION & SORTING <i>Hazard: Contamination from lubricants, cleaning chemicals, or sanitizers.</i>	PARAMETER: PRESENCE OF FOREIGN MATERIAL No foreign material (e.g., metal, plastic, glass, wood, stones) detected in the product during visual inspection.	<i>Foreign materials pose a significant physical hazard, potentially causing injury such as choking, dental damage, or lacerations to consumers. Visual inspection at this critical step is essential to identify and remove such contaminants before further processing or packaging, as no subsequent step will eliminate them.</i>	21 CFR 117.110(b) - Hazard Analysis and Risk-Based Preventive Controls (Preventive Controls for Food Safety)
	PARAMETER: INTEGRITY OF PHYSICAL SEPARATION DEVICES All screens and filters are intact, properly installed, and free from damage (e.g., tears, holes) with specified mesh size maintained.	<i>Screens and filters are designed to physically block and remove foreign materials from the product stream. Their integrity is paramount; any damage or improper installation compromises their effectiveness, allowing contaminants to pass through. Regular inspection ensures they function as intended to prevent physical hazards.</i>	21 CFR 117.35(a) - Current Good Manufacturing Practice in Manufacturing, Packing, or Holding Human Food (Sanitation)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 22 Step 8: VISUAL INSPECTION & SORTING <i>Hazard: Contamination from lubricants, cleaning chemicals, or sanitizers.</i>	PARAMETER: LUBRICANT TYPE Only NSF H1 or H3 certified food-grade lubricants used on equipment with potential food contact.	<i>Using NSF H1 (incidental food contact) or H3 (direct food contact) certified lubricants ensures that any accidental contact with the product does not introduce harmful chemical contaminants. Non-food grade lubricants can contain toxic compounds that pose a chemical hazard if ingested.</i>	NSF International Standards for Nonfood Compounds
	PARAMETER: CHEMICAL RESIDUE ON FOOD CONTACT SURFACES No detectable chemical residues (e.g., cleaning agents, sanitizers) on food contact surfaces after rinsing, verified by visual inspection and/or appropriate testing (e.g., pH strips, ATP swabs).	<i>Effective rinsing is critical to remove cleaning and sanitizing agents, which can be harmful if ingested. Detectable residues indicate inadequate rinsing, posing a chemical hazard to the consumer. Visual inspection and/or chemical tests confirm the absence of these contaminants.</i>	21 CFR 117.35(a) - Current Good Manufacturing Practice in Manufacturing, Packing, or Holding Human Food (Sanitation)
CCP 23 Step 8: VISUAL INSPECTION & SORTING <i>Hazard: Foreign material (e.g., metal fragments from equipment wear, plastic pieces, loose parts from machinery)</i>	PARAMETER: VISUAL INSPECTION FOR FOREIGN MATERIAL No visible foreign material present in product or on equipment surfaces.	<i>The absence of visible foreign material ensures the product meets safety and quality standards, preventing physical injury or choking hazards for consumers. Any visible foreign material would render the product unsafe or unacceptable, requiring immediate corrective action.</i>	21 CFR Part 117 Subpart B (Current Good Manufacturing Practice)
	PARAMETER: PERSONNEL ADHERENCE TO GMPs All personnel handling product must wear appropriate protective clothing (hairnets, beard nets if applicable, clean uniforms), remove all jewelry (except plain wedding bands), and not carry personal items (e.g., pens, cell phones) into the processing area.	<i>Adherence to Good Manufacturing Practices (GMPs) for personnel hygiene is fundamental in preventing the introduction of physical contaminants such as hair, jewelry, or personal items into the food product. This critical limit directly mitigates the risk of foreign material contamination from human sources during handling.</i>	21 CFR Part 117.10 (Personnel)
	PARAMETER: EQUIPMENT INTEGRITY All food contact surfaces of weighing and portioning equipment must be free from damage (e.g., cracks, chips, rust, loose parts) that could lead to foreign material contamination.	<i>Damaged equipment surfaces can shed fragments of plastic, metal, or other materials into the product, posing a significant physical hazard. Regular inspection and maintenance ensure equipment integrity, preventing the introduction of foreign material from equipment breakdown or wear.</i>	21 CFR Part 117.40 (Equipment and Utensils)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 24 Step 8: VISUAL INSPECTION & SORTING Hazard: Foreign material (e.g., metal fragments from equipment wear, plastic pieces, loose parts from machinery)	PARAMETER: PRESENCE OF FOREIGN MATERIAL No foreign material (e.g., metal, plastic, glass, wood, stones) detected in the product during visual inspection.	Foreign materials pose a significant physical hazard, potentially causing injury such as choking, dental damage, or lacerations to consumers. Visual inspection at this critical step is essential to identify and remove such contaminants before further processing or packaging, as no subsequent step will eliminate them.	21 CFR 117.110(b) - Hazard Analysis and Risk-Based Preventive Controls (Preventive Controls for Food Safety)
	PARAMETER: INTEGRITY OF PHYSICAL SEPARATION DEVICES All screens and filters are intact, properly installed, and free from damage (e.g., tears, holes) with specified mesh size maintained.	Screens and filters are designed to physically block and remove foreign materials from the product stream. Their integrity is paramount; any damage or improper installation compromises their effectiveness, allowing contaminants to pass through. Regular inspection ensures they function as intended to prevent physical hazards.	21 CFR 117.35(a) - Current Good Manufacturing Practice in Manufacturing, Packing, or Holding Human Food (Sanitation)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 25 Step 9: WEIGHING & PORTIONING Hazard: Foreign material (e.g., hair, jewelry, personal items from personnel; plastic/metal from equipment; plant debris missed in previous steps)	PARAMETER: PRODUCT TEMPERATURE DURING PACKAGING Product temperature must not exceed 4°C (40°F) during the entire packaging process.	Maintaining product temperature below 4°C (40°F) significantly slows the growth of most pathogenic bacteria, including <i>Listeria monocytogenes</i> , <i>Salmonella</i> spp., and <i>E. coli</i> O157:H7, which are psychrotrophic or mesophilic. This critical limit prevents microbial proliferation during the packaging step, ensuring the product remains safe until it reaches the consumer.	FDA Food Code 2022, 3-401.11 (Temperature and Time Control for Safety)
	PARAMETER: PACKAGING MATERIAL INTEGRITY AND CLEANLINESS All primary packaging materials (PE bags, PP clam-shells) must be food-grade, intact, clean, and free from visible defects or contamination prior to use.	Primary packaging materials must be food-grade, intact, and clean to prevent the introduction of biological hazards to the product. This ensures the product is protected from environmental contamination and maintains its safety and integrity throughout its shelf life.	21 CFR Part 174 (Indirect Food Additives: General), 21 CFR Part 177 (Indirect Food Additives: Polymers)
	PARAMETER: ENVIRONMENTAL HYGIENE IN PACKAGING AREA All food contact surfaces of packaging equipment and surrounding environment must be cleaned and sanitized prior to and during packaging operations as necessary, verified by visual inspection and ATP testing (or equivalent).	A clean and sanitized packaging environment is crucial to prevent cross-contamination from surfaces, air, or personnel to the product during the packaging process. This control is particularly important for preventing the reintroduction of pathogens like <i>Listeria monocytogenes</i> , which can persist in processing environments.	21 CFR Part 117.35 (Sanitation), 21 CFR Part 117.80 (Processes and Controls)
	PARAMETER: MODIFIED ATMOSPHERE PACKAGING (MAP) GAS COMPOSITION (IF USED FOR SAFETY) Oxygen (O2) concentration in packaged product headspace must be maintained below 2% and Carbon Dioxide (CO2) concentration must be maintained between 5-15%.	Specific gas compositions in Modified Atmosphere Packaging (MAP) can inhibit the growth of aerobic spoilage organisms and some pathogenic bacteria, thereby extending shelf life and contributing to product safety. Low oxygen levels reduce the growth of aerobic pathogens, while carbon dioxide can have bacteriostatic effects, provided anaerobic conditions that favor <i>Clostridium botulinum</i> are avoided.	Industry Best Practices for MAP in Fresh Produce (e.g., IFPA, CPMA guidance)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 26 Step 9: WEIGHING & PORTIONING <i>Hazard: Foreign material (e.g., hair, jewelry, personal items from personnel; plastic/metal from equipment; plant debris missed in previous steps)</i>	PARAMETER: VISUAL INSPECTION FOR FOREIGN MATERIAL No visible foreign material present in product or on equipment surfaces.	<i>The absence of visible foreign material ensures the product meets safety and quality standards, preventing physical injury or choking hazards for consumers. Any visible foreign material would render the product unsafe or unacceptable, requiring immediate corrective action.</i>	21 CFR Part 117 Subpart B (Current Good Manufacturing Practice)
	PARAMETER: PERSONNEL ADHERENCE TO GMPS All personnel handling product must wear appropriate protective clothing (hairnets, beard nets if applicable, clean uniforms), remove all jewelry (except plain wedding bands), and not carry personal items (e.g., pens, cell phones) into the processing area.	<i>Adherence to Good Manufacturing Practices (GMPs) for personnel hygiene is fundamental in preventing the introduction of physical contaminants such as hair, jewelry, or personal items into the food product. This critical limit directly mitigates the risk of foreign material contamination from human sources during handling.</i>	21 CFR Part 117.10 (Personnel)
	PARAMETER: EQUIPMENT INTEGRITY All food contact surfaces of weighing and portioning equipment must be free from damage (e.g., cracks, chips, rust, loose parts) that could lead to foreign material contamination.	<i>Damaged equipment surfaces can shed fragments of plastic, metal, or other materials into the product, posing a significant physical hazard. Regular inspection and maintenance ensure equipment integrity, preventing the introduction of foreign material from equipment breakdown or wear.</i>	21 CFR Part 117.40 (Equipment and Utensils)
CCP 27 Step 10: PRIMARY PACKAGING <i>Hazard: Pathogenic bacteria (e.g., Salmonella spp., E. coli O157:H7, Listeria monocytogenes), viruses (e.g., Norovirus), parasites (e.g., Cyclospora cayetanensis)</i>	PARAMETER: PRODUCT TEMPERATURE DURING PACKAGING <= 4°C (40°F)	<i>Maintaining the product temperature at or below 4°C (40°F) during the packaging process is critical to prevent the growth and proliferation of psychrotrophic pathogens like Listeria monocytogenes and other spoilage microorganisms. This temperature control minimizes microbial activity, especially given the high water activity and neutral pH of sliced cabbage, thereby ensuring the microbiological safety and quality of the product.</i>	FDA Food Code 2022, Chapter 3 - Food
	PARAMETER: PACKAGING ENVIRONMENT SANITATION (FOOD CONTACT SURFACES) ATP swab results < 200 RLU (Relative Light Units)	<i>ATP (Adenosine Triphosphate) testing provides a rapid and objective measure of organic residue on food contact surfaces, indicating the effectiveness of cleaning and sanitation procedures. A reading below 200 RLU signifies a clean surface, significantly reducing the risk of transferring pathogenic bacteria from the environment to the product during packaging. This directly contributes to preventing the introduction of pathogens like Listeria and Salmonella.</i>	21 CFR Part 117 - Current Good Manufacturing Practice

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 28 Step 10: PRIMARY PACKAGING Hazard: Pathogenic bacteria (e.g., Salmonella spp., E. coli O157:H7, Listeria monocytogenes), viruses (e.g., Norovirus), parasites (e.g., Cyclospora cayetanensis)	PARAMETER: PRODUCT TEMPERATURE DURING PACKAGING Product temperature must not exceed 4°C (40°F) during the entire packaging process.	Maintaining product temperature below 4°C (40°F) significantly slows the growth of most pathogenic bacteria, including Listeria monocytogenes, Salmonella spp., and E. coli O157:H7, which are psychrotrophic or mesophilic. This critical limit prevents microbial proliferation during the packaging step, ensuring the product remains safe until it reaches the consumer.	FDA Food Code 2022, 3-401.11 (Temperature and Time Control for Safety)
	PARAMETER: PACKAGING MATERIAL INTEGRITY AND CLEANLINESS All primary packaging materials (PE bags, PP clam-shells) must be food-grade, intact, clean, and free from visible defects or contamination prior to use.	Primary packaging materials must be food-grade, intact, and clean to prevent the introduction of biological hazards to the product. This ensures the product is protected from environmental contamination and maintains its safety and integrity throughout its shelf life.	21 CFR Part 174 (Indirect Food Additives: General), 21 CFR Part 177 (Indirect Food Additives: Polymers)
	PARAMETER: ENVIRONMENTAL HYGIENE IN PACKAGING AREA All food contact surfaces of packaging equipment and surrounding environment must be cleaned and sanitized prior to and during packaging operations as necessary, verified by visual inspection and ATP testing (or equivalent).	A clean and sanitized packaging environment is crucial to prevent cross-contamination from surfaces, air, or personnel to the product during the packaging process. This control is particularly important for preventing the reintroduction of pathogens like Listeria monocytogenes, which can persist in processing environments.	21 CFR Part 117.35 (Sanitation), 21 CFR Part 117.80 (Processes and Controls)
	PARAMETER: MODIFIED ATMOSPHERE PACKAGING (MAP) GAS COMPOSITION (IF USED FOR SAFETY) Oxygen (O2) concentration in packaged product headspace must be maintained below 2% and Carbon Dioxide (CO2) concentration must be maintained between 5-15%.	Specific gas compositions in Modified Atmosphere Packaging (MAP) can inhibit the growth of aerobic spoilage organisms and some pathogenic bacteria, thereby extending shelf life and contributing to product safety. Low oxygen levels reduce the growth of aerobic pathogens, while carbon dioxide can have bacteriostatic effects, provided anaerobic conditions that favor Clostridium botulinum are avoided.	Industry Best Practices for MAP in Fresh Produce (e.g., IFPA, CPMA guidance)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 29 Step 11: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING <i>Hazard: Introduction or proliferation of pathogenic bacteria (e.g., Listeria monocytogenes, Salmonella spp.) due to unsanitary packaging environment, contaminated packaging materials, or temperature abuse during packaging.</i>	PARAMETER: PACKAGING MATERIAL COMPLIANCE Packaging materials must be certified food-grade and comply with all applicable regulatory standards for direct food contact.	<i>Ensuring that all primary packaging materials are food-grade and meet regulatory specifications is essential to prevent the migration of non-food-grade chemicals into the sliced cabbage. This critical limit safeguards against chemical contamination, protecting consumer health from potentially harmful substances that could leach from unsuitable packaging.</i>	21 CFR Part 174 - Indirect Food Additives: General
	PARAMETER: ABSENCE OF CLEANING CHEMICAL RESIDUE Zero detectable residue of cleaning and sanitizing chemicals on food contact surfaces after rinsing.	<i>Thorough rinsing after cleaning and sanitizing is paramount to remove all chemical residues from food contact surfaces of packaging equipment. The presence of residual cleaning chemicals can directly contaminate the product, posing a chemical hazard to consumers. Verification of zero detectable residue ensures the safety of the product from chemical contamination.</i>	21 CFR Part 117 - Current Good Manufacturing Practice
	PARAMETER: LUBRICANT TYPE Only NSF H1 certified food-grade lubricants are permitted for use on packaging machinery where there is a potential for incidental food contact.	<i>The use of non-food-grade lubricants on machinery with potential food contact points presents a significant risk of chemical contamination. NSF H1 certified lubricants are specifically formulated to be safe for incidental contact with food, preventing the introduction of harmful chemicals into the product. This critical limit ensures that any lubricant contact does not compromise food safety.</i>	21 CFR Part 178.3570 - Lubricants with incidental food contact
CCP 30 Step 11: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING <i>Hazard: Introduction or proliferation of pathogenic bacteria (e.g., Listeria monocytogenes, Salmonella spp.) due to unsanitary packaging environment, contaminated packaging materials, or temperature abuse during packaging.</i>	PARAMETER: PRODUCT TEMPERATURE DURING PACKAGING <= 4°C (40°F)	<i>Maintaining the product temperature at or below 4°C (40°F) during the packaging process is critical to prevent the growth and proliferation of psychrotrophic pathogens like Listeria monocytogenes and other spoilage microorganisms. This temperature control minimizes microbial activity, especially given the high water activity and neutral pH of sliced cabbage, thereby ensuring the microbiological safety and quality of the product.</i>	FDA Food Code 2022, Chapter 3 - Food
	PARAMETER: PACKAGING ENVIRONMENT SANITATION (FOOD CONTACT SURFACES) ATP swab results < 200 RLU (Relative Light Units)	<i>ATP (Adenosine Triphosphate) testing provides a rapid and objective measure of organic residue on food contact surfaces, indicating the effectiveness of cleaning and sanitation procedures. A reading below 200 RLU signifies a clean surface, significantly reducing the risk of transferring pathogenic bacteria from the environment to the product during packaging. This directly contributes to preventing the introduction of pathogens like Listeria and Salmonella.</i>	21 CFR Part 117 - Current Good Manufacturing Practice

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 31 Step 11: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING <i>Hazard: Chemical contamination from packaging materials (e.g., migration of non-food grade chemicals), cleaning chemicals (if not properly rinsed/sanitized), or lubricants.</i>	PARAMETER: PRODUCT STORAGE TEMPERATURE <= 4°C (40°F)	Maintaining product temperature at or below 4°C significantly inhibits the growth of psychrotrophic pathogens like <i>Listeria monocytogenes</i> and prevents the growth and toxin production of <i>Clostridium botulinum</i>, which is a particular concern in low-oxygen MAP environments where competing spoilage organisms are suppressed.	FDA Food Code 2022, Chapter 3-401.11 (Temperature and Time Control for Safety)
	PARAMETER: MAP OXYGEN (O2) CONCENTRATION >= 3%	Maintaining a minimum oxygen concentration of 3% in the package headspace helps to prevent the creation of strictly anaerobic conditions that could favor the growth and toxin production of <i>Clostridium botulinum</i> if temperature control is lost. While temperature is the primary control, this acts as a secondary barrier.	Industry Best Practices for MAP Fresh Produce, Scientific Literature on <i>Clostridium botulinum</i> in MAP
	PARAMETER: MAP CARBON DIOXIDE (CO2) CONCENTRATION 5-10%	A CO2 concentration in this range effectively inhibits the growth of many spoilage microorganisms and facultative anaerobic pathogens, such as <i>Listeria monocytogenes</i>, thereby extending shelf life and enhancing product safety.	Industry Best Practices for MAP Fresh Produce, Scientific Literature on Antimicrobial Effects of CO2
	PARAMETER: PACKAGE SEAL INTEGRITY No leaks detected (e.g., via vacuum leak test or bubble test)	An intact and strong seal is critical to maintain the intended modified atmosphere, prevent ingress of atmospheric oxygen, and protect the product from external contamination, thus ensuring the efficacy of MAP and product safety.	ASTM F2096 (Standard Test Method for Detecting Gross Leaks in Packaging by Internal Pressurization (Bubble Test)), Industry Best Practices

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 32 Step 11: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING <i>Hazard: Chemical contamination from packaging materials (e.g., migration of non-food grade chemicals), cleaning chemicals (if not properly rinsed/sanitized), or lubricants.</i>	PARAMETER: PACKAGING MATERIAL COMPLIANCE Packaging materials must be certified food-grade and comply with all applicable regulatory standards for direct food contact.	<i>Ensuring that all primary packaging materials are food-grade and meet regulatory specifications is essential to prevent the migration of non-food-grade chemicals into the sliced cabbage. This critical limit safeguards against chemical contamination, protecting consumer health from potentially harmful substances that could leach from unsuitable packaging.</i>	21 CFR Part 174 - Indirect Food Additives: General
	PARAMETER: ABSENCE OF CLEANING CHEMICAL RESIDUE Zero detectable residue of cleaning and sanitizing chemicals on food contact surfaces after rinsing.	<i>Thorough rinsing after cleaning and sanitizing is paramount to remove all chemical residues from food contact surfaces of packaging equipment. The presence of residual cleaning chemicals can directly contaminate the product, posing a chemical hazard to consumers. Verification of zero detectable residue ensures the safety of the product from chemical contamination.</i>	21 CFR Part 117 - Current Good Manufacturing Practice
	PARAMETER: LUBRICANT TYPE Only NSF H1 certified food-grade lubricants are permitted for use on packaging machinery where there is a potential for incidental food contact.	<i>The use of non-food-grade lubricants on machinery with potential food contact points presents a significant risk of chemical contamination. NSF H1 certified lubricants are specifically formulated to be safe for incidental contact with food, preventing the introduction of harmful chemicals into the product. This critical limit ensures that any lubricant contact does not compromise food safety.</i>	21 CFR Part 178.3570 - Lubricants with incidental food contact
CCP 33 Step 12: METAL DETECTION <i>Hazard: Growth of anaerobic or facultative anaerobic pathogens (e.g., Clostridium botulinum, Listeria monocytogenes) due to improper gas mixture, seal integrity failure, or temperature abuse.</i>	PARAMETER: MAP GAS FOOD-GRADE CERTIFICATION 100% food-grade certified gases (e.g., CO ₂ , N ₂ , O ₂)	<i>Using only food-grade certified gases ensures that the gases meet purity standards and are free from harmful contaminants or impurities that could chemically adulterate the product and pose a health risk to consumers.</i>	FDA 21 CFR Part 173 (Secondary Direct Food Additives Permitted in Food for Human Consumption), e.g., 173.340 for Nitrogen, 184.1240 for Carbon Dioxide
	PARAMETER: APPROVED GAS SUPPLIER STATUS MAP gases sourced exclusively from suppliers approved through the company's vendor qualification program	<i>A robust vendor qualification program ensures that gas suppliers consistently provide food-grade gases, adhere to quality specifications, and maintain appropriate handling and delivery protocols, thereby mitigating the risk of chemical contamination.</i>	21 CFR Part 117, Subpart G (Supply-Chain Program), GFSI Recognized Standards (e.g., BRCGS, SQF)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 34 Step 12: METAL DETECTION <i>Hazard: Growth of anaerobic or facultative anaerobic pathogens (e.g., Clostridium botulinum, Listeria monocytogenes) due to improper gas mixture, seal integrity failure, or temperature abuse.</i>	PARAMETER: PRODUCT STORAGE TEMPERATURE <= 4°C (40°F)	Maintaining product temperature at or below 4°C significantly inhibits the growth of psychrotrophic pathogens like <i>Listeria monocytogenes</i> and prevents the growth and toxin production of <i>Clostridium botulinum</i> , which is a particular concern in low-oxygen MAP environments where competing spoilage organisms are suppressed.	FDA Food Code 2022, Chapter 3-401.11 (Temperature and Time Control for Safety)
	PARAMETER: MAP OXYGEN (O2) CONCENTRATION >= 3%	Maintaining a minimum oxygen concentration of 3% in the package headspace helps to prevent the creation of strictly anaerobic conditions that could favor the growth and toxin production of <i>Clostridium botulinum</i> if temperature control is lost. While temperature is the primary control, this acts as a secondary barrier.	Industry Best Practices for MAP Fresh Produce, Scientific Literature on Clostridium botulinum in MAP
	PARAMETER: MAP CARBON DIOXIDE (CO2) CONCENTRATION 5-10%	A CO2 concentration in this range effectively inhibits the growth of many spoilage microorganisms and facultative anaerobic pathogens, such as <i>Listeria monocytogenes</i> , thereby extending shelf life and enhancing product safety.	Industry Best Practices for MAP Fresh Produce, Scientific Literature on Antimicrobial Effects of CO2
	PARAMETER: PACKAGE SEAL INTEGRITY No leaks detected (e.g., via vacuum leak test or bubble test)	An intact and strong seal is critical to maintain the intended modified atmosphere, prevent ingress of atmospheric oxygen, and protect the product from external contamination, thus ensuring the efficacy of MAP and product safety.	ASTM F2096 (Standard Test Method for Detecting Gross Leaks in Packaging by Internal Pressurization (Bubble Test)), Industry Best Practices
CCP 35 Step 12: METAL DETECTION <i>Hazard: Contamination from MAP gases (e.g., non-food grade gases, impurities).</i>	PARAMETER: LABEL ACCURACY AND COMPLETENESS All labels applied to sliced cabbage product must accurately reflect the product contents (100% cabbage, no allergens) and include all mandatory information, including 'WASH BEFORE USE' instruction.	Mislabeling with allergen-containing product labels can lead to severe allergic reactions in sensitive consumers. Providing clear instructions like 'WASH BEFORE USE' is crucial for raw produce to inform consumers about necessary preparation steps to reduce potential microbiological risks, as the product is not ready-to-eat. This ensures consumers are aware of the product's raw nature and the need for further preparation.	FDA Food Allergen Labeling and Consumer Protection Act (FALCPA) of 2004; 21 CFR Part 101 (General Food Labeling)
	PARAMETER: LABEL APPLICATION VERIFICATION 100% of product units must be verified to have the correct label applied for 'sliced cabbage' and contain all mandatory information prior to release from the labeling station.	Verification ensures that the correct label is consistently applied to the correct product, preventing mislabeling errors. This direct check at the point of application is the last opportunity to prevent the distribution of mislabeled product or product lacking critical safety instructions. This critical limit ensures that all products leaving the labeling step meet the required safety communication standards.	Codex Alimentarius CAC/RCP 1-1969, Rev. 2020 (General Principles of Food Hygiene); 21 CFR Part 117 (Preventive Controls for Human Food)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 36 Step 12: METAL DETECTION Hazard: Contamination from MAP gases (e.g., non-food grade gases, impurities).	PARAMETER: MAP GAS FOOD-GRADE CERTIFICATION 100% food-grade certified gases (e.g., CO ₂ , N ₂ , O ₂)	Using only food-grade certified gases ensures that the gases meet purity standards and are free from harmful contaminants or impurities that could chemically adulterate the product and pose a health risk to consumers.	FDA 21 CFR Part 173 (Secondary Direct Food Additives Permitted in Food for Human Consumption), e.g., 173.340 for Nitrogen, 184.1240 for Carbon Dioxide
	PARAMETER: APPROVED GAS SUPPLIER STATUS MAP gases sourced exclusively from suppliers approved through the company's vendor qualification program	A robust vendor qualification program ensures that gas suppliers consistently provide food-grade gases, adhere to quality specifications, and maintain appropriate handling and delivery protocols, thereby mitigating the risk of chemical contamination.	21 CFR Part 117, Subpart G (Supply-Chain Program), GFSI Recognized Standards (e.g., BRCGS, SQF)
CCP 37 Step 13: LABELING Hazard: Undeclared allergens due to mislabeling (e.g., applying a label from an allergen-containing product to sliced cabbage) or incorrect product information (e.g., missing 'WASH BEFORE USE' instruction).	PARAMETER: STORAGE TEMPERATURE Product storage temperature must be maintained at or below 4°C (40°F).	Maintaining temperatures at or below 4°C (40°F) significantly inhibits the growth of most foodborne pathogens, including psychrotrophic bacteria like <i>Listeria monocytogenes</i> . This temperature range is widely recognized in food safety regulations for perishable foods to control microbial proliferation and extend shelf life. This critical limit ensures the product remains safe throughout storage.	FDA Food Code 2022 (Section 3-501.16, Time/Temperature Control for Safety Food)
	PARAMETER: MAXIMUM TEMPERATURE EXCURSION No single temperature excursion above 4°C (40°F) is permitted for more than 2 hours, and under no circumstances shall the temperature exceed 7°C (45°F).	Brief temperature excursions can occur during loading or unloading operations. Limiting these excursions to a short duration (e.g., 2 hours) and a maximum temperature (e.g., 7°C) minimizes the opportunity for significant pathogen growth, particularly for psychrotrophic organisms. This provides a practical operational limit while maintaining product safety and quality.	FDA Food Code 2022 (Section 3-501.16, Time/Temperature Control for Safety Food); Industry Best Practices for Cold Chain Management
CCP 38 Step 13: LABELING Hazard: Undeclared allergens due to mislabeling (e.g., applying a label from an allergen-containing product to sliced cabbage) or incorrect product information (e.g., missing 'WASH BEFORE USE' instruction).	PARAMETER: LABEL ACCURACY AND COMPLETENESS All labels applied to sliced cabbage product must accurately reflect the product contents (100% cabbage, no allergens) and include all mandatory information, including 'WASH BEFORE USE' instruction.	Mislabeling with allergen-containing product labels can lead to severe allergic reactions in sensitive consumers. Providing clear instructions like 'WASH BEFORE USE' is crucial for raw produce to inform consumers about necessary preparation steps to reduce potential microbiological risks, as the product is not ready-to-eat. This ensures consumers are aware of the product's raw nature and the need for further preparation.	FDA Food Allergen Labeling and Consumer Protection Act (FALCPA) of 2004; 21 CFR Part 101 (General Food Labeling)
	PARAMETER: LABEL APPLICATION VERIFICATION 100% of product units must be verified to have the correct label applied for 'sliced cabbage' and contain all mandatory information prior to release from the labeling station.	Verification ensures that the correct label is consistently applied to the correct product, preventing mislabeling errors. This direct check at the point of application is the last opportunity to prevent the distribution of mislabeled product or product lacking critical safety instructions. This critical limit ensures that all products leaving the labeling step meet the required safety communication standards.	Codex Alimentarius CAC/RCP 1-1969, Rev. 2020 (General Principles of Food Hygiene); 21 CFR Part 117 (Preventive Controls for Human Food)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 39 Step 15: FINISHED PRODUCT COLD STORAGE <i>Hazard: Pathogen growth (e.g., Listeria monocytogenes, psychrotrophic spoilage organisms) due to temperature abuse during storage.</i>	PARAMETER: PRODUCT TEMPERATURE DURING LOADING/TRANSIT ≤ 4°C (40°F)	Maintaining product temperature at or below 4°C (40°F) is critical to prevent the rapid multiplication of psychrotrophic pathogens like <i>Listeria monocytogenes</i> and mesophilic pathogens such as <i>Salmonella</i> spp. and pathogenic <i>E. coli</i>. This temperature range significantly inhibits microbial growth, ensuring the microbiological safety of the raw, perishable sliced cabbage throughout distribution. Exceeding this limit allows for potential pathogen proliferation to unsafe levels, as no subsequent step will reduce the hazard.	FDA Food Code 2022, Section 3-401.11 (Temperature and Time Control for Safety)
CCP 40 Step 15: FINISHED PRODUCT COLD STORAGE <i>Hazard: Pathogen growth (e.g., Listeria monocytogenes, psychrotrophic spoilage organisms) due to temperature abuse during storage.</i>	PARAMETER: STORAGE TEMPERATURE Product storage temperature must be maintained at or below 4°C (40°F).	Maintaining temperatures at or below 4°C (40°F) significantly inhibits the growth of most foodborne pathogens, including psychrotrophic bacteria like <i>Listeria monocytogenes</i>. This temperature range is widely recognized in food safety regulations for perishable foods to control microbial proliferation and extend shelf life. This critical limit ensures the product remains safe throughout storage.	FDA Food Code 2022 (Section 3-501.16, Time/Temperature Control for Safety Food)
	PARAMETER: MAXIMUM TEMPERATURE EXCURSION No single temperature excursion above 4°C (40°F) is permitted for more than 2 hours, and under no circumstances shall the temperature exceed 7°C (45°F).	Brief temperature excursions can occur during loading or unloading operations. Limiting these excursions to a short duration (e.g., 2 hours) and a maximum temperature (e.g., 7°C) minimizes the opportunity for significant pathogen growth, particularly for psychrotrophic organisms. This provides a practical operational limit while maintaining product safety and quality.	FDA Food Code 2022 (Section 3-501.16, Time/Temperature Control for Safety Food); Industry Best Practices for Cold Chain Management
CCP 41 Step 16: DISPATCH <i>Hazard: Pathogenic bacterial growth (e.g., Listeria monocytogenes, Salmonella spp., pathogenic E. coli) due to temperature abuse during loading or transit.</i>	PARAMETER: TRANSPORT VEHICLE INTERIOR CONDITION Interior surfaces and cargo area must be clean, dry, and free from visible residues, spills, or odors of non-food grade chemicals, lubricants, or fuel.	The presence of non-food grade chemicals, lubricants, or fuel in the transport vehicle poses a severe chemical hazard. These substances can contaminate product packaging, and potentially the product itself if packaging integrity is compromised, leading to consumer illness or injury. This critical limit ensures that the transport environment is free from direct or indirect chemical contaminants, safeguarding product safety during transit. Verification is typically done via visual inspection and olfactory assessment prior to loading.	FDA Food Safety Modernization Act (FSMA) - Sanitary Transportation of Human and Animal Food Rule (21 CFR Part 1 Subpart O)

CCP	CRITICAL LIMITS	SCIENTIFIC JUSTIFICATION	REFERENCE
CCP 42 Step 16: DISPATCH Hazard: Pathogenic bacterial growth (e.g., <i>Listeria monocytogenes</i>, <i>Salmonella</i> spp., pathogenic <i>E. coli</i>) due to temperature abuse during loading or transit.	PARAMETER: PRODUCT TEMPERATURE DURING LOADING/TRANSIT ≤ 4°C (40°F)	Maintaining product temperature at or below 4°C (40°F) is critical to prevent the rapid multiplication of psychrotrophic pathogens like <i>Listeria monocytogenes</i> and mesophilic pathogens such as <i>Salmonella</i> spp. and pathogenic <i>E. coli</i>. This temperature range significantly inhibits microbial growth, ensuring the microbiological safety of the raw, perishable sliced cabbage throughout distribution. Exceeding this limit allows for potential pathogen proliferation to unsafe levels, as no subsequent step will reduce the hazard.	FDA Food Code 2022, Section 3-401.11 (Temperature and Time Control for Safety)
CCP 43 Step 16: DISPATCH Hazard: Contamination with cleaning chemicals, lubricants, or fuel from the transport vehicle.	PARAMETER: TRANSPORT VEHICLE INTERIOR CONDITION Interior surfaces and cargo area must be clean, dry, and free from visible residues, spills, or odors of non-food grade chemicals, lubricants, or fuel.	The presence of non-food grade chemicals, lubricants, or fuel in the transport vehicle poses a severe chemical hazard. These substances can contaminate product packaging, and potentially the product itself if packaging integrity is compromised, leading to consumer illness or injury. This critical limit ensures that the transport environment is free from direct or indirect chemical contaminants, safeguarding product safety during transit. Verification is typically done via visual inspection and olfactory assessment prior to loading.	FDA Food Safety Modernization Act (FSMA) - Sanitary Transportation of Human and Animal Food Rule (21 CFR Part 1 Subpart O)

Step 6: Monitoring Procedures

CCP 1

Step 1: RECEIVING RAW CABBAGE

B HAZARD

CRITICAL LIMIT

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Verification of supplier approval status.

HOW TO MONITOR

Review of supplier's Certificate of Analysis (COA), Letter of Guarantee, or cross-reference with the company's Approved Supplier List.

WHEN (FREQUENCY)

Upon receipt of each delivery.

WHO (RESPONSIBILITY)

Receiving Clerk / Quality Assurance (QA) Technician.

MONITORING RECORD

[Ref: Receiving Log / Approved Supplier Verification Form.](#)

CCP 1

Step 2: RECEIVING RAW CABBAGE

B HAZARD

CRITICAL LIMIT

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Internal temperature of raw cabbage.

HOW TO MONITOR

Calibrated probe thermometer (e.g., thermocouple or bimetallic stem thermometer) inserted into the core of representative cabbage heads (at least 3 measurements from different locations within the load).

WHEN (FREQUENCY)

Upon receipt of each delivery.

WHO (RESPONSIBILITY)

Receiving Clerk / Quality Assurance (QA) Technician.

MONITORING RECORD

[Ref: Receiving Log / Temperature Monitoring Log.](#)

CCP 1 Step 3: RECEIVING RAW CABBAGE**B HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Visual condition of incoming raw cabbage (absence of spoilage, pests, foreign material, excessive dirt, mold, or damaged packaging).

HOW TO MONITOR

Direct visual inspection of each incoming lot/delivery of raw cabbage, including a representative sample of individual heads/bags, against established acceptance criteria.

WHEN (FREQUENCY)

Upon receipt of each delivery/lot of raw cabbage, prior to acceptance into storage or processing.

WHO (RESPONSIBILITY)

Receiving Personnel or Quality Control Technician.

MONITORING RECORD

[Ref: Raw Material Receiving Log / Visual Inspection Checklist.](#)

CCP 1 Step 4: RECEIVING RAW CABBAGE**B HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Presence and validity of COA for chemical residues (pesticides, heavy metals).

HOW TO MONITOR

Review of supplier-provided COA against regulatory limits and company specifications.

WHEN (FREQUENCY)

Upon receipt of each raw cabbage lot.

WHO (RESPONSIBILITY)

Receiving Clerk / Quality Control Technician

MONITORING RECORD

[Ref: Raw Material Receiving Log / COA Verification Record](#)

CCP 1 Step 5: RECEIVING RAW CABBAGE**B HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Cleanliness of transport vehicle, absence of chemical residues/odors, and proper segregation of raw cabbage from allergens/incompatible materials.

HOW TO MONITOR

Visual inspection of the transport vehicle interior and exterior, and the raw cabbage product upon arrival. Olfactory check for unusual odors. Verification of loading manifest for co-transported items.

WHEN (FREQUENCY)

Upon receipt of each raw cabbage lot, prior to unloading.

WHO (RESPONSIBILITY)

Receiving Clerk / Quality Control Technician

MONITORING RECORD

[Ref: Raw Material Receiving Inspection Checklist](#)

CCP 2 Step 6: RECEIVING RAW CABBAGE**C HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Verification that the raw cabbage supplier is on the company's Approved Supplier List.

HOW TO MONITOR

Review of supplier documentation (e.g., delivery manifest, Certificate of Analysis, Letter of Guarantee) and cross-referencing with the current Approved Supplier List.

WHEN (FREQUENCY)

Upon receipt of each delivery/lot of raw cabbage, prior to acceptance into storage or processing.

WHO (RESPONSIBILITY)

Receiving Personnel or Quality Control Technician.

MONITORING RECORD

[Ref: Raw Material Receiving Log / Approved Supplier Verification Record.](#)

CCP 2 Step 7: RECEIVING RAW CABBAGE**C HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Presence and content of Certificate of Analysis (COA) for pesticide residues and heavy metals.

HOW TO MONITOR

Review of supplier-provided COA to verify compliance with regulatory limits for pesticide residues and heavy metals.

WHEN (FREQUENCY)

Upon receipt of each lot of raw cabbage.

WHO (RESPONSIBILITY)

Receiving Clerk / Quality Assurance Technician.

MONITORING RECORD

[Ref: Raw Material Receiving Log / COA Verification Record.](#)

CCP 2 Step 8: RECEIVING RAW CABBAGE**C HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Cleanliness of transport vehicle, absence of chemical residues/odors, and segregation of raw cabbage from known allergens or incompatible materials.

HOW TO MONITOR

Visual inspection of the interior of the transport vehicle and the raw cabbage load upon arrival. Olfactory check for unusual odors.

WHEN (FREQUENCY)

Upon receipt of each transport vehicle carrying raw cabbage, prior to unloading.

WHO (RESPONSIBILITY)

Receiving Clerk / Quality Assurance Technician.

MONITORING RECORD

[Ref: Raw Material Receiving Log / Vehicle Inspection Checklist.](#)

CCP 2 Step 9: RECEIVING RAW CABBAGE**B HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Internal temperature of raw cabbage

HOW TO MONITOR

Calibrated probe thermometer (e.g., thermocouple or digital thermometer) inserted into the core of representative cabbage heads from each incoming lot/delivery.

WHEN (FREQUENCY)

Upon receipt of each delivery/lot of raw cabbage.

WHO (RESPONSIBILITY)

Receiving Personnel or Quality Control Technician.

MONITORING RECORD

[Ref: Raw Material Receiving Log / Temperature Monitoring Record](#)

CCP 2 Step 10: RECEIVING RAW CABBAGE**B HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Visible signs of spoilage, pest infestation, foreign material, and unsanitary conditions on raw cabbage.

HOW TO MONITOR

Thorough visual inspection of representative samples from each incoming lot/delivery, including outer leaves and general condition.

WHEN (FREQUENCY)

Upon receipt of each delivery/lot of raw cabbage.

WHO (RESPONSIBILITY)

Receiving Personnel or Quality Control Technician.

MONITORING RECORD

[Ref: Raw Material Receiving Log / Visual Inspection Checklist](#)

CCP 2 Step 11: RECEIVING RAW CABBAGE**B HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Temperature of raw cabbage upon receipt and/or transport vehicle air temperature.

HOW TO MONITOR

Use a calibrated infrared thermometer to check surface temperature of multiple units/pallets and/or a calibrated probe thermometer to check air temperature inside the transport vehicle.

WHEN (FREQUENCY)

Upon arrival of each delivery vehicle/lot, prior to unloading.

WHO (RESPONSIBILITY)

Receiving Clerk or Quality Control Technician

MONITORING RECORD

[Ref: Raw Material Receiving Temperature Log](#)

CCP 3 Step 12: RAW MATERIAL COLD STORAGE

B HAZARD

CRITICAL LIMIT

Cumulative time product is held above 4°C (40°F) must not exceed 4 hours.

WHAT TO MONITOR

Temperature of the raw material cold storage unit/area.

HOW TO MONITOR

Calibrated continuous temperature monitoring device (e.g., data logger) with probes placed at representative locations within the storage unit. Manual verification with a calibrated thermometer.

WHEN (FREQUENCY)

Continuous monitoring with alarms for excursions.
Manual verification once per shift (minimum daily).

WHO (RESPONSIBILITY)

Cold Storage Operator / QA Technician

MONITORING RECORD

[Ref: Raw Material Cold Storage Temperature Log / Continuous Temperature Monitoring System Record](#)

CCP 3 Step 13: RAW MATERIAL COLD STORAGE

B HAZARD

CRITICAL LIMIT

Cumulative time product is held above 4°C (40°F) must not exceed 4 hours.

WHAT TO MONITOR

Sanitizer concentration and contact time on food contact surfaces in raw material cold storage.

HOW TO MONITOR

Test strips or chemical titration kit for sanitizer concentration. Stopwatch for contact time.

WHEN (FREQUENCY)

Prior to storage of raw materials, after cleaning and sanitizing, or at a specified frequency (e.g., daily/per shift) for critical surfaces.

WHO (RESPONSIBILITY)

Sanitation Team / Quality Control Technician.

MONITORING RECORD

[Ref: Sanitation Log / Pre-Operational Inspection Checklist.](#)

CCP 4 Step 14: RECEIVING RAW CABBAGE

B HAZARD

CRITICAL LIMIT

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Cumulative time the product (sliced cabbage) is held at temperatures above 4°C (40°F) during pre-processing preparation.

WHEN (FREQUENCY)

Start and end times of product removal from cold storage and return to cold storage/processing are recorded for each batch. Product temperature checked hourly if held at ambient.

HOW TO MONITOR

Visual observation and time tracking using a timer or batch log. Temperature measurement of product or ambient air with a calibrated thermometer if product is held outside refrigeration for extended periods.

WHO (RESPONSIBILITY)

Production Line Operator / Production Supervisor

MONITORING RECORD

[Ref: Pre-processing Product Holding Time/Temperature Log](#)

CCP 4 Step 15: RECEIVING RAW CABBAGE

B HAZARD

CRITICAL LIMIT

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Sanitizer concentration (e.g., free chlorine) and contact time on all food contact surfaces.

WHEN (FREQUENCY)

Before start of production, after any major interruption, and every 4 hours during continuous operation.

HOW TO MONITOR

Chemical test strips or digital colorimeter for concentration measurement; stopwatch for contact time verification.

WHO (RESPONSIBILITY)

Sanitation Team Lead / Quality Control Technician.

MONITORING RECORD

[Ref: CCP 4 Sanitation Monitoring Log](#)

CCP 4 Step 16: RECEIVING RAW CABBAGE**C HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Verification of Certificate of Analysis (COA) for pesticide residues and heavy metals.

HOW TO MONITOR

Review the COA document provided by the supplier to ensure it meets specified limits and regulatory requirements. Check for supplier name, lot number, test results, and signature.

WHEN (FREQUENCY)

Upon arrival of each delivery lot, prior to acceptance and unloading.

WHO (RESPONSIBILITY)

Receiving Clerk or Quality Control Technician

MONITORING RECORD

[Ref: Raw Material COA Verification Log](#)

CCP 4 Step 17: RECEIVING RAW CABBAGE**C HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Cleanliness of transport vehicle, absence of chemical residues/odors, and segregation of raw cabbage from allergens/incompatible materials.

HOW TO MONITOR

Visual inspection of the interior of the transport vehicle and the raw cabbage load upon arrival. Verification of shipping documents for previous loads if applicable.

WHEN (FREQUENCY)

Upon arrival of each raw cabbage delivery.

WHO (RESPONSIBILITY)

Receiving Personnel / Quality Control Technician.

MONITORING RECORD

[Ref: Raw Material Receiving Log / Vehicle Inspection Checklist.](#)

CCP 4 Step 18: RECEIVING RAW CABBAGE**B HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Visible residue on food contact surfaces (e.g., conveyors, cutting blades, bins).

HOW TO MONITOR

Direct visual inspection of all food contact surfaces.

WHEN (FREQUENCY)

Prior to start of production, after cleaning and sanitizing, and every 4 hours during continuous operation.

WHO (RESPONSIBILITY)

Production Line Operator / Quality Control Technician.

MONITORING RECORD

[Ref: Pre-operational Inspection Checklist / Sanitation Verification Log.](#)

CCP 4 Step 19: RECEIVING RAW CABBAGE**B HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Microbial cleanliness of food contact surfaces (ATP levels).

HOW TO MONITOR

ATP luminometer using validated ATP swabs on designated food contact surfaces.

WHEN (FREQUENCY)

After cleaning and sanitizing, prior to start of production, and randomly once per shift per processing line.

WHO (RESPONSIBILITY)

Quality Control Technician.

MONITORING RECORD

[Ref: ATP Swab Monitoring Log.](#)

CCP 4 Step 20: RECEIVING RAW CABBAGE**B HAZARD****CRITICAL LIMIT**

Raw cabbage must be sourced from an approved supplier.

WHAT TO MONITOR

Presence of specified allergens on food contact surfaces after cleaning.

HOW TO MONITOR

Rapid allergen test kits (e.g., specific allergen swabs) or laboratory analysis of swab samples.

WHEN (FREQUENCY)

After cleaning and before production begins, or after any product changeover involving allergens.

WHO (RESPONSIBILITY)

Quality Control Technician or trained Production Personnel.

MONITORING RECORD

[Ref: Allergen Cleaning Verification Log](#)

CCP 5 Step 21: PRE-PROCESSING PREPARATION**C HAZARD****CRITICAL LIMIT**

No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.

WHAT TO MONITOR

Presence of any visible residue (cleaning chemicals, lubricants, or previous product material) on food contact surfaces.

HOW TO MONITOR

Thorough visual inspection of all food contact surfaces under adequate lighting.

WHEN (FREQUENCY)

Before start of production, after any cleaning/sanitation cycle, and before product contact.

WHO (RESPONSIBILITY)

Production Line Operator / Quality Control Technician.

MONITORING RECORD

[Ref: CCP 5 Pre-Operational Inspection Checklist](#)

CCP 5 Step 22: PRE-PROCESSING PREPARATION**C HAZARD****CRITICAL LIMIT**

No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.

WHAT TO MONITOR

Cleanliness of food contact surfaces (presence of organic residue)

HOW TO MONITOR

ATP luminometer with calibrated swabs, following manufacturer's instructions for swabbing technique and reading.

WHEN (FREQUENCY)

After cleaning and sanitizing, prior to the start of production or before changeover to a different product.

WHO (RESPONSIBILITY)

Quality Control Technician or trained Sanitation/Production Supervisor

MONITORING RECORD

[*Ref: Pre-Operational Sanitation Verification Log \(ATP Monitoring\)*](#)

CCP 5 Step 23: PRE-PROCESSING PREPARATION**C HAZARD****CRITICAL LIMIT**

No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.

WHAT TO MONITOR

Presence of specified allergens on food contact surfaces

HOW TO MONITOR

Rapid allergen test kits (e.g., lateral flow devices) or laboratory analysis (e.g., ELISA) for relevant allergens, following manufacturer's instructions for swabbing and testing.

WHEN (FREQUENCY)

After cleaning and sanitizing, prior to the start of production or before changeover to a different product, specifically when an allergen-free product follows an allergen-containing product.

WHO (RESPONSIBILITY)

Quality Control Technician

MONITORING RECORD

[*Ref: Pre-Operational Allergen Verification Log*](#)

CCP 5 Step 24: PRE-PROCESSING PREPARATION

B HAZARD

CRITICAL LIMIT

No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.

WHAT TO MONITOR

Integrity and mesh size of all screens and sieves.

HOW TO MONITOR

Visual inspection of screens/sieves for damage (tears, holes, deformation, fraying) and verification of mesh size against specifications using a calibrated mesh gauge or template.

WHEN (FREQUENCY)

Prior to start of production, at each product changeover, and hourly during continuous operation.

WHO (RESPONSIBILITY)

Production Operator / Quality Control Technician

MONITORING RECORD

[Ref: CCP 5 Pre-Processing Equipment Inspection Log](#)

CCP 5 Step 25: PRE-PROCESSING PREPARATION

B HAZARD

CRITICAL LIMIT

No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.

WHAT TO MONITOR

Metal detector's ability to detect specified sizes of ferrous, non-ferrous, and stainless steel test pieces.

HOW TO MONITOR

Pass certified test pieces (ferrous, non-ferrous, stainless steel) of specified sizes through the metal detector aperture with product. Ensure rejection mechanism activates.

WHEN (FREQUENCY)

At the start of each production shift, every 2 hours during continuous operation, at each product changeover, and at the end of the production run. Also, after any maintenance or adjustment.

WHO (RESPONSIBILITY)

Production Operator / Quality Control Technician

MONITORING RECORD

[Ref: CCP 5 Metal Detector Monitoring Log](#)

CCP 6 Step 26: RAW MATERIAL COLD STORAGE**P HAZARD****CRITICAL LIMIT****<= 4°C (40°F)****WHAT TO MONITOR**

Visual presence of foreign material in raw cabbage upon receipt and on processing lines/equipment surfaces.

WHEN (FREQUENCY)

Upon receipt of each lot of raw cabbage; Continuously during pre-processing (e.g., trimming, coring, initial washing); Before start-up and after any maintenance on processing lines; Hourly during operation.

HOW TO MONITOR

Direct visual inspection by trained personnel of incoming raw cabbage lots and continuous visual monitoring of product on conveyor belts/inspection tables and processing equipment.

WHO (RESPONSIBILITY)

Receiving Personnel (for raw material inspection); Line Operators/Inspectors (during processing); Quality Control Technician (for verification checks).

MONITORING RECORD

[Ref: Pre-Processing Visual Inspection Log / Foreign Material Monitoring Record](#)

CCP 6 Step 27: RAW MATERIAL COLD STORAGE**P HAZARD****CRITICAL LIMIT****<= 4°C (40°F)****WHAT TO MONITOR**

Physical condition (intactness, absence of damage) and mesh size of screens and sieves used in the pre-processing area (e.g., for washing, dewatering).

WHEN (FREQUENCY)

Before start-up of each shift/production run; After any cleaning or maintenance involving screens/sieves; Weekly during operation (or more frequently based on wear assessment).

HOW TO MONITOR

Visual inspection for tears, holes, deformation, or wear. Verification of mesh size against specifications using a gauge or comparing to a standard.

WHO (RESPONSIBILITY)

Maintenance Personnel (for initial setup and repair); Line Operators/Supervisors (for pre-shift checks); Quality Control Technician (for periodic verification).

MONITORING RECORD

[Ref: Equipment Pre-Operational Check Log / Screen/Sieve Integrity Log](#)

CCP 6 Step 28: RAW MATERIAL COLD STORAGE**P HAZARD****CRITICAL LIMIT****<= 4°C (40°F)****WHAT TO MONITOR**

The ability of the metal detector to detect specified sizes of ferrous, non-ferrous, and stainless steel test pieces.

HOW TO MONITOR

Pass certified test pieces (ferrous 1.5mm, non-ferrous 2.0mm, stainless steel 2.5mm) through the metal detector aperture with product. The detector must successfully detect and reject the test piece.

WHEN (FREQUENCY)

At the start of each production shift, every 2 hours during continuous operation, at the end of the production run, and after any maintenance or adjustment to the metal detector.

WHO (RESPONSIBILITY)

Production Operator / Quality Control Technician

MONITORING RECORD

[Ref: Metal Detector Monitoring Log](#)

CCP 6 Step 29: RAW MATERIAL COLD STORAGE**B HAZARD****CRITICAL LIMIT****<= 4°C (40°F)****WHAT TO MONITOR**

Microbiological quality of process water.

HOW TO MONITOR

Collect water samples and send to an accredited laboratory for E. coli and Total Coliforms analysis.

WHEN (FREQUENCY)

Weekly, and after any maintenance or repair to the water system.

WHO (RESPONSIBILITY)

Quality Control Technician/Manager.

MONITORING RECORD

[Ref: Process Water Potability Log / Laboratory Analysis Report](#)

CCP 7 Step 30: SLICING/SHREDDING**B HAZARD****CRITICAL LIMIT****≤ 4 mg/L (ppm)****WHAT TO MONITOR**

Microbiological quality of the process water used for slicing/shredding.

HOW TO MONITOR

Collect aseptic water samples from designated sampling points (e.g., point of use at slicing/shredding equipment) and send to an accredited laboratory for analysis of E. coli and Total Coliforms.

WHEN (FREQUENCY)

Weekly, or more frequently as determined by risk assessment (e.g., after plumbing repairs, changes in water source).

WHO (RESPONSIBILITY)

**Quality Control Technician (sample collection);
Accredited external laboratory (analysis)**

MONITORING RECORD

[Ref: Process Water Potability Test Results Log / Laboratory Reports](#)

CCP 7 Step 31: SLICING/SHREDDING**B HAZARD****CRITICAL LIMIT****≤ 4 mg/L (ppm)****WHAT TO MONITOR**

Levels of regulated heavy metals and industrial contaminants in process water.

HOW TO MONITOR

Collection of process water samples and submission to an accredited third-party laboratory for analysis against established Maximum Contaminant Levels (MCLs).

WHEN (FREQUENCY)

Annually, or more frequently if there are changes to the water source, municipal water quality reports indicate issues, or regulatory requirements dictate.

WHO (RESPONSIBILITY)

Quality Assurance Manager (responsible for coordinating testing) / Third-Party Accredited Laboratory (performing analysis).

MONITORING RECORD

[Ref: Process Water Chemical Analysis Report](#)

CCP 7 Step 32: SLICING/SHREDDING

B HAZARD

CRITICAL LIMIT

≤ 4 mg/L (ppm)

WHAT TO MONITOR

Presence of cleaning chemicals in process water.

HOW TO MONITOR

Chemical test strips or analytical test kit specific for common cleaning chemicals (e.g., pH indicator, surfactant test). Visual inspection for foaming or unusual odor.

WHEN (FREQUENCY)

At the start of each production shift and after any maintenance or cleaning of the water system.

WHO (RESPONSIBILITY)

Quality Control Technician or Production Operator.

MONITORING RECORD

[Ref: Process Water Quality Log](#)

CCP 8 Step 33: PRE-PROCESSING PREPARATION

C HAZARD

CRITICAL LIMIT

Cumulative time product is held above 4°C (40°F) must not exceed 4 hours.

WHAT TO MONITOR

Free chlorine residual level in the process water used for slicing/shredding.

HOW TO MONITOR

Using a calibrated chlorine test kit (e.g., DPD method) or an inline chlorine sensor. Verification of test kit calibration or sensor accuracy.

WHEN (FREQUENCY)

At the start of each production shift, every 4 hours during continuous operation, and after any water line maintenance or chemical replenishment.

WHO (RESPONSIBILITY)

Quality Control Technician or Production Operator.

MONITORING RECORD

[Ref: Process Water Chlorine Monitoring Log](#)

CCP 8 Step 34: PRE-PROCESSING PREPARATION

C HAZARD

CRITICAL LIMIT

Cumulative time product is held above 4°C (40°F) must not exceed 4 hours.

WHAT TO MONITOR

Presence and concentration of regulated heavy metals and industrial contaminants in the process water.

HOW TO MONITOR

Sending water samples to an accredited external laboratory for analysis (e.g., ICP-MS for heavy metals, GC-MS for organic contaminants). Reviewing municipal water quality reports if applicable and available.

WHEN (FREQUENCY)

Annually, or semi-annually, and whenever there is a change in water source, significant upstream environmental event, or regulatory requirement. Review municipal water reports monthly/quarterly.

WHO (RESPONSIBILITY)

Quality Assurance Manager.

MONITORING RECORD

[Ref: Process Water Contaminant Analysis Report / Municipal Water Quality Report Review Log](#)

CCP 8 Step 35: PRE-PROCESSING PREPARATION

C HAZARD

CRITICAL LIMIT

Cumulative time product is held above 4°C (40°F) must not exceed 4 hours.

WHAT TO MONITOR

Presence of residual cleaning chemicals (e.g., detergents, sanitizers) in the process water used for slicing/shredding.

HOW TO MONITOR

Visual inspection of process water for foaming, turbidity, or unusual discoloration. Olfactory check for chemical odors. pH test strips to confirm neutral pH (6.0-8.0) indicating absence of acidic/alkaline cleaning agents.

WHEN (FREQUENCY)

Prior to the start of each production shift, and after any cleaning, sanitation, or maintenance of water lines/equipment.

WHO (RESPONSIBILITY)

Production Operator / Quality Control Technician

MONITORING RECORD

[Ref: Process Water Quality Check Log](#)

CCP 8 Step 36: PRE-PROCESSING PREPARATION**B HAZARD****CRITICAL LIMIT**

Cumulative time product is held above 4°C (40°F) must not exceed 4 hours.

WHAT TO MONITOR

Sanitizer concentration and contact time on food contact surfaces.

HOW TO MONITOR

Chemical test strips or titration kit for sanitizer concentration; stopwatch for contact time.

WHEN (FREQUENCY)

Before production begins, after each major cleaning cycle, and every 4 hours during continuous operation.

WHO (RESPONSIBILITY)

Production Operator or Quality Control Technician.

MONITORING RECORD

[Ref: Sanitation Log / Chemical Concentration Record](#)

CCP 8 Step 37: PRE-PROCESSING PREPARATION**B HAZARD****CRITICAL LIMIT**

Cumulative time product is held above 4°C (40°F) must not exceed 4 hours.

WHAT TO MONITOR

Free chlorine concentration in the process water used for slicing/shredding.

HOW TO MONITOR

Calibrated chlorine test strips or a digital colorimeter/photometer specifically designed for free chlorine measurement.

WHEN (FREQUENCY)

Every 2 hours during continuous operation, or at the start of each batch/production run if intermittent.

WHO (RESPONSIBILITY)

Production Operator or Quality Control Technician.

MONITORING RECORD

[Ref: Sanitizer Concentration Log](#)

CCP 8 Step 38: PRE-PROCESSING PREPARATION**B HAZARD****CRITICAL LIMIT**

Cumulative time product is held above 4°C (40°F) must not exceed 4 hours.

WHAT TO MONITOR

pH of the process water used for slicing/shredding.

HOW TO MONITOR

Calibrated pH meter or pH test strips.

WHEN (FREQUENCY)

At the start of each production shift and every 4 hours thereafter, or more frequently if water source changes or issues are suspected.

WHO (RESPONSIBILITY)

Production Operator or Quality Control Technician.

MONITORING RECORD

[Ref: Water Quality Monitoring Log - pH](#)

CCP 8 Step 39: PRE-PROCESSING PREPARATION**B HAZARD****CRITICAL LIMIT****Cumulative time product is held above 4°C (40°F) must not exceed 4 hours.****WHAT TO MONITOR**

Compliance of process water with EPA National Primary Drinking Water Regulations.

WHEN (FREQUENCY)

Annually for municipal reports; quarterly or semi-annually for in-house lab analysis, or as required by local regulations/supplier changes.

HOW TO MONITOR

Review of municipal water quality reports or laboratory analysis of water samples for specified contaminants.

WHO (RESPONSIBILITY)

Quality Assurance Manager or designated personnel.

MONITORING RECORD

[Ref: Water Potability Test Records / Municipal Water Report Review Log](#)

CCP 9 Step 40: POST-SLICING WASHING & RINSING**B HAZARD****CRITICAL LIMIT****Product internal temperature $\leq$ 4°C (40°F)****WHAT TO MONITOR**

Free chlorine concentration in the wash water.

WHEN (FREQUENCY)

At the start of each production shift, every 2 hours during continuous operation, and after any water replenishment or chemical dosing adjustment.

HOW TO MONITOR

Using a calibrated free chlorine test kit (e.g., test strips, digital colorimeter, or photometer) according to manufacturer's instructions.

WHO (RESPONSIBILITY)

Production Operator / Quality Control Technician

MONITORING RECORD

[Ref: Wash Water Sanitizer Concentration Log](#)

CCP 9 Step 41: POST-SLICING WASHING & RINSING**B HAZARD****CRITICAL LIMIT****Product internal temperature $\leq$ 4°C (40°F)****WHAT TO MONITOR**

pH of the wash water in the post-slicing washing and rinsing tank.

WHEN (FREQUENCY)

At the start of each production shift, every 2 hours during continuous operation, and after any water change or chemical addition.

HOW TO MONITOR

Calibrated pH meter or pH test strips.

WHO (RESPONSIBILITY)

Line Operator or Quality Control Technician.

MONITORING RECORD

[Ref: Wash Water Quality Log / CCP 9 Monitoring Record](#)

CCP 9 Step 42: POST-SLICING WASHING & RINSING**B HAZARD****CRITICAL LIMIT****Product internal temperature $\leq 4^{\circ}\text{C}$ (40°F)****WHAT TO MONITOR**

Microbiological quality of the incoming water supply used for washing and rinsing.

HOW TO MONITOR

Laboratory analysis of water samples for coliforms, E. coli, and other relevant indicators as per EPA regulations.

WHEN (FREQUENCY)

Annually, or quarterly, and after any repairs or changes to the water supply system.

WHO (RESPONSIBILITY)

Quality Control Manager (for scheduling and reviewing results), Third-party accredited laboratory (for analysis).

MONITORING RECORD

[Ref: Water Potability Test Reports / Laboratory Certificates of Analysis](#)

CCP 9 Step 43: POST-SLICING WASHING & RINSING**B HAZARD****CRITICAL LIMIT****Product internal temperature $\leq 4^{\circ}\text{C}$ (40°F)****WHAT TO MONITOR**

ATP swab readings on food contact surfaces post-sanitation.

HOW TO MONITOR

ATP luminometer with calibrated swabs.

WHEN (FREQUENCY)

After sanitation, prior to start of production, and after any major equipment breakdown or cleaning event.

WHO (RESPONSIBILITY)

Quality Control Technician / Sanitation Supervisor.

MONITORING RECORD

[Ref: Sanitation Verification Log / ATP Monitoring Record](#)

CCP 10 Step 44: PRE-PROCESSING PREPARATION**B HAZARD****CRITICAL LIMIT****No visible residue of cleaning chemicals, lubricants, or previous product material on food contact surfaces.****WHAT TO MONITOR**

Internal core temperature of incoming sliced cabbage product.

HOW TO MONITOR

Calibrated digital thermometer with a probe, inserted into the core of a representative sample of product from each lot/delivery.

WHEN (FREQUENCY)

Upon receipt of each lot/delivery, and at the start of the shift for product held in the processing area prior to sorting.

WHO (RESPONSIBILITY)

Receiving Personnel / Production Line Operator

MONITORING RECORD

[Ref: Receiving Temperature Log / In-Process Temperature Monitoring Log](#)

CCP 10

Step 45: PRE-PROCESSING PREPARATION

B HAZARD

CRITICAL LIMIT

No visible residue of cleaning chemicals, lubricants, or previous product material on food contact surfaces.

WHAT TO MONITOR

Cleanliness of food contact surfaces (e.g., conveyor belts, sorting tables, bins) after sanitation.

HOW TO MONITOR

For ATP: ATP luminometer with calibrated swabs, following manufacturer's instructions. For pathogens: Sterile environmental swabs collected from food contact surfaces, sent to an accredited laboratory for pathogen testing (e.g., *Listeria monocytogenes*).

WHEN (FREQUENCY)

For ATP: After each sanitation cycle, prior to the start of production. For pathogens: Weekly or bi-weekly, as part of an environmental monitoring program.

WHO (RESPONSIBILITY)

Quality Control Technician / Sanitation Supervisor (for ATP); Quality Assurance Manager (for pathogen sampling and lab coordination).

MONITORING RECORD

[Ref: Sanitation Verification Log \(ATP\) / Environmental Monitoring Log \(Pathogen Swabs\)](#)

CCP 10

Step 46: PRE-PROCESSING PREPARATION

C HAZARD

CRITICAL LIMIT

No visible residue of cleaning chemicals, lubricants, or previous product material on food contact surfaces.

WHAT TO MONITOR

Cleanliness of food contact surfaces, specifically the presence of organic residue.

HOW TO MONITOR

Using a calibrated ATP luminometer and ATP swabs on designated food contact surfaces (e.g., conveyor belts, slicer blades, hoppers) according to manufacturer instructions.

WHEN (FREQUENCY)

After cleaning and sanitizing, prior to the start of production or changeover to a new product.

WHO (RESPONSIBILITY)

Quality Control Technician or trained Production Lead.

MONITORING RECORD

[Ref: Pre-Operational Sanitation Verification Log](#)

CCP 10

Step 47: PRE-PROCESSING PREPARATION

C HAZARD

CRITICAL LIMIT

No visible residue of cleaning chemicals, lubricants, or previous product material on food contact surfaces.

WHAT TO MONITOR

Absence of specified allergens on food contact surfaces.

HOW TO MONITOR

Using rapid allergen test kits (e.g., lateral flow devices or ELISA-based swabs) for relevant allergens on designated food contact surfaces, following manufacturer's instructions.

WHEN (FREQUENCY)

After cleaning and sanitizing, prior to the start of production or changeover to a new product, especially when processing an allergen-free product after an allergen-containing one.

WHO (RESPONSIBILITY)

Quality Control Technician or trained Production Lead.

MONITORING RECORD

[Ref: Allergen Cleaning Verification Log](#)

CCP 10

Step 48: PRE-PROCESSING PREPARATION

B HAZARD

CRITICAL LIMIT

No visible residue of cleaning chemicals, lubricants, or previous product material on food contact surfaces.

WHAT TO MONITOR

Type and certification of lubricants used on equipment with potential food contact during visual inspection and sorting.

HOW TO MONITOR

Verification of lubricant product labels, Safety Data Sheets (SDS), and supplier certificates against NSF H1 or H3 certification. Visual inspection of lubricant containers.

WHEN (FREQUENCY)

Upon receipt of new lubricants, annually for existing lubricants, and during pre-operational checks or equipment maintenance.

WHO (RESPONSIBILITY)

Maintenance Manager / Quality Assurance Technician

MONITORING RECORD

[Ref: Food-Grade Lubricant Verification Log / Maintenance Records](#)

CCP 10

Step 49: PRE-PROCESSING PREPARATION

B HAZARD

CRITICAL LIMIT

No visible residue of cleaning chemicals, lubricants, or previous product material on food contact surfaces.

WHAT TO MONITOR

Presence of chemical residues on food contact surfaces.

HOW TO MONITOR

Visual inspection for residue and odor. Use of pH strips or specific chemical residue test kits (e.g., for sanitizer concentration) on representative food contact surfaces.

WHEN (FREQUENCY)

After final rinse of sanitation cycle, prior to start of production, and after any equipment breakdown or reassembly.

WHO (RESPONSIBILITY)

Sanitation Supervisor or Quality Control Technician.

MONITORING RECORD

[Ref: Pre-Operational Inspection Checklist / Chemical Residue Monitoring Log.](#)

CCP 11

Step 50: VISUAL INSPECTION & SORTING

C HAZARD

CRITICAL LIMIT

No foreign material (e.g., metal, plastic, glass, wood, stones) detected in the product during visual inspection.

WHAT TO MONITOR

Type of lubricant used on equipment with potential food contact.

HOW TO MONITOR

Visually inspect lubricant labels for NSF H1 or H3 certification. Verify against Approved Lubricant List and purchasing records.

WHEN (FREQUENCY)

Upon receipt of new lubricants, during equipment maintenance, and at the start of each production shift.

WHO (RESPONSIBILITY)

Maintenance Supervisor / Production Supervisor

MONITORING RECORD

[Ref: Equipment Maintenance Log / Approved Lubricant List / Pre-operational Inspection Checklist](#)

CCP 11

Step 51: VISUAL INSPECTION & SORTING

C HAZARD

CRITICAL LIMIT

No foreign material (e.g., metal, plastic, glass, wood, stones) detected in the product during visual inspection.

WHAT TO MONITOR

Presence of chemical residues (e.g., cleaning agents, sanitizers) on food contact surfaces after rinsing.

HOW TO MONITOR

Visual inspection for residue, film, or odor. Use pH strips to test for residual alkalinity/acidity on representative food contact surfaces. ATP swabs may be used as an additional indicator of overall cleanliness.

WHEN (FREQUENCY)

Prior to the start of production, after each major cleaning and sanitizing cycle, and after any equipment breakdown requiring cleaning.

WHO (RESPONSIBILITY)

Quality Control Technician / Sanitation Supervisor / Production Line Operator

MONITORING RECORD

[Ref: Pre-operational Inspection Checklist / Sanitation Verification Log](#)

CCP 11

Step 52: VISUAL INSPECTION & SORTING

B HAZARD

CRITICAL LIMIT

No foreign material (e.g., metal, plastic, glass, wood, stones) detected in the product during visual inspection.

WHAT TO MONITOR

Integrity and proper installation of all physical separation devices (screens and filters) on processing lines.

HOW TO MONITOR

Visual inspection of screens and filters for any damage (tears, holes, dislodgement) and verification of correct installation and mesh size.

WHEN (FREQUENCY)

Prior to start of production, and hourly during operation, or after any maintenance affecting these devices.

WHO (RESPONSIBILITY)

Line Operator / Quality Control Technician

MONITORING RECORD

[Ref: CCP Monitoring Log - Physical Separation Device Integrity](#)

CCP 12

Step 53: PRE-PROCESSING PREPARATION

P HAZARD

CRITICAL LIMIT

No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.

WHAT TO MONITOR

Presence of foreign material (metal, plastic, glass, wood, stones) in the sliced cabbage product.

HOW TO MONITOR

Direct visual inspection of product on conveyor belts or sorting tables by trained personnel under adequate lighting.

WHEN (FREQUENCY)

Continuously during the visual inspection and sorting process.

WHO (RESPONSIBILITY)

Trained Production Operator/Sorter.

MONITORING RECORD

[Ref: Foreign Material Detection Log / Visual Inspection Log](#)

CCP 12

Step 54: PRE-PROCESSING PREPARATION

P HAZARD

CRITICAL LIMIT

No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.

WHAT TO MONITOR

Condition and proper installation of physical separation devices (screens and filters), including mesh integrity, absence of tears/holes, and correct mesh size.

HOW TO MONITOR

Physical inspection of screens and filters by trained maintenance or production personnel. Verification against equipment specifications and visual check for damage.

WHEN (FREQUENCY)

At the start of each shift and after any maintenance activity involving the devices.

WHO (RESPONSIBILITY)

Maintenance Technician / Production Supervisor.

MONITORING RECORD

[Ref: Equipment Pre-Operational Check / Screen & Filter Integrity Log](#)

CCP 12 Step 55: PRE-PROCESSING PREPARATION

P HAZARD

CRITICAL LIMIT

No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.

WHAT TO MONITOR

Physical condition and mesh size of all screens and sieves.

HOW TO MONITOR

Visual inspection of screens and sieves for tears, holes, damage, or incorrect mesh size. Use a calibrated gauge or template to verify mesh opening size if necessary.

WHEN (FREQUENCY)

Prior to start of production, at each shift change, and hourly during continuous operation.

WHO (RESPONSIBILITY)

Production Operator/Supervisor.

MONITORING RECORD

Ref: [Screen/Sieve Inspection Log](#)

CCP 12 Step 56: PRE-PROCESSING PREPARATION

P HAZARD

CRITICAL LIMIT

No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.

WHAT TO MONITOR

Functionality and sensitivity of the metal detector.

HOW TO MONITOR

Pass certified test pieces (ferrous, non-ferrous, stainless steel) of specified sizes through the metal detector at the designated check points.

WHEN (FREQUENCY)

At the start of each production shift, every 2 hours during continuous operation, and at the end of the shift. Also, after any maintenance or adjustment.

WHO (RESPONSIBILITY)

Production Operator / Quality Control Technician.

MONITORING RECORD

Ref: [Metal Detector Monitoring Log](#)

CCP 12

Step 57: PRE-PROCESSING PREPARATION

B HAZARD

CRITICAL LIMIT

No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.

WHAT TO MONITOR

Presence of visible foreign material in sliced cabbage product and on processing equipment surfaces.

HOW TO MONITOR

Direct visual inspection by trained personnel on the production line and during pre-operational checks.

WHEN (FREQUENCY)

Continuously during sorting and processing operations; prior to start-up for equipment surfaces.

WHO (RESPONSIBILITY)

Production Line Operator / Quality Control Technician

MONITORING RECORD

[Ref: Foreign Material Visual Inspection Log / Pre-operational Inspection Checklist](#)

CCP 12

Step 58: PRE-PROCESSING PREPARATION

B HAZARD

CRITICAL LIMIT

No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.

WHAT TO MONITOR

Personnel adherence to GMPs regarding protective clothing, jewelry, and personal items in the processing area.

HOW TO MONITOR

Direct visual observation of personnel by supervisor or QC staff.

WHEN (FREQUENCY)

At the start of each shift, upon re-entry to the processing area, and periodically throughout the shift (e.g., hourly checks).

WHO (RESPONSIBILITY)

Production Supervisor / Quality Control Technician

MONITORING RECORD

[Ref: GMP Adherence Checklist / Personnel Hygiene Monitoring Log](#)

CCP 12

Step 59: PRE-PROCESSING PREPARATION

B HAZARD

CRITICAL LIMIT

No visible foreign material (e.g., insects, excessive plant debris, plastic, metal, personal effects) in raw cabbage or on processing lines.

WHAT TO MONITOR

Integrity of food contact surfaces of weighing and portioning equipment.

HOW TO MONITOR

Visual inspection of all food contact surfaces for damage (cracks, chips, rust, loose parts).

WHEN (FREQUENCY)

Prior to start of production and every 4 hours during operation.

WHO (RESPONSIBILITY)

Equipment Operator / Quality Control Technician.

MONITORING RECORD

[Ref: Equipment Pre-Operational Check & Integrity Log.](#)

CCP 13 Step 60: WEIGHING & PORTIONING

P HAZARD

CRITICAL LIMIT

Product temperature must not exceed 4°C (40°F) during the entire packaging process.

WHAT TO MONITOR

Presence of visible foreign material in sliced cabbage product and on weighing/portioning equipment surfaces.

WHEN (FREQUENCY)

Before start of production shift; at the beginning of each new product batch; every 30 minutes during continuous operation; after any equipment adjustment or maintenance.

HOW TO MONITOR

Direct visual inspection of product batches and all accessible equipment surfaces (e.g., scales, conveyors, hoppers, contact surfaces).

WHO (RESPONSIBILITY)

Production Operator / Quality Control Technician

MONITORING RECORD

[Ref: Foreign Material Visual Inspection Log](#)

CCP 13 Step 61: WEIGHING & PORTIONING

P HAZARD

CRITICAL LIMIT

Product temperature must not exceed 4°C (40°F) during the entire packaging process.

WHAT TO MONITOR

Personnel compliance with Good Manufacturing Practices (GMPs) regarding protective clothing, jewelry, and personal items in the weighing & portioning area.

WHEN (FREQUENCY)

At the start of each shift for all personnel entering the area; random spot checks at least hourly during production; after breaks or re-entry into the processing area.

HOW TO MONITOR

Direct visual observation of personnel by supervisor or Quality Control staff upon entry to the area and during production.

WHO (RESPONSIBILITY)

Production Supervisor / Quality Control Technician

MONITORING RECORD

[Ref: GMP Personnel Compliance Checklist / Log](#)

CCP 13 Step 62: WEIGHING & PORTIONING

P HAZARD

CRITICAL LIMIT

Product temperature must not exceed 4°C (40°F) during the entire packaging process.

WHAT TO MONITOR

The physical condition and integrity of all food contact surfaces of weighing and portioning equipment.

HOW TO MONITOR

Visual inspection of all food contact surfaces (e.g., scales, hoppers, conveyors) for any signs of damage such as cracks, chips, rust, or loose parts. A checklist will be used to ensure all critical areas are examined.

WHEN (FREQUENCY)

Prior to the start of each production shift and after any equipment maintenance or cleaning.

WHO (RESPONSIBILITY)

Production Line Operator or Quality Control Technician.

MONITORING RECORD

[Ref: CCP 13 Equipment Integrity Check Sheet](#)

CCP 13 Step 63: WEIGHING & PORTIONING

B HAZARD

CRITICAL LIMIT

Product temperature must not exceed 4°C (40°F) during the entire packaging process.

WHAT TO MONITOR

Condition and cleanliness of primary packaging materials (PE bags, PP clam-shells)

HOW TO MONITOR

Visual inspection of packaging materials for tears, holes, foreign material, and general cleanliness. Verification of food-grade certification upon receipt.

WHEN (FREQUENCY)

At the start of each packaging run, periodically (e.g., hourly) during the run, and upon receipt of new packaging material batches

WHO (RESPONSIBILITY)

Packaging Line Operator or Quality Control Technician

MONITORING RECORD

[Ref: Packaging Material Inspection Log](#)

CCP 13 Step 64: WEIGHING & PORTIONING

B HAZARD

CRITICAL LIMIT

Product temperature must not exceed 4°C (40°F) during the entire packaging process.

WHAT TO MONITOR

Cleanliness and sanitation of food contact surfaces and surrounding environment in the packaging area.

HOW TO MONITOR

Visual inspection for cleanliness and absence of debris. ATP testing (or equivalent rapid hygiene test) on food contact surfaces.

WHEN (FREQUENCY)

Prior to start of packaging operations, and every 4 hours during continuous operation, or after any significant interruption/maintenance.

WHO (RESPONSIBILITY)

Quality Control Technician / Production Supervisor

MONITORING RECORD

Ref: [Pre-operational & In-process Hygiene Check Log](#)

CCP 13 Step 65: WEIGHING & PORTIONING

B HAZARD

CRITICAL LIMIT

Product temperature must not exceed 4°C (40°F) during the entire packaging process.

WHAT TO MONITOR

O2 and CO2 concentration in the headspace of packaged product.

HOW TO MONITOR

Headspace gas analyzer (e.g., portable O2/CO2 analyzer) for random samples. In-line gas analyzer for continuous monitoring (if available).

WHEN (FREQUENCY)

Every 30 minutes or at the start of each new roll of film/batch, and after any machine adjustment or film change.

WHO (RESPONSIBILITY)

Packaging Operator / Quality Control Technician

MONITORING RECORD

Ref: [MAP Gas Composition Log](#)

CCP 14 Step 66: SLICING/SHREDDING

B HAZARD

CRITICAL LIMIT

E. coli absent in 100 mL; Total Coliforms absent in 100 mL

WHAT TO MONITOR

Internal temperature of the sliced cabbage product as it is being packaged.

HOW TO MONITOR

Using a calibrated, sanitized probe thermometer (e.g., thermocouple) to measure the internal temperature of randomly selected product samples directly from the packaging line. The probe must be inserted into the product mass to obtain an accurate reading.

WHEN (FREQUENCY)

At the start of the packaging run, then every 30 minutes, and at the end of the packaging run. More frequent monitoring if there are any deviations or changes in the packaging environment.

WHO (RESPONSIBILITY)

Packaging Line Operator or Quality Control Technician.

MONITORING RECORD

[Ref: CCP 14 Product Temperature Monitoring Log](#)

CCP 14 Step 67: SLICING/SHREDDING

B HAZARD

CRITICAL LIMIT

E. coli absent in 100 mL; Total Coliforms absent in 100 mL

WHAT TO MONITOR

Visual integrity and cleanliness of primary packaging materials (PE bags, PP clam-shells).

HOW TO MONITOR

Visual inspection of each roll/stack of packaging material upon receipt and before loading onto packaging equipment. Random spot checks during packaging operations.

WHEN (FREQUENCY)

Upon receipt of each new lot/delivery of packaging materials; prior to loading onto packaging equipment for each production run; hourly during packaging operations.

WHO (RESPONSIBILITY)

Quality Control Technician / Packaging Line Operator

MONITORING RECORD

[Ref: Packaging Material Inspection Log](#)

CCP 14

Step 68: SLICING/SHREDDING

B HAZARD

CRITICAL LIMIT

E. coli absent in 100 mL; Total Coliforms absent in 100 mL

WHAT TO MONITOR

Cleanliness and sanitation of food contact surfaces of packaging equipment and surrounding environment.

HOW TO MONITOR

Visual inspection for cleanliness (no visible debris, product residue, or foreign matter). ATP swab testing (or equivalent rapid hygiene test) on selected food contact surfaces.

WHEN (FREQUENCY)

Visual inspection: Prior to start of each shift/production run, and every 2-4 hours during operation. ATP testing: Daily, prior to start of first production run, or after major cleaning and sanitization events.

WHO (RESPONSIBILITY)

Packaging Line Supervisor / Quality Control Technician

MONITORING RECORD

[Ref: Environmental Hygiene Monitoring Log / ATP Testing Log](#)

CCP 14

Step 69: SLICING/SHREDDING

B HAZARD

CRITICAL LIMIT

E. coli absent in 100 mL; Total Coliforms absent in 100 mL

WHAT TO MONITOR

O2 and CO2 concentration in the headspace of packaged sliced cabbage

HOW TO MONITOR

Using a calibrated headspace gas analyzer. Analyzer must be calibrated daily or as per manufacturer's instructions. Sample a representative package.

WHEN (FREQUENCY)

At the start of each production run, every 30-60 minutes during continuous operation, and after any equipment adjustment or changeover.

WHO (RESPONSIBILITY)

Packaging Machine Operator or Quality Control Technician

MONITORING RECORD

[Ref: MAP Gas Composition Log](#)

CCP 14

Step 70: SLICING/SHREDDING

B HAZARD

CRITICAL LIMIT

E. coli absent in 100 mL; Total Coliforms absent in 100 mL

WHAT TO MONITOR

Internal product temperature of sliced cabbage.

HOW TO MONITOR

Using a calibrated probe thermometer, insert into the core of a representative product package.

WHEN (FREQUENCY)

At the start of each packaging run and every 30 minutes thereafter.

WHO (RESPONSIBILITY)

Packaging Line Operator / Quality Control Technician

MONITORING RECORD

[Ref: CCP Product Temperature Log - Primary Packaging](#)

CCP 14

Step 71: SLICING/SHREDDING

B HAZARD

CRITICAL LIMIT

E. coli absent in 100 mL; Total Coliforms absent in 100 mL

WHAT TO MONITOR

ATP levels on food contact surfaces of primary packaging equipment (e.g., conveyor belts, hoppers, scales).

HOW TO MONITOR

ATP luminometer using validated ATP swabs, following manufacturer's instructions and company SOP.

WHEN (FREQUENCY)

Prior to the start of each production shift, and every 4 hours during continuous operation, or after any major equipment breakdown/maintenance.

WHO (RESPONSIBILITY)

Quality Control Technician or trained Production Operator.

MONITORING RECORD

[Ref: Environmental Monitoring Log - ATP Swab Results](#)

CCP 15

Step 72: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING

B HAZARD

CRITICAL LIMIT

Packaging materials must be certified food-grade and comply with all applicable regulatory standards for direct food contact.

WHAT TO MONITOR

Internal temperature of sliced cabbage product immediately prior to or during packaging

HOW TO MONITOR

Using a calibrated, sanitized probe thermometer (e.g., thermocouple or RTD) inserted into a representative sample of product.

WHEN (FREQUENCY)

At the start of each production run, every 30-60 minutes during continuous operation, and after any line stoppage exceeding 15 minutes.

WHO (RESPONSIBILITY)

Packaging Machine Operator or Quality Control Technician

MONITORING RECORD

[Ref: Product Temperature Log - Packaging](#)

CCP 15

Step 73: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING

B HAZARD

CRITICAL LIMIT

Packaging materials must be certified food-grade and comply with all applicable regulatory standards for direct food contact.

WHAT TO MONITOR

Sanitation level of food contact surfaces on MAP packaging equipment (e.g., conveyor belts, product contact plates, sealing jaws).

HOW TO MONITOR

ATP bioluminescence swab test using a calibrated luminometer. Swabs are taken from designated critical food contact points after sanitation and before production starts.

WHEN (FREQUENCY)

Before the start of each production shift, after any major equipment breakdown or maintenance, and after any unscheduled stop exceeding 4 hours.

WHO (RESPONSIBILITY)

Quality Control Technician or trained Sanitation Supervisor.

MONITORING RECORD

Ref: [Pre-Operational Sanitation Verification Log / ATP Monitoring Record](#)

CCP 15

Step 74: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING

B HAZARD

CRITICAL LIMIT

Packaging materials must be certified food-grade and comply with all applicable regulatory standards for direct food contact.

WHAT TO MONITOR

Presence/absence of cleaning chemical residue on food contact surfaces of the MAP machine and sealing equipment.

HOW TO MONITOR

Visual inspection and/or chemical residue test kits (e.g., pH strips, specific chemical tests) on swabbed surfaces.

WHEN (FREQUENCY)

After each cleaning and sanitizing cycle, prior to the start of packaging operations.

WHO (RESPONSIBILITY)

Sanitation Supervisor or Quality Control Technician.

MONITORING RECORD

Ref: [Pre-Operational Sanitation Check Record / Chemical Residue Test Log](#)

CCP 15

Step 75: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING

B HAZARD

CRITICAL LIMIT

Packaging materials must be certified food-grade and comply with all applicable regulatory standards for direct food contact.

WHAT TO MONITOR

Type of lubricant used on packaging machinery where incidental food contact is possible.

HOW TO MONITOR

Visual inspection of lubricant product labels and Safety Data Sheets (SDS) or Certificate of Conformance (CoC) from supplier. Verification against approved lubricant list.

WHEN (FREQUENCY)

Prior to initial use of new lubricant, at the start of each production week, and during routine maintenance checks.

WHO (RESPONSIBILITY)

Maintenance Supervisor or Quality Control Technician

MONITORING RECORD

[Ref: Approved Lubricant Verification Log](#)

CCP 16

Step 76: SLICING/SHREDDING

C HAZARD

CRITICAL LIMIT

≤ 4 mg/L (ppm)

WHAT TO MONITOR

Certification and specifications of incoming primary packaging materials (e.g., PE bags, PP clam-shell containers).

HOW TO MONITOR

Review of Certificates of Conformance (CoC) or Letters of Guarantee (LoG) from packaging material suppliers, verifying food-grade status and compliance with relevant regulatory standards (e.g., FDA, EU regulations). Visual inspection of packaging materials upon receipt for damage or contamination.

WHEN (FREQUENCY)

Upon receipt of each new lot/shipment of packaging materials. Annually, or whenever there is a change in supplier or material specification, review supplier documentation.

WHO (RESPONSIBILITY)

Receiving Clerk (for visual inspection), Quality Assurance Manager (for documentation review and approval).

MONITORING RECORD

[Ref: Incoming Packaging Material Log / Supplier CoC/LoG File](#)

CCP 16

Step 77: SLICING/SHREDDING

C HAZARD

CRITICAL LIMIT

≤ 4 mg/L (ppm)

WHAT TO MONITOR

Presence/absence of cleaning and sanitizing chemical residues on food contact surfaces of MAP/sealing equipment.

WHEN (FREQUENCY)

After completion of cleaning and sanitizing procedures, prior to the start of packaging operations, and after any unscheduled cleaning.

HOW TO MONITOR

Visual inspection for any visible residue. For critical areas, use pH test strips or specific chemical test kits (e.g., chlorine, quat) to confirm absence of residue after final rinse. Swab testing for ATP or specific chemical residues may be used for verification.

WHO (RESPONSIBILITY)

Sanitation Supervisor or QA Technician.

MONITORING RECORD

[Ref: Pre-Operational Inspection Log / Cleaning & Sanitation Verification Record](#)

CCP 16

Step 78: SLICING/SHREDDING

C HAZARD

CRITICAL LIMIT

≤ 4 mg/L (ppm)

WHAT TO MONITOR

Type and certification status of lubricants used on MAP/sealing machinery where incidental food contact is possible.

WHEN (FREQUENCY)

At the start of each production shift, during routine preventative maintenance checks, and upon receipt of new lubricant supplies. Annual review of lubricant inventory and usage.

HOW TO MONITOR

Verify lubricant labels for NSF H1 certification mark. Cross-reference with the facility's approved lubricant list and corresponding Safety Data Sheets (SDS). Visually inspect lubricant containers in use.

WHO (RESPONSIBILITY)

Maintenance Supervisor or Production Supervisor.

MONITORING RECORD

[Ref: Lubricant Verification Log / Maintenance Log](#)

CCP 16

Step 79: SLICING/SHREDDING

C HAZARD

CRITICAL LIMIT

 $\leq 4 \text{ mg/L (ppm)}$

WHAT TO MONITOR

Levels of regulated heavy metals and industrial contaminants in process water.

HOW TO MONITOR

Laboratory analysis of water samples by an accredited external laboratory.

WHEN (FREQUENCY)

Annually, or whenever there is a change in water source or system, or as required by local regulations.

WHO (RESPONSIBILITY)

Quality Assurance Manager

MONITORING RECORD

[Ref: Water Quality Test Report - Heavy Metals & Industrial Contaminants](#)

CCP 16

Step 80: SLICING/SHREDDING

C HAZARD

CRITICAL LIMIT

 $\leq 4 \text{ mg/L (ppm)}$

WHAT TO MONITOR

Presence/absence of cleaning chemicals in process water.

HOW TO MONITOR

Visual inspection for foaming/discoloration, sensory evaluation (smell), and pH testing using a calibrated pH meter or test strips.

WHEN (FREQUENCY)

At the start of each production shift, after any cleaning/sanitizing of water lines, and periodically throughout the shift (e.g., every 4 hours).

WHO (RESPONSIBILITY)

Production Operator / Quality Control Technician

MONITORING RECORD

[Ref: Process Water Quality Check Log](#)

CCP 16

Step 81: SLICING/SHREDDING

B HAZARD

CRITICAL LIMIT

 $\leq 4 \text{ mg/L (ppm)}$

WHAT TO MONITOR

Temperature of the sliced cabbage product in the holding area or storage prior to entering the MAP packaging line.

HOW TO MONITOR

Calibrated digital thermometer with a probe inserted into a representative product container or a temperature data logger placed within the product mass.

WHEN (FREQUENCY)

At the start of each production shift, and every 2 hours during continuous packaging operations.

WHO (RESPONSIBILITY)

Production Operator or Quality Control Technician.

MONITORING RECORD

[Ref: Product Temperature Log - Pre-MAP Packaging](#)

CCP 16

Step 82: SLICING/SHREDDING

B HAZARD

CRITICAL LIMIT

≤ 4 mg/L (ppm)

WHAT TO MONITOR

Oxygen (O2) concentration in the headspace of packaged sliced cabbage.

WHEN (FREQUENCY)

At the start of each production run, every 30 minutes during continuous operation, and after any equipment adjustment or changeover.

HOW TO MONITOR

Calibrated headspace gas analyzer (e.g., O2/CO2 analyzer).

WHO (RESPONSIBILITY)

Packaging Line Operator

MONITORING RECORD

[Ref: MAP Gas Concentration Log](#)

CCP 16

Step 83: SLICING/SHREDDING

B HAZARD

CRITICAL LIMIT

≤ 4 mg/L (ppm)

WHAT TO MONITOR

Carbon Dioxide (CO2) concentration in the headspace of packaged sliced cabbage.

WHEN (FREQUENCY)

At the start of each production run, every 30 minutes during continuous operation, and after any equipment adjustment or changeover.

HOW TO MONITOR

Calibrated headspace gas analyzer (e.g., O2/CO2 analyzer).

WHO (RESPONSIBILITY)

Packaging Line Operator

MONITORING RECORD

[Ref: MAP Gas Concentration Log](#)

CCP 16

Step 84: SLICING/SHREDDING

B HAZARD

CRITICAL LIMIT

≤ 4 mg/L (ppm)

WHAT TO MONITOR

Package seal integrity (absence of leaks, creases, or trapped product in seal area)

WHEN (FREQUENCY)

At the start of each production run, every 30-60 minutes during continuous operation, and after any machine adjustments or material changes.

HOW TO MONITOR

Visual inspection of seals for defects (creases, wrinkles, trapped product), followed by a vacuum leak test or bubble test on a representative sample.

WHO (RESPONSIBILITY)

Packaging Line Operator / Quality Control Technician

MONITORING RECORD

[Ref: MAP Seal Integrity Monitoring Log](#)

CCP 17 Step 85: METAL DETECTION

B HAZARD

CRITICAL LIMIT

100% food-grade certified gases (e.g., CO2, N2, O2)

WHAT TO MONITOR

Internal temperature of the cold storage facility/refrigerated area and representative product temperature.

WHEN (FREQUENCY)

Continuously monitored with alarms for excursions. Manual verification of facility temperature at least twice per shift (e.g., beginning and middle). Product temperature checked at receiving and before dispatch.

HOW TO MONITOR

Calibrated continuous temperature monitoring system with probes placed in critical areas of the storage facility and/or calibrated digital thermometer for spot checks of product temperature.

WHO (RESPONSIBILITY)

Cold Storage Operator / QA Technician

MONITORING RECORD

[Ref: Cold Storage Temperature Log / Product Temperature Verification Record](#)

CCP 17 Step 86: METAL DETECTION

B HAZARD

CRITICAL LIMIT

100% food-grade certified gases (e.g., CO2, N2, O2)

WHAT TO MONITOR

Oxygen (O2) concentration within sealed Modified Atmosphere Packaging (MAP) bags/containers.

WHEN (FREQUENCY)

At the start of each production run, after any machine adjustments, and every 30-60 minutes or per lot/batch (e.g., 3-5 packages per hour).

HOW TO MONITOR

Calibrated headspace gas analyzer, sampling from a representative number of finished packages.

WHO (RESPONSIBILITY)

Packaging Line Operator / QA Technician

MONITORING RECORD

[Ref: MAP Gas Concentration Log](#)

CCP 17 Step 87: METAL DETECTION

B HAZARD

CRITICAL LIMIT

100% food-grade certified gases (e.g., CO2, N2, O2)

WHAT TO MONITOR

The concentration of Carbon Dioxide (CO2) within the Modified Atmosphere Packaging (MAP) for sliced cabbage.

WHEN (FREQUENCY)

At the start of each production run, after any machine adjustments, and then every 30-60 minutes, or at least once per pallet/lot, whichever is more frequent.

HOW TO MONITOR

Using a calibrated headspace gas analyzer to measure CO2 concentration from a representative sample of sealed packages.

WHO (RESPONSIBILITY)

Packaging Line Operator or Quality Control Technician

MONITORING RECORD

[Ref: MAP Gas Concentration Log](#)

CCP 17 Step 88: METAL DETECTION

B HAZARD

CRITICAL LIMIT

100% food-grade certified gases (e.g., CO2, N2, O2)

WHAT TO MONITOR

The integrity of the seals on the Modified Atmosphere Packaging (MAP) for sliced cabbage.

WHEN (FREQUENCY)

Visual inspection continuously by packaging line operators. Destructive leak test at the start of each production run, after any machine adjustments, and then every 30-60 minutes, or at least once per pallet/lot, whichever is more frequent.

HOW TO MONITOR

Visual inspection for obvious defects (e.g., creases, trapped product, tears) and a destructive leak test (e.g., vacuum leak test where packages are submerged in water and vacuum applied, observing for bubbles).

WHO (RESPONSIBILITY)

Packaging Line Operator or Quality Control Technician

MONITORING RECORD

[Ref: Package Seal Integrity Log](#)

CCP 17

Step 89: METAL DETECTION

B HAZARD

CRITICAL LIMIT

100% food-grade certified gases (e.g., CO2, N2, O2)

WHAT TO MONITOR

Verification of MAP gas supplier approval status and documentation.

HOW TO MONITOR

Review of current Approved Supplier List, supplier qualification records, and Certificates of Conformance/Analysis for incoming gas deliveries.

WHEN (FREQUENCY)

Upon receipt of each new gas cylinder/delivery and annually during supplier re-qualification.

WHO (RESPONSIBILITY)

Quality Assurance Manager or Receiving Supervisor.

MONITORING RECORD

[Ref: Approved Supplier Log / Incoming Material Inspection Record](#)

CCP 18

Step 90: POST-SLICING WASHING & RINSING

C HAZARD

CRITICAL LIMIT

50-200 ppm free chlorine (or other approved sanitizer at manufacturer's recommended effective concentration)

WHAT TO MONITOR

Verification of food-grade certification for all MAP gases used.

HOW TO MONITOR

Review of Certificates of Analysis (COA) or Letters of Guarantee from gas suppliers, confirming compliance with food-grade standards (e.g., EIGA, FDA, national food safety regulations).

WHEN (FREQUENCY)

Upon receipt of each new gas cylinder/delivery, or at a minimum, annually for standing contracts, and whenever a new supplier or gas type is introduced.

WHO (RESPONSIBILITY)

Quality Assurance (QA) Manager or Receiving Supervisor.

MONITORING RECORD

[Ref: MAP Gas Certification Log / Supplier COA File](#)

CCP 18

Step 91: POST-SLICING WASHING & RINSING

C HAZARD

CRITICAL LIMIT

50-200 ppm free chlorine (or other approved sanitizer at manufacturer's recommended effective concentration)

WHAT TO MONITOR

Confirmation that the MAP gas supplier is on the company's Approved Supplier List.

HOW TO MONITOR

Cross-reference supplier name on delivery documents/invoices with the official Approved Supplier List maintained by the company.

WHEN (FREQUENCY)

Upon receipt of each new gas cylinder/delivery.

WHO (RESPONSIBILITY)

Receiving Supervisor or QA Technician.

MONITORING RECORD

[Ref: Receiving Log / Approved Supplier Verification Form](#)

CCP 18

Step 92: POST-SLICING WASHING & RINSING

B HAZARD

CRITICAL LIMIT

50-200 ppm free chlorine (or other approved sanitizer at manufacturer's recommended effective concentration)

WHAT TO MONITOR

pH of the wash water containing chlorine-based sanitizer.

HOW TO MONITOR

Calibrated pH meter or pH test strips.

WHEN (FREQUENCY)

At the start of each shift/batch, and every 2 hours during continuous operation.

WHO (RESPONSIBILITY)

Production Operator / Quality Control Technician.

MONITORING RECORD

[Ref: Wash Water Quality Log / CCP Monitoring Log](#)

CCP 18

Step 93: POST-SLICING WASHING & RINSING

B HAZARD

CRITICAL LIMIT

50-200 ppm free chlorine (or other approved sanitizer at manufacturer's recommended effective concentration)

WHAT TO MONITOR

Potability of water used for post-slicing washing and rinsing.

HOW TO MONITOR

Review of municipal water quality reports or in-house microbiological testing (e.g., coliforms, E. coli). Visual inspection for clarity, odor, and absence of foreign matter.

WHEN (FREQUENCY)

Annually for municipal reports; weekly/monthly for in-house testing (if applicable); daily visual inspection.

WHO (RESPONSIBILITY)

Quality Assurance Manager / Production Supervisor

MONITORING RECORD

[Ref: Water Potability Log / Municipal Water Quality Report File](#)

CCP 18

Step 94: POST-SLICING WASHING & RINSING

B HAZARD

CRITICAL LIMIT

50-200 ppm free chlorine (or other approved sanitizer at manufacturer's recommended effective concentration)

WHAT TO MONITOR

Accuracy and completeness of all mandatory label information, including product name, ingredients, 'WASH BEFORE USE' instruction, net weight, date code, storage instructions, and manufacturer details.

HOW TO MONITOR

Visually compare applied labels against the approved master label specification for the specific product. Verify presence and correctness of all mandatory elements, especially 'WASH BEFORE USE' and ingredient declaration (100% cabbage, no allergens).

WHEN (FREQUENCY)

At the start of each production run, at every product changeover, and hourly during continuous production.

WHO (RESPONSIBILITY)

Packaging Line Operator or Quality Control Technician.

MONITORING RECORD

[Ref: CCP 18 Label Verification Log](#)

CCP 18 Step 95: POST-SLICING WASHING & RINSING

B HAZARD

CRITICAL LIMIT

50-200 ppm free chlorine (or other approved sanitizer at manufacturer's recommended effective concentration)

WHAT TO MONITOR

Correct label application and presence of all mandatory information on product units.

WHEN (FREQUENCY)

At the start of each production run, every 30 minutes during operation, and at product changeovers.

HOW TO MONITOR

Visual inspection of product units by packaging line operator. Automated vision system check (if applicable).

WHO (RESPONSIBILITY)

Packaging Line Operator / Quality Control Technician

MONITORING RECORD

[Ref: Label Verification Log](#)

CCP 19 Step 96: LABELING

C HAZARD

CRITICAL LIMIT

Product storage temperature must be maintained at or below 4°C (40°F).

WHAT TO MONITOR

Verification of label content for accuracy (100% cabbage, no allergens) and completeness (all mandatory information, including 'WASH BEFORE USE' instruction).

WHEN (FREQUENCY)

At the start of each production run, when a new roll of labels is loaded, and after any label changeover.

HOW TO MONITOR

Visual inspection of a sample label from each new roll or batch of labels, comparing against the approved label specification and product formulation.

WHO (RESPONSIBILITY)

Quality Control Technician / Line Supervisor

MONITORING RECORD

[Ref: Label Content Verification Log](#)

CCP 19

Step 97: LABELING

C HAZARD

CRITICAL LIMIT

Product storage temperature must be maintained at or below 4°C (40°F).

WHAT TO MONITOR

Verification that the correct 'sliced cabbage' label is applied to each product unit and contains all mandatory information.

HOW TO MONITOR

Visual inspection of a statistically significant sample of labeled units from each production lot/pallet. If an automated vision system is in place, continuous monitoring of 100% of units. Comparison against approved label specification.

WHEN (FREQUENCY)

At the start of each production run, every 30 minutes (or specified interval based on line speed), and prior to release of each pallet/lot from the labeling station. If automated system, continuously.

WHO (RESPONSIBILITY)

Line Operator / Quality Control Technician

MONITORING RECORD

[Ref: Label Application Verification Log / Finished Product Release Record](#)

CCP 19

Step 98: LABELING

B HAZARD

CRITICAL LIMIT

Product storage temperature must be maintained at or below 4°C (40°F).

WHAT TO MONITOR

Product temperature during storage and distribution.

HOW TO MONITOR

Continuous monitoring using calibrated data loggers placed with product, and/or manual checks with calibrated thermometers in storage/transport units.

WHEN (FREQUENCY)

Continuously (data loggers); at least once per shift for storage units; upon loading and unloading for transport units.

WHO (RESPONSIBILITY)

Warehouse Supervisor, Logistics Coordinator, or QA Technician

MONITORING RECORD

[Ref: Temperature Monitoring Log \(Storage/Transport\), Data Logger Records Review](#)

CCP 20

Step 99: VISUAL INSPECTION & SORTING

B HAZARD

CRITICAL LIMIT

Product internal temperature $\leq 4^{\circ}\text{C}$ (40°F)

WHAT TO MONITOR

The ambient air temperature within the finished product cold storage unit.

HOW TO MONITOR

Calibrated continuous temperature monitoring system (e.g., data logger with probes) with alarms, verified by a calibrated handheld thermometer.

WHEN (FREQUENCY)

Continuously recorded by data logger. Manual verification at least twice per shift (e.g., beginning and end of shift) or every 4 hours, whichever is more frequent.

WHO (RESPONSIBILITY)

Cold Storage Operator / Quality Control Technician.

MONITORING RECORD

[Ref: Finished Product Cold Storage Temperature Log / Continuous Temperature Monitoring System Record](#)

CCP 20

Step 100: VISUAL INSPECTION & SORTING

B HAZARD

CRITICAL LIMIT

Product internal temperature $\leq 4^{\circ}\text{C}$ (40°F)

WHAT TO MONITOR

Duration and peak temperature of any temperature excursion above 4°C within the finished product cold storage unit.

HOW TO MONITOR

Continuous temperature monitoring system (data logger) with programmable alarms for both duration and peak temperature limits. Review of data logger records.

WHEN (FREQUENCY)

Continuously monitored by data logger. Review of data logger records at least once per shift or daily, and immediate investigation upon alarm activation.

WHO (RESPONSIBILITY)

Cold Storage Operator / Quality Control Supervisor.

MONITORING RECORD

[Ref: Finished Product Cold Storage Excursion Log / Continuous Temperature Monitoring System Alarm Report](#)

CCP 20 Step 101: VISUAL INSPECTION & SORTING

B HAZARD

CRITICAL LIMIT

Product internal temperature <= 4°C (40°F)

WHAT TO MONITOR

Microbiological cleanliness of food contact surfaces (ATP levels and/or pathogen presence).

HOW TO MONITOR

ATP luminometer with calibrated swabs for RLU readings on critical food contact surfaces. For pathogen detection, sterile environmental swabs collected and sent to an accredited laboratory.

WHEN (FREQUENCY)

ATP testing: Daily, after sanitation and prior to start of production, on a rotating schedule for all critical food contact surfaces. Pathogen swabbing: Weekly or bi-weekly, on a rotating schedule, as per environmental monitoring program.

WHO (RESPONSIBILITY)

Quality Control Technician or Sanitation Supervisor.

MONITORING RECORD

[Ref: ATP Monitoring Log / Environmental Swabbing Record.](#)

CCP 20 Step 102: VISUAL INSPECTION & SORTING

B HAZARD

CRITICAL LIMIT

Product internal temperature <= 4°C (40°F)

WHAT TO MONITOR

Internal product temperature of a representative sample of finished product during loading and transit.

HOW TO MONITOR

Calibrated probe thermometer (e.g., thermocouple) inserted into the product.

WHEN (FREQUENCY)

At the start and end of each loading process, and upon arrival at transit points/destination.

WHO (RESPONSIBILITY)

Shipping/Receiving Personnel or Logistics Coordinator.

MONITORING RECORD

[Ref: Finished Product Loading/Transit Temperature Log.](#)

CCP 21 Step 103: DISPATCH

B HAZARD

CRITICAL LIMIT

Interior surfaces and cargo area must be clean, dry, and free from visible residues, spills, or odors of non-food grade chemicals, lubricants, or fuel.

WHAT TO MONITOR

Internal temperature of a representative sample of product packages and/or air temperature within the transport vehicle's cargo area.

HOW TO MONITOR

Calibrated probe thermometer for product internal temperature; calibrated infrared thermometer for surface temperature; verification of transport vehicle's refrigeration unit display and/or temperature logger data.

WHEN (FREQUENCY)

Prior to loading the first pallet/batch, and verification of continuous temperature logging for the duration of transit.

WHO (RESPONSIBILITY)

Dispatch Supervisor or Loading Personnel

MONITORING RECORD

[Ref: Dispatch Temperature Log / Shipping Temperature Record](#)

CCP 22 Step 104: VISUAL INSPECTION & SORTING

C HAZARD

CRITICAL LIMIT

Only NSF H1 or H3 certified food-grade lubricants used on equipment with potential food contact.

WHAT TO MONITOR

Visual cleanliness and odor of the transport vehicle's interior cargo area (walls, floor, ceiling).

HOW TO MONITOR

Visual inspection and olfactory assessment (smell test) of the cargo area.

WHEN (FREQUENCY)

Prior to loading any product into the vehicle.

WHO (RESPONSIBILITY)

Dispatch Supervisor or Loading Personnel

MONITORING RECORD

[Ref: Pre-shipment Vehicle Inspection Log / Transport Vehicle Condition Checklist](#)

CCP 22 Step 105: VISUAL INSPECTION & SORTING

C HAZARD

CRITICAL LIMIT

Only NSF H1 or H3 certified food-grade lubricants used on equipment with potential food contact.

WHAT TO MONITOR

Presence of chemical residues on all food contact surfaces after final rinse.

HOW TO MONITOR

Visual inspection for residue or odor, and/or use of pH strips or ATP swabs on randomly selected food contact surfaces to confirm absence of chemical residues.

WHEN (FREQUENCY)

Prior to start of production, after cleaning and sanitizing procedures, and after any equipment breakdown requiring re-cleaning.

WHO (RESPONSIBILITY)

Sanitation Supervisor / Quality Control Technician

MONITORING RECORD

[Ref: CCP Monitoring Log - Chemical Residue Verification](#)

CCP 24 Step 106: VISUAL INSPECTION & SORTING

P HAZARD

CRITICAL LIMIT

No foreign material (e.g., metal, plastic, glass, wood, stones) detected in the product during visual inspection.

WHAT TO MONITOR

Absence of foreign material in the sliced cabbage product.

HOW TO MONITOR

Continuous visual inspection of product on conveyor belt/sorting table by trained personnel.

WHEN (FREQUENCY)

Continuously during the visual inspection and sorting process.

WHO (RESPONSIBILITY)

Sorting Line Operator / Quality Control Inspector.

MONITORING RECORD

[Ref: Foreign Material Detection Log / Visual Inspection Record.](#)

CCP 24 Step 107: VISUAL INSPECTION & SORTING

P HAZARD

CRITICAL LIMIT

No foreign material (e.g., metal, plastic, glass, wood, stones) detected in the product during visual inspection.

WHAT TO MONITOR

Integrity and condition of physical separation devices (screens, filters)

HOW TO MONITOR

Visual inspection of screens and filters for tears, holes, proper installation, and specified mesh size

WHEN (FREQUENCY)

Prior to start of production, at each shift change, and after any maintenance or adjustment

WHO (RESPONSIBILITY)

Quality Control Technician or Production Supervisor

MONITORING RECORD

[Ref: Physical Separation Device Inspection Log](#)

CCP 26 Step 108: WEIGHING & PORTIONING

P HAZARD

CRITICAL LIMIT

No visible foreign material present in product or on equipment surfaces.

WHAT TO MONITOR

Presence of visible foreign material in product and on equipment surfaces.

HOW TO MONITOR

Direct visual inspection of product flow, weighing/portioning equipment, and surrounding work area.

WHEN (FREQUENCY)

Continuously during operation and at the start of each shift/batch.

WHO (RESPONSIBILITY)

Line Operator / Quality Control Technician

MONITORING RECORD

[Ref: Foreign Material Visual Inspection Log](#)

CCP 26 Step 109: WEIGHING & PORTIONING**P HAZARD****CRITICAL LIMIT****No visible foreign material present in product or on equipment surfaces.****WHAT TO MONITOR**

Personnel compliance with GMPs regarding protective clothing, jewelry, and personal items.

HOW TO MONITOR

Direct visual observation of personnel by supervisor or QC.

WHEN (FREQUENCY)

At the start of each shift and periodically throughout the shift (e.g., hourly checks).

WHO (RESPONSIBILITY)

Production Supervisor / Quality Control Technician

MONITORING RECORD

[Ref: GMP Personnel Monitoring Checklist](#)

CCP 26 Step 110: WEIGHING & PORTIONING**P HAZARD****CRITICAL LIMIT****No visible foreign material present in product or on equipment surfaces.****WHAT TO MONITOR**

Physical condition and integrity of food contact surfaces of weighing and portioning equipment.

HOW TO MONITOR

Visual inspection of all food contact surfaces for damage (cracks, chips, rust, loose parts).

WHEN (FREQUENCY)

Prior to start of production, and every 4 hours during operation.

WHO (RESPONSIBILITY)

Production Line Operator / Quality Control Technician

MONITORING RECORD

[Ref: CCP Equipment Integrity Check Log](#)

CCP 28 Step 111: PRIMARY PACKAGING**B HAZARD****CRITICAL LIMIT****Product temperature must not exceed 4°C (40°F) during the entire packaging process.****WHAT TO MONITOR**

Internal product temperature of sliced cabbage during the packaging process.

HOW TO MONITOR

Calibrated digital thermometer with a penetration probe, inserted into the core of the product.

WHEN (FREQUENCY)

At the start of packaging, every 30 minutes during continuous operation, and at the end of the packaging run.

WHO (RESPONSIBILITY)

Production Operator or Quality Control Technician.

MONITORING RECORD

[Ref: Product Temperature Monitoring Log - Packaging](#)

CCP 28 Step 112: PRIMARY PACKAGING**B HAZARD****CRITICAL LIMIT**

Product temperature must not exceed 4°C (40°F) during the entire packaging process.

WHAT TO MONITOR

Primary packaging materials (PE bags, PP clam-shells) for food-grade status, intactness, cleanliness, and absence of visible defects or contamination.

WHEN (FREQUENCY)

Prior to use of each new batch/delivery of packaging materials. Visually inspect a representative sample at the start of each production run and hourly during operations.

HOW TO MONITOR

Visual inspection of a representative sample of packaging materials. Verification of supplier Certificates of Conformance for food-grade status.

WHO (RESPONSIBILITY)

Packaging Line Operator or Quality Control Technician

MONITORING RECORD

[Ref: Packaging Material Inspection Log](#)

CCP 28 Step 113: PRIMARY PACKAGING**B HAZARD****CRITICAL LIMIT**

Product temperature must not exceed 4°C (40°F) during the entire packaging process.

WHAT TO MONITOR

Cleanliness and sanitation of food contact surfaces of packaging equipment and surrounding environment.

WHEN (FREQUENCY)

Prior to the start of packaging operations. Visually inspect periodically (e.g., every 2-4 hours) during operations. ATP testing daily or per shift after sanitation.

HOW TO MONITOR

Visual inspection for cleanliness. ATP testing (or equivalent rapid hygiene test) on food contact surfaces.

WHO (RESPONSIBILITY)

Sanitation Team Leader, Packaging Line Supervisor, or Quality Control Technician

MONITORING RECORD

[Ref: Sanitation Verification Log / ATP Monitoring Record](#)

CCP 28 Step 114: PRIMARY PACKAGING**B HAZARD****CRITICAL LIMIT**

Product temperature must not exceed 4°C (40°F) during the entire packaging process.

WHAT TO MONITOR

O2 and CO2 concentration in the packaged product headspace.

WHEN (FREQUENCY)

At the start of each production run, every 30-60 minutes during continuous operation, and after any machine adjustments or material changes.

HOW TO MONITOR

Calibrated headspace gas analyzer.

WHO (RESPONSIBILITY)

Packaging Line Operator or Quality Control Technician.

MONITORING RECORD

[Ref: MAP Gas Composition Log](#)

CCP 30

Step 115: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING

B HAZARD

CRITICAL LIMIT

<= 4°C (40°F)

WHAT TO MONITOR

Internal temperature of sliced cabbage product immediately prior to packaging.

HOW TO MONITOR

Using a calibrated, sanitized thermocouple probe or digital thermometer, insert into the center of a representative product sample from the packaging line.

WHEN (FREQUENCY)

At the start of each packaging run and every 30 minutes thereafter, or at a minimum of once per batch.

WHO (RESPONSIBILITY)

Packaging Line Operator or Quality Control Technician

MONITORING RECORD

[Ref: Packaging Product Temperature Log](#)

CCP 30

Step 116: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING

B HAZARD

CRITICAL LIMIT

<= 4°C (40°F)

WHAT TO MONITOR

ATP levels on food contact surfaces of packaging equipment (e.g., conveyor belts, product contact areas of the filler/sealer).

HOW TO MONITOR

ATP luminometer using calibrated ATP swabs according to manufacturer's instructions.

WHEN (FREQUENCY)

Before start of production, after any major cleaning or sanitation event, and every 4 hours during continuous operation.

WHO (RESPONSIBILITY)

Quality Control Technician or Sanitation Supervisor.

MONITORING RECORD

[Ref: ATP Swab Monitoring Log - Packaging Surfaces](#)

CCP 32

Step 117: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING

C HAZARD

CRITICAL LIMIT

Packaging materials must be certified food-grade and comply with all applicable regulatory standards for direct food contact.

WHAT TO MONITOR

Certification and compliance of packaging materials (e.g., film, trays)

HOW TO MONITOR

Review of Certificates of Conformance (CoC) or Letters of Guarantee (LoG) from suppliers, verifying against internal specifications and regulatory requirements.

WHEN (FREQUENCY)

Upon receipt of each new lot/shipment of packaging material, and annually for supplier re-qualification.

WHO (RESPONSIBILITY)

Quality Assurance Manager / Purchasing Manager

MONITORING RECORD

[Ref: Packaging Material Compliance Records / Supplier CoC Log](#)

CCP 32 Step 118: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING

C HAZARD

CRITICAL LIMIT

Packaging materials must be certified food-grade and comply with all applicable regulatory standards for direct food contact.

WHAT TO MONITOR

Presence/absence of cleaning chemical residue on food contact surfaces of packaging machinery.

WHEN (FREQUENCY)

Prior to start of packaging operations, after any cleaning/sanitizing cycle, and at shift change.

HOW TO MONITOR

Visual inspection for residue; pH testing or chemical residue test strips on rinsed surfaces.

WHO (RESPONSIBILITY)

Sanitation Supervisor / QA Technician

MONITORING RECORD

Ref: [Pre-Operational Inspection Checklist / Sanitation Verification Log](#)

CCP 32 Step 119: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING

C HAZARD

CRITICAL LIMIT

Packaging materials must be certified food-grade and comply with all applicable regulatory standards for direct food contact.

WHAT TO MONITOR

Type and certification of lubricants used on packaging machinery where there is potential for incidental food contact.

WHEN (FREQUENCY)

Upon receipt of new lubricant batches, during routine maintenance checks, and annually as part of the HACCP review.

HOW TO MONITOR

Review of lubricant product labels, Safety Data Sheets (SDS), and purchasing records to confirm NSF H1 certification. Visual inspection of lubricant containers.

WHO (RESPONSIBILITY)

Maintenance Manager / QA Manager

MONITORING RECORD

Ref: [Approved Lubricant List / Maintenance Log / Purchasing Records](#)

CCP 34

Step 120: METAL DETECTION

B HAZARD

CRITICAL LIMIT

<= 4°C (40°F)

WHAT TO MONITOR

Internal product storage temperature in refrigerated units/vehicles.

HOW TO MONITOR

Using a calibrated digital thermometer or continuous temperature data logger placed within the product mass or at the warmest point of the storage area.

WHEN (FREQUENCY)

At least once per shift for static storage units; continuously for refrigerated transport with data logger download upon arrival/departure.

WHO (RESPONSIBILITY)

Cold Storage Operator / Warehouse Supervisor / Transport Driver.

MONITORING RECORD

[Ref: Refrigeration Temperature Log / Cold Chain Monitoring Record](#)

CCP 34

Step 121: METAL DETECTION

B HAZARD

CRITICAL LIMIT

<= 4°C (40°F)

WHAT TO MONITOR

MAP Oxygen (O2) concentration in packaged sliced cabbage

HOW TO MONITOR

Using a calibrated headspace gas analyzer to sample gas from sealed packages

WHEN (FREQUENCY)

At the start of each production run, every 30 minutes during production, and at the end of the run (or minimum 3 packages per hour per line)

WHO (RESPONSIBILITY)

Packaging Line Operator or Quality Control Technician

MONITORING RECORD

[Ref: MAP Gas Composition Monitoring Log](#)

CCP 34

Step 122: METAL DETECTION

B HAZARD

CRITICAL LIMIT

<= 4°C (40°F)

WHAT TO MONITOR

MAP Carbon Dioxide (CO2) concentration in packaged sliced cabbage

HOW TO MONITOR

Using a calibrated headspace gas analyzer to sample gas from sealed packages

WHEN (FREQUENCY)

At the start of each production run, every 30 minutes during production, and at the end of the run (or minimum 3 packages per hour per line)

WHO (RESPONSIBILITY)

Packaging Line Operator or Quality Control Technician

MONITORING RECORD

[Ref: MAP Gas Composition Monitoring Log](#)

CCP 34

Step 123: METAL DETECTION

B HAZARD

CRITICAL LIMIT

<= 4°C (40°F)

WHAT TO MONITOR

Package seal integrity (absence of leaks, tears, or incomplete seals).

HOW TO MONITOR

Conduct a vacuum leak test or bubble test on a representative sample of 3-5 packages. Visually inspect all packages for obvious seal defects (creases, folds, tears, incomplete seals).

WHEN (FREQUENCY)

At the start of each production run, every 2 hours during continuous production, and after any packaging machine adjustment or film roll change.

WHO (RESPONSIBILITY)

Packaging Line Operator or Quality Control Technician.

MONITORING RECORD

[Ref: CCP 34 Packaging Seal Integrity Log](#)

CCP 36

Step 124: METAL DETECTION

C HAZARD

CRITICAL LIMIT

100% food-grade certified gases (e.g., CO₂, N₂, O₂)

WHAT TO MONITOR

Food-grade certification for all MAP gases (CO₂, N₂, O₂) used in packaging.

HOW TO MONITOR

Review of Certificates of Analysis (CoA) or Certificates of Conformance (CoC) from gas suppliers. Verification of supplier approval status.

WHEN (FREQUENCY)

Upon receipt of each new gas delivery (cylinder/tanker) and annually for supplier certification.

WHO (RESPONSIBILITY)

Receiving Personnel / Quality Assurance Manager

MONITORING RECORD

[Ref: MAP Gas Certification Log / Supplier CoA File](#)

CCP 36

Step 125: METAL DETECTION

C HAZARD

CRITICAL LIMIT

100% food-grade certified gases (e.g., CO₂, N₂, O₂)

WHAT TO MONITOR

Verification of MAP gas supplier approval status and gas quality documentation.

HOW TO MONITOR

Review of current Approved Supplier List, supplier certificates of analysis (COA) for each gas delivery, and purchasing records.

WHEN (FREQUENCY)

Upon receipt of each new batch/delivery of MAP gases and annually during supplier re-evaluation.

WHO (RESPONSIBILITY)

Quality Assurance Manager or Purchasing Manager

MONITORING RECORD

[Ref: Approved Supplier List, MAP Gas COA Log, Supplier Audit Records](#)

CCP 38 Step 126: LABELING

C HAZARD

CRITICAL LIMIT

All labels applied to sliced cabbage product must accurately reflect the product contents (100% cabbage, no allergens) and include all mandatory information, including 'WASH BEFORE USE' instruction.

WHAT TO MONITOR

Content of the label, verifying '100% cabbage, no allergens' and 'WASH BEFORE USE' instruction.

HOW TO MONITOR

Visual comparison of a sample label against the approved master label specification.

WHEN (FREQUENCY)

At the start of each production run, after any label roll change, and at least once per shift.

WHO (RESPONSIBILITY)

Quality Control Technician or Production Supervisor

MONITORING RECORD

[Ref: Pre-Operational Label Verification Log](#)

CCP 38 Step 127: LABELING

C HAZARD

CRITICAL LIMIT

All labels applied to sliced cabbage product must accurately reflect the product contents (100% cabbage, no allergens) and include all mandatory information, including 'WASH BEFORE USE' instruction.

WHAT TO MONITOR

Correct 'sliced cabbage' label is applied to each product unit, legible, and intact.

HOW TO MONITOR

Visual inspection of product units on the line by trained personnel. Automated label verification system (e.g., barcode scanner, vision system) may be used.

WHEN (FREQUENCY)

Continuously by the line operator during production, with periodic documented checks by Quality Control (e.g., every 30 minutes or per pallet).

WHO (RESPONSIBILITY)

Line Operator (continuous), Quality Control Technician (periodic)

MONITORING RECORD

[Ref: In-Process Label Application Check Log](#)

CCP 40 Step 128: FINISHED PRODUCT COLD STORAGE

B HAZARD

CRITICAL LIMIT

Product storage temperature must be maintained at or below 4°C (40°F).

WHAT TO MONITOR

Ambient air temperature of the finished product cold storage unit.

HOW TO MONITOR

Calibrated continuous temperature recording device (data logger) with visual verification of a primary thermometer.

WHEN (FREQUENCY)

Continuously recorded by data logger, with visual verification at least once per shift.

WHO (RESPONSIBILITY)

Cold Storage Operator or Quality Control Technician.

MONITORING RECORD

[Ref: Cold Storage Temperature Log / Continuous Temperature Recorder Chart.](#)

CCP 40

Step 129: FINISHED PRODUCT COLD STORAGE

B HAZARD

CRITICAL LIMIT

Product storage temperature must be maintained at or below 4°C (40°F).

WHAT TO MONITOR

Ambient temperature of the finished product cold storage unit and representative product temperature.

HOW TO MONITOR

Calibrated continuous temperature monitoring system (data logger) with alarms, verified by a calibrated probe thermometer for spot checks of product temperature.

WHEN (FREQUENCY)

Continuous monitoring of ambient temperature with daily verification of data logger readings. Product temperature checked at least once per shift.

WHO (RESPONSIBILITY)

Cold Storage Operator / Quality Control Technician

MONITORING RECORD

[Ref: Finished Product Cold Storage Temperature Log / Continuous Temperature Monitoring Chart](#)

CCP 42

Step 130: DISPATCH

B HAZARD

CRITICAL LIMIT

≤ 4°C (40°F)

WHAT TO MONITOR

Internal temperature of finished product pallets/cases prior to loading and ambient temperature of the transport vehicle during transit.

HOW TO MONITOR

Calibrated probe thermometer for product temperature. Calibrated continuous temperature monitoring device (data logger) within the transport vehicle.

WHEN (FREQUENCY)

Product temperature checked for the first and last pallet/case of each load, and at least 3 representative pallets/cases per vehicle. Transport vehicle temperature continuously monitored during transit.

WHO (RESPONSIBILITY)

Dispatch Supervisor / Loading Personnel

MONITORING RECORD

[Ref: Dispatch Temperature Log / Shipping Vehicle Temperature Record](#)

CRITICAL LIMIT

Interior surfaces and cargo area must be clean, dry, and free from visible residues, spills, or odors of non-food grade chemicals, lubricants, or fuel.

WHAT TO MONITOR

Cleanliness, dryness, and absence of non-food grade chemical residues, spills, or odors in the transport vehicle's interior and cargo area.

WHEN (FREQUENCY)

Prior to loading each transport vehicle.

HOW TO MONITOR

Visual inspection of interior surfaces and cargo area for cleanliness, dryness, and absence of residues/spills. Olfactory inspection for any non-food grade chemical or fuel odors.

WHO (RESPONSIBILITY)

Dispatch Supervisor or Quality Control Technician.

MONITORING RECORD

[Ref: Transport Vehicle Pre-Load Inspection Log](#)

Step 7: Corrective Action Plans

CCP 1 Step 1: RECEIVING RAW CABBAGE

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Segregate and hold the entire lot of raw cabbage from the non-approved supplier. Do not accept or unload the product. Clearly label the product as 'HOLD - Non-Approved Supplier'. 2. PROCESS CORRECTION: Reject the entire shipment and return it to the supplier. Update the Approved Supplier List if the supplier was mistakenly removed or if a new supplier needs to be evaluated. Reinforce training for receiving personnel on the importance of verifying supplier status against the Approved Supplier List. 3. INVESTIGATION: QA Manager/Purchasing Manager to investigate why the non-approved supplier was used. This includes reviewing purchasing orders, communication logs, and training records of receiving personnel. Evaluate the supplier for potential future approval if they meet all company and regulatory requirements, or confirm their permanent disqualification. Assess the risk of any previously accepted product from this supplier if applicable.

ACTION RESPONSIBLE

Receiving Clerk / QA Manager / Purchasing Manager

Step 7 Compliance

CCP 1 Step 2: RECEIVING RAW CABBAGE

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Segregate and hold the entire lot of raw cabbage that exceeds the critical limit. Do not accept or unload the product. Clearly label the product as 'HOLD - Temperature Deviation'. 2. PROCESS CORRECTION: Evaluate the extent and duration of the temperature abuse. If the temperature deviation is minor and short-term (e.g., slightly above 4°C for a very brief period, and product quality is not compromised), the QA Manager may approve acceptance for immediate processing, but this must be a documented decision with justification. Otherwise, reject the entire shipment and return it to the supplier. If accepted, prioritize its use and ensure it is processed quickly. 3. INVESTIGATION: QA Manager/Logistics Manager to investigate the cause of the temperature deviation (e.g., supplier's pre-cooling, transport refrigeration failure, delayed unloading). Communicate findings to the supplier and request corrective actions from them. Review supplier performance records and consider re-evaluation if temperature deviations are recurring. Review internal receiving procedures and equipment (e.g., thermometer calibration) to ensure they are not contributing factors.

ACTION RESPONSIBLE

Receiving Clerk / QA Manager / Logistics Manager

Step 7 Compliance

CCP 1 Step 3: RECEIVING RAW CABBAGE

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: Hold and segregate the entire non-conforming lot of raw cabbage. Reject the lot and return to the supplier or dispose of it according to company policy for contaminated produce. PROCESS CORRECTION: Retrain Receiving Personnel on visual inspection criteria and critical limits for raw cabbage. Communicate the non-conformance to the supplier, requesting an investigation into their quality control processes and corrective actions. Increase inspection frequency or stringency for subsequent deliveries from the same supplier until confidence is restored. INVESTIGATION: Determine the extent of the problem (isolated incident vs. systemic issue). Investigate if any non-conforming product inadvertently entered the process; if so, identify and hold any potentially affected finished product. Review the supplier's previous performance records for similar issues.

ACTION RESPONSIBLE

Quality Control Technician / Receiving Manager

Step 7 Compliance

CCP 2 Step 4: RECEIVING RAW CABBAGE

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: Hold and segregate the entire lot of raw cabbage received from the non-approved supplier. Reject the lot and return to the supplier or dispose of it according to company policy. Do not accept any further deliveries from this supplier until they are formally approved. PROCESS CORRECTION: Retrain Receiving Personnel on the importance of verifying supplier approval status and the procedure for checking documentation against the Approved Supplier List. Review the Approved Supplier Program to ensure it is robust and up-to-date. Initiate the supplier approval process for the non-approved supplier if they wish to become an approved vendor, or formally disqualify them. INVESTIGATION: Determine why the non-approved supplier was used (e.g., emergency order, oversight, lack of training, outdated list). Assess if any product from this non-approved supplier has been previously accepted and entered the production stream; if so, identify and hold any potentially affected finished product. Review the effectiveness of the Approved Supplier Program and internal communication regarding supplier status.

ACTION RESPONSIBLE**Quality Control Technician / Purchasing Manager**

Step 7 Compliance

CCP 2 Step 5: RECEIVING RAW CABBAGE

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Isolate and hold the entire lot of raw cabbage. Do not accept or unload the product. Tag the lot as 'HOLD - NON-CONFORMING'. 2. PROCESS CORRECTION: Contact the supplier immediately to obtain the missing COA or to address the non-conforming results. If the COA confirms non-compliance, reject the entire lot and arrange for its return or destruction. Update supplier approval status if repeated issues occur. 3. INVESTIGATION: Investigate the reason for the missing/non-compliant COA. Review supplier performance records. Determine if the issue is systemic with the supplier or an isolated incident. Re-evaluate supplier approval if necessary.

ACTION RESPONSIBLE**Receiving Clerk / Quality Assurance Technician / Purchasing Manager**

Step 7 Compliance

CCP 2 Step 6: RECEIVING RAW CABBAGE

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Isolate and hold the entire lot of raw cabbage. Do not accept or unload the product. Tag the lot as 'HOLD - NON-CONFORMING'. If direct contamination (e.g., chemical spill, allergen contact) is evident, immediately reject the entire lot. 2. PROCESS CORRECTION: Contact the supplier/carrier immediately to report the non-conformance. Reject the entire lot of raw cabbage if there is any risk of contamination. Request a clean, appropriate transport vehicle for future deliveries. Update carrier/supplier approval status if repeated issues occur. 3. INVESTIGATION: Investigate the cause of the non-conforming transport vehicle (e.g., carrier cleaning procedures, loading practices, segregation protocols). Review supplier/carrier performance records. Determine if the issue is systemic or isolated. Re-evaluate carrier/supplier approval if necessary.

ACTION RESPONSIBLE**Receiving Clerk / Quality Assurance Technician / Purchasing Manager**

Step 7 Compliance

CCP 3 Step 7: RAW MATERIAL COLD STORAGE

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Isolate and hold all affected raw material lots. Immediately move product to a properly functioning cold storage unit to bring temperature back within critical limits. If the temperature deviation was prolonged (e.g., >4 hours above 4°C or any exposure above 7°C), the product must be rejected and disposed of. Notify QA and Production Management. 2. PROCESS CORRECTION: Verify the functionality of the cold storage unit (e.g., check thermostat settings, door seals, refrigeration unit performance). Adjust or repair the cold storage unit as necessary. Retrain cold storage operators on proper temperature monitoring procedures and critical limit adherence. 3. INVESTIGATION: Determine the root cause of the temperature deviation (e.g., equipment malfunction, human error, power outage, improper loading, door left open). Review temperature monitoring records for trends. Evaluate the impact on product safety based on the duration and temperature of abuse. Implement preventive measures to avoid recurrence (e.g., scheduled maintenance, alarm systems, revised SOPs).

ACTION RESPONSIBLE**Cold Storage Operator / QA Technician / Maintenance Manager**

Step 7 Compliance

CCP 4 Step 8: RECEIVING RAW CABBAGE

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately cease processing of the affected batch. Isolate and hold the affected product. If the product has exceeded the 4-hour cumulative time limit above 4°C, it must be rejected and disposed of due to the significant risk of pathogen proliferation in raw, high-water activity produce. Notify QA and Production Supervisor. 2. PROCESS CORRECTION: Review and reinforce training for production line operators and supervisors on time tracking procedures and critical limit adherence. Adjust production schedules or line speeds to ensure product moves through pre-processing within the critical time limit. Ensure adequate refrigeration or chilling is available for product staging areas if delays are unavoidable. 3. INVESTIGATION: Determine the root cause of the time deviation (e.g., equipment breakdown, staffing issues, inefficient workflow, lack of awareness, improper use of timers). Review production logs and time tracking records. Implement preventive measures to avoid recurrence (e.g., workflow optimization, additional training, equipment maintenance, revised SOPs for staging).

ACTION RESPONSIBLE**Production Line Operator / Production Supervisor / QA Supervisor**

Step 7 Compliance

CCP 4 Step 9: RECEIVING RAW CABBAGE

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Stop processing. Hold all product processed since the last compliant check. Re-sanitize the equipment/surfaces immediately using the correct sanitizer concentration and contact time. If product was exposed to inadequate sanitation, assess its disposition: if the deviation is minor and quickly corrected, and subsequent microbial testing (if applicable) shows no issue, product may be released; if the deviation is significant or prolonged, product must be diverted for cooking (if suitable and safe) or destroyed. 2. PROCESS CORRECTION: Adjust sanitizer dosing system or manually add sanitizer to achieve the critical limit. Re-train personnel on proper sanitizer preparation, application, and contact time procedures. Verify calibration of monitoring equipment (test strips, colorimeter, stopwatch). 3. INVESTIGATION: Determine the root cause of the deviation (e.g., incorrect chemical mixing, faulty dosing pump, expired sanitizer, inadequate training, human error, equipment malfunction). Review sanitation logs and training records. Implement preventive measures to avoid recurrence (e.g., scheduled maintenance for dosing equipment, refresher training, updated SOPs).

ACTION RESPONSIBLE**Sanitation Team Lead / Quality Control Technician / Production Manager**

Step 7 Compliance

CCP 5 Step 10: PRE-PROCESSING PREPARATION

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Stop production immediately. Hold all product processed since the last compliant inspection. Clean and re-sanitize the affected food contact surfaces until no visible residue remains. Assess the held product for contamination: if chemical/lubricant residue is found, product must be destroyed; if allergen residue is found, product must be clearly identified as containing the allergen and segregated, or destroyed if not suitable for rework/relabeling. 2. PROCESS CORRECTION: Re-clean and re-sanitize the affected equipment/surfaces. Re-inspect to ensure critical limit is met. If the residue is from a previous product, ensure proper allergen cleaning procedures are followed for shared lines. 3. INVESTIGATION: Determine the root cause of the residue (e.g., inadequate cleaning procedure, insufficient rinsing, incorrect cleaning chemicals, equipment design flaw, lack of training, rushed cleaning). Review cleaning logs, training records, and cleaning chemical inventory. Implement preventive measures (e.g., revise cleaning SOPs, provide refresher training, verify cleaning chemical concentrations, schedule equipment maintenance).

ACTION RESPONSIBLE**Production Line Operator / Quality Control Technician / Sanitation Team Lead / Production Manager****Step 7
Compliance****CCP 5 Step 11: PRE-PROCESSING PREPARATION**

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Stop production on the affected line immediately. Isolate all product processed since the last acceptable monitoring result; hold for evaluation and potential disposition (rework, re-clean, or discard). Re-clean and re-sanitize the identified food contact surfaces. Re-test ATP levels on the affected surfaces. 2. PROCESS CORRECTION: Do not resume production until ATP levels are within the critical limit (< 200 RLU). Review cleaning and sanitation procedures with the sanitation team, providing re-training if necessary on proper chemical concentrations, contact times, and application methods. Verify proper functioning of cleaning equipment. 3. INVESTIGATION: Investigate the root cause of the high ATP reading (e.g., inadequate cleaning procedure, insufficient chemical concentration, improper rinsing, equipment malfunction, human error, missed areas). Review sanitation logs and chemical usage records. Assess if other lines or equipment are affected.

ACTION RESPONSIBLE**Quality Control Technician / Sanitation Supervisor / Production Manager****Step 7 Compliance****CCP 5 Step 12: PRE-PROCESSING PREPARATION**

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Stop production on the affected line immediately. Isolate all product processed since the last acceptable monitoring result; hold for evaluation and potential disposition (rework, re-clean, or discard). Re-clean and re-sanitize the identified food contact surfaces using a validated allergen cleaning procedure. Re-test for the specific allergen on the affected surfaces. 2. PROCESS CORRECTION: Do not resume production until allergen test results are negative ($< \text{Limit of Detection}$). Review allergen cleaning and sanitation procedures with the sanitation team, providing re-training if necessary on proper chemical concentrations, contact times, and application methods specifically for allergen removal. Verify proper functioning of cleaning equipment. Ensure proper segregation and scheduling of allergen-containing products. 3. INVESTIGATION: Investigate the root cause of the positive allergen test (e.g., inadequate cleaning procedure, insufficient chemical concentration, improper rinsing, equipment malfunction, human error, missed areas, improper scheduling, cross-contact during cleaning). Review sanitation logs, allergen cleaning validation records, and production schedules. Assess if other lines or equipment are affected or if the allergen source is external.

ACTION RESPONSIBLE**Quality Control Technician / Sanitation Supervisor / Production Manager****Step 7 Compliance**

CCP 6 Step 13: RAW MATERIAL COLD STORAGE

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Stop the line immediately. Isolate and hold all affected product (from the last good check to the point of discovery). Visually inspect the isolated product; if foreign material is confirmed, reject the affected lot/batch or re-sort/re-trim if feasible and safe, ensuring no foreign material remains. Clean and sanitize the affected area of the processing line. 2. PROCESS CORRECTION: Identify the source of the foreign material (e.g., raw material, equipment, personnel, environment). If from raw material, communicate with the supplier and review supplier specifications. If from equipment, repair or replace damaged parts. If from personnel, retrain staff on GMPs and foreign material control. Increase monitoring frequency for a defined period. 3. INVESTIGATION: Conduct a thorough investigation to determine the root cause. Review monitoring records, maintenance logs, training records, and supplier documentation. Implement long-term preventive measures (e.g., enhanced raw material inspection protocols, equipment upgrades, improved personnel training). Assess the risk to other products or batches.

ACTION RESPONSIBLE**Line Supervisor / Quality Control Technician**

Step 7 Compliance

CCP 6 Step 14: RAW MATERIAL COLD STORAGE Log: Non-Conformance & Corrective Action Report / Maintenance Work Order**CORRECTIVE ACTIONS TO BE TAKEN**

1. IMMEDIATE ACTION: Stop the processing line immediately. Isolate and hold all product processed since the last successful check of the screen/sieve. Inspect the screen/sieve and remove/repair/replace the damaged component. Conduct a thorough inspection of the isolated product for potential foreign material contamination (e.g., pieces of screen/sieve, or foreign material that may have passed through); if contamination is found, dispose of the affected product or re-sort if feasible and safe. 2. PROCESS CORRECTION: Ensure the replacement screen/sieve meets specifications (material, mesh size). Reinforce pre-shift inspection procedures for screens/sieves with operators/supervisors. Review and update preventative maintenance schedules for screens/sieves. 3. INVESTIGATION: Investigate the root cause of the screen/sieve damage (e.g., impact, material fatigue, improper handling, incorrect installation). Review maintenance logs, operator training, and equipment specifications. Implement preventive measures such as using more robust materials, improving operator training, or adjusting maintenance frequency. Assess the risk to other products or batches and verify the effectiveness of the repair/replacement.

ACTION RESPONSIBLE**Maintenance Manager / Production Supervisor / Quality Control Technician**

Step 7 Compliance

CCP 6 Step 15: RAW MATERIAL COLD STORAGE

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Stop the production line immediately. Isolate all product processed since the last successful metal detector check. Hold the isolated product for re-inspection, re-processing through a functional metal detector, or disposal based on risk assessment. 2. PROCESS CORRECTION: Identify the cause of the metal detector failure (e.g., calibration issue, equipment malfunction, incorrect settings). Adjust, repair, or recalibrate the metal detector. Re-test the metal detector with certified test pieces to ensure it is functioning correctly and meets critical limits before resuming production. 3. INVESTIGATION: Review maintenance logs and operator training records. Determine if there are any systemic issues with equipment maintenance or operator procedures. Implement preventive measures such as enhanced maintenance schedules, operator re-training, or equipment upgrades to prevent recurrence. Assess the risk to previously produced product if the failure duration is unknown or extended.

ACTION RESPONSIBLE**Production Supervisor / Maintenance Manager / QA Supervisor**

Step 7 Compliance

CCP 7

Step 16: SLICING/SHREDDING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately cease use of the contaminated water source for direct product contact. Identify and segregate all product processed using the non-compliant water since the last satisfactory test result. Hold the affected product for disposition (e.g., diversion for cooking, reprocessing if feasible and safe, or destruction) based on risk assessment. 2. PROCESS CORRECTION: Investigate the source of water contamination (e.g., cross-connection, pipe damage, filter failure, inadequate chlorination). Implement immediate measures to rectify the issue (e.g., repair pipes, replace filters, shock chlorination). Flush the water system thoroughly. Re-sample and re-test the water until results confirm it meets critical limits before resuming use for direct product contact. 3. INVESTIGATION: Conduct a comprehensive review of the water treatment system, plumbing, and monitoring program. Assess the integrity of the entire water distribution system. Determine if there are any systemic failures in water quality management. Implement long-term preventive measures such as enhanced maintenance, improved filtration, regular system audits, or increased monitoring frequency. Review the product's intended use and consumer base to determine the severity of the risk posed by the contaminated product.

ACTION RESPONSIBLE

Quality Assurance Manager / Plant Manager / Maintenance Manager

Step 7 Compliance

CCP 8

Step 17: PRE-PROCESSING PREPARATION

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: Stop production immediately. Isolate and hold all product processed since the last acceptable monitoring result. Flush the water lines until chlorine levels are within the critical limit. Re-test chlorine levels to confirm compliance before resuming production. PROCESS CORRECTION: Adjust the chlorine dosing system (e.g., reduce flow rate of chlorine solution, adjust chlorinator settings). Inspect the chlorine dosing equipment for malfunction or incorrect calibration. Review and update SOPs for chlorine dosing and monitoring if necessary. INVESTIGATION: Determine the root cause of the elevated chlorine level (e.g., equipment malfunction, human error in setting/monitoring, incorrect chemical concentration, water pressure fluctuations). Review maintenance logs for the chlorine dosing system. Assess the potential impact on product safety and quality of the held product. Implement preventive measures to avoid recurrence.

ACTION RESPONSIBLE

Quality Control Technician / Production Supervisor / QA Manager

Step 7 Compliance

CCP 8 Step 18: PRE-PROCESSING PREPARATION

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: Stop production immediately. Isolate and hold all product processed since the last acceptable monitoring result. Immediately cease using the contaminated water source for direct product contact. Switch to an alternative, verified safe water source if available, or use bottled/filtered water for critical processes until the issue is resolved. Notify relevant regulatory authorities and municipal water suppliers. PROCESS CORRECTION: Implement a temporary alternative water supply or install appropriate filtration/treatment systems to remove the identified contaminants. If the source is municipal, work with the municipal water authority to understand and resolve the contamination. If the source is internal (e.g., well water), implement necessary treatment or find an alternative source. INVESTIGATION: Determine the root cause of the contamination (e.g., municipal water supply issue, cross-connection, pipe corrosion, environmental contamination of well water, improper use/storage of cleaning chemicals). Conduct further testing of the water source and potentially the product to assess the extent of contamination. Review all water system schematics, maintenance records, and chemical usage logs. Develop and implement long-term solutions, which may include source remediation, advanced water treatment, or a permanent change in water supply. Evaluate the safety of the held product based on laboratory analysis and expert risk assessment.

ACTION RESPONSIBLE**Quality Assurance Manager / Plant Manager****Step 7 Compliance****CCP 8 Step 19: PRE-PROCESSING PREPARATION**

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately stop the slicing/shredding line. Isolate and hold all product processed since the last acceptable monitoring check. Divert contaminated water and flush the water lines until visual, olfactory, and pH checks confirm the absence of cleaning chemicals and pH is within limits. 2. PROCESS CORRECTION: Identify the source of the chemical contamination (e.g., incomplete rinsing after sanitation, cross-connection, faulty valve). Rectify the issue (e.g., re-train sanitation crew on proper rinsing, repair/replace faulty equipment). Re-verify water quality before restarting production. 3. INVESTIGATION: Review sanitation records, maintenance logs, and operator training records. Determine the root cause of the chemical contamination. Assess the impact on affected product and determine disposition (e.g., re-wash, discard). Implement preventive measures to avoid recurrence, such as enhanced pre-operational checks or automated flushing systems.

ACTION RESPONSIBLE**Production Manager / QA Supervisor****Step 7 Compliance****CCP 9 Step 20: POST-SLICING WASHING & RINSING**

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately stop the washing/rinsing process. Isolate and hold all product processed since the last acceptable monitoring check. Adjust the sanitizer dosing system to achieve the critical limit (50-200 ppm free chlorine). Re-test and confirm the sanitizer concentration is within limits before resuming operations. 2. PROCESS CORRECTION: Identify the reason for the deviation (e.g., empty sanitizer tank, clogged dosing line, faulty pump, incorrect calibration of dosing system, operator error). Rectify the issue (e.g., refill tank, clear line, repair/replace pump, re-calibrate, re-train operator). 3. INVESTIGATION: Review sanitizer usage logs, maintenance records for dosing equipment, and operator training. Determine the root cause of the sanitizer deviation. Assess the microbiological risk of the affected product; conduct microbial testing if necessary, and determine disposition (e.g., re-wash with effective sanitizer, divert to cooking-only use, discard). Implement preventive measures such as automated dosing systems with alarms, more frequent checks, or enhanced training.

ACTION RESPONSIBLE**Production Manager / QA Supervisor****Step 7 Compliance**

CCP 9

Step 21: POST-SLICING WASHING & RINSING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately stop the washing line. Hold all product processed since the last compliant check. Adjust sanitizer concentration to bring pH back within 6.0-7.5. Re-test pH to confirm compliance. If pH cannot be corrected immediately, drain and refill the wash tank with fresh water and sanitizer, ensuring correct concentration and pH. Segregate and re-evaluate affected product; product processed during the deviation must be re-washed with a correctly prepared sanitizer solution, diverted to a cooking-only stream, or disposed of. 2. PROCESS CORRECTION: Identify why the pH deviated (e.g., incorrect sanitizer dosing, equipment malfunction, water quality change). Retrain the line operator/QC technician on proper sanitizer preparation, pH monitoring, and adjustment procedures. Verify calibration of the pH meter/test strips. Implement enhanced monitoring frequency until confidence in process control is restored. 3. INVESTIGATION: Review production records, sanitizer usage logs, and previous pH monitoring records to determine the duration of the deviation and the extent of affected product. Investigate potential causes for pH deviation (e.g., pump malfunction, human error, raw material variability). Determine if the deviation indicates a systemic issue requiring changes to SOPs, equipment maintenance, or training programs. Assess the risk to public health based on the deviation and disposition of affected product.

ACTION RESPONSIBLE

Production Manager / QA Supervisor

Step 7 Compliance

CCP 9

Step 22: POST-SLICING WASHING & RINSING

Log: Non-Conformance & Corrective Action Report / Water Quality Incident Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately cease all operations using the affected water supply. Hold all product processed since the last confirmed potable water test. Switch to an alternative, verified potable water source if available, or use bottled potable water for critical operations. Notify local health authorities and water utility provider. Segregate and re-evaluate affected product; product processed with non-potable water must be diverted to a cooking-only stream (if deemed safe after risk assessment), re-washed with verified potable water and sanitizer, or disposed of. 2. PROCESS CORRECTION: Identify the source of contamination in the water supply (e.g., broken pipe, cross-connection, municipal water issue, well contamination). Implement immediate repairs or remediation actions to restore water potability. Flush the water system thoroughly after repairs. Conduct repeat water testing to confirm potability before resuming operations. Increase frequency of water testing until confidence in the water supply is restored. 3. INVESTIGATION: Review all water testing records to determine the duration and extent of the non-potable water use. Investigate the root cause of the water contamination (e.g., infrastructure failure, external contamination event, inadequate treatment). Assess the potential impact on product safety and public health. Determine if a product recall is necessary based on risk assessment. Update water management plan, maintenance schedules, or supplier agreements as needed.

ACTION RESPONSIBLE

Quality Control Manager / Plant Manager

Step 7 Compliance

CCP 10

Step 23: PRE-PROCESSING PREPARATION

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Isolate and hold the affected lot of product. Re-check temperature of multiple samples from the same lot to confirm deviation. If confirmed, reject the entire lot from the supplier or divert for immediate cooking/processing if deemed safe and feasible, otherwise dispose of. Notify supplier of non-conformance. 2. PROCESS CORRECTION: Verify calibration of the thermometer used. Inspect receiving area and transport vehicle for proper refrigeration and temperature control. Review receiving procedures with personnel to ensure proper temperature monitoring techniques. 3. INVESTIGATION: Determine the root cause of the temperature deviation (e.g., supplier transport issue, delayed unloading, refrigeration unit malfunction). Review supplier's temperature logs and transport records. Implement preventive measures (e.g., adjust supplier specifications, improve receiving logistics, maintenance on refrigeration). Assess potential impact on food safety based on duration and extent of temperature abuse.

ACTION RESPONSIBLE**Receiving Personnel / QA Supervisor / Production Manager**

Step 7 Compliance

CCP 10

Step 24: PRE-PROCESSING PREPARATION

Log: Sanitation Verification Log / Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Stop production on the affected line/equipment. Isolate and hold all product processed on the affected line/equipment since the last acceptable sanitation verification. Re-clean and re-sanitize the identified food contact surfaces and surrounding areas. Re-test ATP or re-swab for pathogens to verify effectiveness of re-sanitation. Do not resume production until results are acceptable. 2. PROCESS CORRECTION: Review sanitation procedures and chemical concentrations. Retrain sanitation personnel on proper cleaning and sanitizing techniques, including disassembly/assembly of equipment. Inspect equipment for harborage points, damage, or areas difficult to clean. Increase frequency of ATP testing or environmental swabbing for a defined period. 3. INVESTIGATION: Determine the root cause of the sanitation failure (e.g., inadequate cleaning time, incorrect chemical concentration, equipment design flaw, personnel training gap, biofilm formation). Review sanitation logs, chemical usage records, and personnel training records. Assess the risk to product held and determine final disposition (e.g., release, re-process, dispose) based on risk assessment and further testing if necessary. Implement long-term preventive measures (e.g., equipment modification, new sanitation chemicals, enhanced training program, revised master sanitation schedule).

ACTION RESPONSIBLE**Sanitation Supervisor / Quality Assurance Manager**

Step 7 Compliance

CCP 11

Step 25: VISUAL INSPECTION & SORTING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: Stop production immediately. Isolate and hold all product produced since the last compliant check. Identify and remove the non-compliant lubricant. Clean and sanitize any food contact surfaces potentially contaminated. Replace with an approved NSF H1 or H3 certified lubricant. PROCESS CORRECTION: Review purchasing procedures to ensure only approved lubricants are ordered and received. Retrain maintenance and purchasing staff on the Approved Lubricant List and verification procedures. Update the Approved Lubricant List if necessary. Increase monitoring frequency for a defined period. INVESTIGATION: Investigate how the non-compliant lubricant entered the facility/was used. Determine the extent of potential contamination and assess the risk to affected product. Review training records and competency of personnel involved. Evaluate the effectiveness of current receiving and inventory control procedures for lubricants. Determine disposition of affected product (e.g., re-inspection, re-processing, destruction).

ACTION RESPONSIBLE**Maintenance Supervisor / Production Supervisor / Quality Assurance Manager**

Step 7 Compliance

CCP 11

Step 26: VISUAL INSPECTION & SORTING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: Stop production immediately. Isolate and hold all product produced since the last compliant check. Re-clean and re-rinse all affected food contact surfaces until no residues are detectable by visual inspection, pH strips, and/or ATP swabs. Verify cleanliness before resuming production. PROCESS CORRECTION: Review and reinforce sanitation Standard Operating Procedures (SOPs), specifically focusing on proper rinsing techniques and verification steps. Retrain sanitation and production personnel on proper chemical application, rinsing, and residue verification methods. Inspect and ensure proper functioning of rinsing equipment. Adjust chemical concentrations if found to be excessively high. Increase monitoring frequency for a defined period. INVESTIGATION: Investigate the cause of the residue (e.g., insufficient rinsing, incorrect chemical concentration, equipment malfunction, lack of training). Determine the extent of potential contamination and assess the risk to affected product. Review sanitation records and personnel training. Evaluate the effectiveness of the current sanitation program. Determine disposition of affected product (e.g., re-inspection, re-processing, destruction).

ACTION RESPONSIBLE

Quality Control Technician / Sanitation Supervisor / Production Manager

Step 7 Compliance

CCP 12

Step 27: PRE-PROCESSING PREPARATION

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately stop the line. Isolate and hold all product produced since the last successful check. Visually inspect the affected product lot; if foreign material is found, segregate and re-sort or dispose of as non-conforming product. 2. PROCESS CORRECTION: Identify and remove the source of the foreign material from the product and processing environment. Repair or adjust equipment if the foreign material originated from machinery. Retrain the operator on proper visual inspection techniques. 3. INVESTIGATION: Conduct a thorough investigation to determine the root cause (e.g., equipment malfunction, operator error, raw material contamination). Implement preventive measures such as scheduled equipment maintenance, improved raw material inspection, or enhanced operator training. Verify effectiveness through subsequent monitoring.

ACTION RESPONSIBLE

Production Supervisor / QA Supervisor

Step 7 Compliance

CCP 12

Step 28: PRE-PROCESSING PREPARATION

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately stop the line. Isolate and hold all product processed since the last successful check of the screen/filter integrity. Inspect the held product for foreign material; if found or suspected, segregate and re-sort, pass through an alternative detection step, or dispose of as non-conforming product. 2. PROCESS CORRECTION: Immediately repair or replace the damaged screen or filter with an intact one of the correct specification. Ensure proper installation and verify integrity of all other screens/filters. 3. INVESTIGATION: Conduct a thorough investigation to determine the root cause of the screen/filter damage (e.g., improper handling, material fatigue, lack of preventive maintenance). Implement preventive measures such as revising maintenance schedules, providing additional operator training, or upgrading screen/filter material. Verify effectiveness through subsequent monitoring.

ACTION RESPONSIBLE

Maintenance Manager / QA Supervisor

Step 7 Compliance

CCP 13 Step 29: WEIGHING & PORTIONING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately stop production on the affected line/equipment. Isolate and hold all product processed since the last compliant check. Inspect the affected product for foreign material; if found, dispose of the product. Thoroughly clean and sanitize the equipment, focusing on the identified source or area of contamination. 2. PROCESS CORRECTION: Re-train operators on visual inspection protocols and foreign material prevention. Implement enhanced cleaning and inspection frequencies for the specific equipment. Review equipment maintenance logs for potential wear and tear. 3. INVESTIGATION: Determine the root cause of the foreign material (e.g., equipment malfunction, inadequate cleaning, raw material issue, personnel error). Review previous monitoring records and conduct a thorough investigation to prevent recurrence. Update SOPs or HACCP plan if necessary.

ACTION RESPONSIBLE**Production Supervisor / Quality Control Manager**

Step 7 Compliance

CCP 13 Step 30: WEIGHING & PORTIONING

Log: Non-Conformance & Corrective Action Report / Employee Training Record

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately remove the non-compliant personnel from the processing area. Ensure the individual corrects their attire/removes prohibited items before re-entering. Inspect any product handled by the non-compliant personnel for potential contamination; if contamination is suspected or observed, hold and dispose of the affected product. 2. PROCESS CORRECTION: Conduct immediate re-education/refresher training for the individual and, if necessary, the entire team on GMPs, personal hygiene, and foreign material prevention policies. Increase supervisory oversight and monitoring frequency of personnel compliance. 3. INVESTIGATION: Determine the root cause of non-compliance (e.g., lack of understanding, complacency, inadequate supervision). Review training records and assess the effectiveness of current training programs. Implement disciplinary action if repeated or willful non-compliance is observed. Update GMP training materials or policies if clarity is an issue.

ACTION RESPONSIBLE**Production Supervisor / Quality Control Manager**

Step 7 Compliance

CCP 13 Step 31: WEIGHING & PORTIONING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Stop the line immediately. Isolate and hold all product processed since the last acceptable monitoring check. Visually inspect the affected equipment to identify the specific damage. If foreign material is found in product, quarantine the affected lot and conduct a thorough investigation to determine the extent of contamination. 2. PROCESS CORRECTION: Repair or replace the damaged equipment part. If repair is not possible or poses a risk, replace the entire piece of equipment. Thoroughly clean and sanitize the repaired/replaced equipment and surrounding area. Re-inspect the equipment to ensure it meets the critical limit before resuming production. If product was contaminated, dispose of or reprocess (if safe and feasible) the affected product under strict supervision. 3. INVESTIGATION: Investigate the root cause of the equipment damage (e.g., improper handling, lack of maintenance, material fatigue, design flaw). Review maintenance logs and operator training records. Assess if similar equipment elsewhere in the facility is at risk. Implement preventive measures such as enhanced preventative maintenance schedules, revised equipment handling SOPs, or additional operator training. Verify the effectiveness of the corrective actions through follow-up inspections.

ACTION RESPONSIBLE**Production Supervisor / Maintenance Manager / QA Manager**

Step 7 Compliance

CCP 14

Step 32: SLICING/SHREDDING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Stop the packaging line immediately. Isolate and hold all product processed since the last acceptable temperature check. Immediately move the non-conforming product to a designated refrigerated holding area (0-4°C) to bring its temperature down rapidly. Re-measure the temperature of the affected product to confirm the excursion and the effectiveness of cooling. 2. PROCESS CORRECTION: Identify and correct the source of the temperature deviation (e.g., malfunctioning refrigeration unit, extended dwell time on the line, ambient temperature too high, insufficient pre-cooling). Adjust process parameters (e.g., reduce line speed, increase cooling capacity, shorten transfer times) to ensure product temperature remains below 4°C. Once the process is corrected and verified, resume packaging. Evaluate the disposition of the held product: if the temperature excursion was minor and duration short, and product can be rapidly cooled back to critical limit, it may be released. If the excursion was significant or prolonged, the product must be diverted for alternative use (e.g., cooking if safe) or disposed of, based on a risk assessment by QA. 3. INVESTIGATION: Investigate the root cause of the temperature excursion (e.g., equipment malfunction, operator error, environmental conditions, process flow issues). Review temperature logs, maintenance records, and operator training. Implement preventive measures such as enhanced equipment maintenance, recalibration of sensors, revised SOPs for line speed or product handling, or additional operator training on temperature control. Verify the effectiveness of the corrective actions through continuous monitoring and follow-up temperature checks.

ACTION RESPONSIBLE

Packaging Supervisor / QA Manager / Maintenance Manager

Step 7 Compliance

CCP 14

Step 33: SLICING/SHREDDING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: 1. Stop packaging operations immediately. 2. Segregate and hold all affected packaging material (the roll/stack where the defect was found, and potentially adjacent rolls/stacks from the same lot). 3. Identify and segregate any product packaged with the non-conforming material; conduct a risk assessment to determine disposition (e.g., rework, disposal). PROCESS CORRECTION: 1. Replace the non-conforming packaging material with verified compliant material. 2. Retrain the Quality Control Technician/Packaging Line Operator on proper visual inspection procedures and critical limits for packaging materials. 3. Increase monitoring frequency for a defined period to ensure compliance. INVESTIGATION: 1. Investigate the root cause of the non-conformance (e.g., supplier issue, handling damage, storage conditions, equipment malfunction causing damage). 2. Contact the packaging material supplier to report the defect and request a root cause analysis and corrective actions from their end. 3. Review supplier COA/COC and internal receiving inspection records. 4. Assess the impact on product safety and quality, including any product already produced or shipped.

ACTION RESPONSIBLE

Quality Control Manager / Production Manager

Step 7 Compliance

CCP 14

Step 34: SLICING/SHREDDING

Log: Non-Conformance & Corrective Action Report / Sanitation Verification Log

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: 1. Stop packaging operations immediately. 2. Segregate and hold all product packaged since the last successful verification (visual inspection/ATP test). 3. Immediately re-clean and re-sanitize all affected food contact surfaces and the surrounding environment. 4. Re-verify cleanliness by visual inspection and ATP testing (or equivalent) before resuming operations. PROCESS CORRECTION: 1. Do not resume packaging until all surfaces pass visual inspection and ATP testing. 2. Retrain the Packaging Line Supervisor/Quality Control Technician and sanitation crew on proper cleaning and sanitization procedures, including chemical concentrations, contact times, and verification methods. 3. Review and update sanitation SOPs if deficiencies are identified. 4. Increase monitoring frequency (visual and ATP) for a defined period to ensure sustained compliance. INVESTIGATION: 1. Investigate the root cause of the failure (e.g., inadequate cleaning procedure, incorrect chemical concentration, insufficient contact time, equipment design flaw, lack of training, environmental contamination source). 2. Review sanitation records, training logs, and equipment maintenance logs. 3. Assess the potential for cross-contamination to other areas or products. 4. Determine the disposition of affected product based on risk assessment.

ACTION RESPONSIBLE

Sanitation Manager / Quality Control Manager / Production Manager

Step 7 Compliance

CCP 14

Step 35: SLICING/SHREDDING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: All product packaged since the last acceptable monitoring result must be immediately placed on hold, identified, and isolated. Evaluate the extent of the deviation; product with O2 levels above 2% or CO2 outside 5-15% must be deemed adulterated due to compromised MAP integrity and potential for pathogen growth. Product must be diverted for non-food use or destroyed. 2. PROCESS CORRECTION: Stop the packaging line immediately. Verify calibration of the headspace gas analyzer and recalibrate if necessary. Check the gas mixture supply and settings on the MAP machine to ensure correct gas ratios are being delivered. Inspect the packaging machine for any issues affecting gas flushing or sealing (e.g., leaks, worn seals, incorrect dwell time/pressure). Adjust machine settings as needed and re-monitor packaging to ensure critical limits are met before resuming production. 3. INVESTIGATION: Investigate the root cause of the deviation (e.g., equipment malfunction, incorrect gas cylinder, operator error, calibration issue, packaging material defect). Review maintenance logs and calibration records. Implement preventive measures such as enhanced operator training, preventative maintenance schedule adjustments, or improved calibration frequency to prevent recurrence. Document all actions.

ACTION RESPONSIBLE

Packaging Machine Operator / QA Supervisor

Step 7 Compliance

CCP
15**Step 36: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING**

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: All product packaged since the last acceptable monitoring result must be immediately placed on hold, identified, and isolated. Evaluate the temperature deviation (magnitude and duration). Product exceeding 4°C for a significant duration must be deemed adulterated due to potential pathogen proliferation. Product must be diverted for non-food use or destroyed. If the excursion was minor and brief, product may be immediately moved to proper refrigeration and re-evaluated by QA. 2. PROCESS CORRECTION: Stop the packaging line immediately. Identify the source of the temperature deviation (e.g., ambient room temperature too high, product held too long on the line, refrigeration unit malfunction upstream, inadequate cooling prior to packaging). Adjust environmental controls, reduce line speed, or ensure product is adequately chilled before reaching packaging. Verify the calibration of the thermometer and recalibrate if necessary. Re-monitor product temperature to ensure critical limits are met before resuming production. 3. INVESTIGATION: Investigate the root cause of the temperature deviation (e.g., equipment malfunction in upstream cooling, HVAC failure, excessive hold times, operator error, inadequate pre-cooling). Review temperature logs for the processing area and upstream cooling steps. Implement preventive measures such as improved temperature monitoring in the packaging area, revised production schedules, enhanced maintenance of refrigeration units, or additional operator training to prevent recurrence. Document all actions.

ACTION RESPONSIBLE**Production Manager / QA Supervisor**

Step 7 Compliance

CCP
15**Step 37: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING**

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: Hold all product packaged since the last acceptable ATP swab result. Stop production on the affected line/equipment immediately. Re-clean and re-sanitize the affected food contact surfaces and packaging environment. Re-swab and verify ATP results are < 200 RLU before resuming production. PROCESS CORRECTION: Review sanitation procedures, chemical concentrations, contact times, and employee training for sanitation. Increase frequency of ATP monitoring if recurring issues are identified. Verify calibration of luminometer and proper swabbing technique. INVESTIGATION: Conduct a root cause analysis to determine why sanitation failed (e.g., inadequate cleaning, equipment malfunction, improper chemical use, training gap). Evaluate the held product for potential microbiological contamination (e.g., indicator organisms, pathogens) if deemed necessary based on risk assessment and duration of exposure. Implement preventive measures based on root cause findings.

ACTION RESPONSIBLE**Quality Control Manager / Sanitation Supervisor**

Step 7 Compliance

CCP 16

Step 38: SLICING/SHREDDING Log: Supplier Non-Conformance Report / Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: Quarantine all affected packaging materials immediately. Do not use them. If non-compliant packaging materials have been used, identify and hold all product packaged with these materials. Notify the supplier of the non-conformance. PROCESS CORRECTION: Reinforce receiving procedures for packaging materials, emphasizing the review of Certificates of Conformance (CoC)/Letters of Guarantee (LoG) and visual inspection. Retrain receiving personnel and QA staff on critical limit requirements and verification methods. Update supplier approval program if necessary, or seek alternative approved suppliers. INVESTIGATION: Conduct a root cause analysis to determine why non-compliant materials were received or used (e.g., supplier error, internal procedural breakdown, lack of training). Assess the risk to held product based on the nature of the non-compliance (e.g., type of chemical, potential for migration). Determine disposition of held product (e.g., rework, destruction) based on risk assessment. Follow up with the supplier for corrective actions on their end.

ACTION RESPONSIBLE

Quality Assurance Manager / Receiving Manager

Step 7 Compliance

CCP 16

Step 39: SLICING/SHREDDING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately stop production. Hold all product packaged since the last acceptable monitoring result. Isolate and re-clean/re-rinse affected food contact surfaces until zero detectable residue is confirmed by re-monitoring. Evaluate affected product for disposition (e.g., rework if safe and feasible, or discard). 2. PROCESS CORRECTION: Review and adjust sanitation procedures (SOPs) to ensure adequate rinsing. Re-train sanitation personnel on proper chemical application, rinsing techniques, and verification methods. Repair or adjust any equipment affecting the rinsing process. Increase monitoring frequency temporarily until compliance is consistently achieved. 3. INVESTIGATION: Determine the root cause of the residue (e.g., insufficient rinse time, incorrect chemical concentration, equipment malfunction, training gap). Review previous sanitation records and monitoring logs for trends. Implement preventive measures such as updated training modules, equipment maintenance schedules, or revised chemical handling protocols to prevent recurrence.

ACTION RESPONSIBLE

Sanitation Supervisor / QA Technician

Step 7 Compliance

CCP 16

Step 40: SLICING/SHREDDING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately stop production. Hold all product packaged since the last acceptable monitoring result. Remove any non-NSF H1 certified lubricant from use and replace it with an approved NSF H1 lubricant. Clean and sanitize any food contact surfaces potentially contaminated by the non-food grade lubricant. Evaluate affected product for disposition (e.g., discard if contamination is confirmed or suspected). 2. PROCESS CORRECTION: Review and update lubricant management procedures (purchasing, storage, application). Re-train maintenance personnel on the exclusive use of approved NSF H1 lubricants and proper application techniques to prevent overspray or drips. Update the facility's approved lubricant list and ensure accessibility for all relevant personnel. Increase monitoring frequency of lubricant use and storage temporarily. 3. INVESTIGATION: Determine the root cause of the non-conformance (e.g., purchasing error, lack of training, unauthorized lubricant brought into facility, mislabeling, incorrect application). Assess the risk to product safety based on the type of lubricant and extent of potential contact. Review maintenance records and purchasing logs. Implement preventive measures such as stricter purchasing controls, enhanced training, and regular audits of lubricant inventory and usage.

ACTION RESPONSIBLE

Maintenance Supervisor / Production Supervisor / QA Supervisor

Step 7 Compliance

CCP 17

Step 41: METAL DETECTION

Log: Non-Conformance & Corrective Action Report / Temperature Deviation Log

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Identify and segregate all product exposed to the temperature deviation. Assess the duration and magnitude of the temperature excursion. If the temperature excursion is minor and brief (e.g., < 2 hours above 4°C but < 7°C), product may be re-cooled and released after QA approval. If the temperature excursion is significant (e.g., > 2 hours above 4°C, or any time above 7°C), the product must be held, evaluated by QA for potential spoilage/pathogen growth, and potentially diverted for cooking (if safe and feasible) or disposed of as waste. 2. PROCESS CORRECTION: Investigate the root cause of the temperature deviation (e.g., refrigeration unit malfunction, door left open, power outage, improper loading/stacking, thermostat setting error). Repair or adjust refrigeration unit. Retrain personnel on proper storage procedures (e.g., door closure, loading practices). Implement preventative maintenance checks on refrigeration equipment. 3. INVESTIGATION: Determine the exact time and duration of the deviation. Review temperature logs and maintenance records. Identify all affected product lots. Implement a plan to prevent recurrence.

ACTION RESPONSIBLE**Cold Storage Operator / QA Technician / Maintenance Manager**

Step 7 Compliance

CCP 17

Step 42: METAL DETECTION

Log: Non-Conformance & Corrective Action Report / MAP Gas Deviation Log

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Identify and segregate all product packaged since the last acceptable monitoring check. Hold affected product. Conduct further testing (e.g., re-analyze headspace gas composition). If the O₂ level is significantly low or the deviation duration is long, the product must be held for QA evaluation, potential diversion for cooking (if safe and feasible), or disposed of as waste. 2. PROCESS CORRECTION: Stop the packaging line. Investigate the root cause of the improper gas mixture (e.g., gas mixer malfunction, incorrect gas cylinder, gas line leak, incorrect machine settings, calibration issue with gas analyzer). Adjust or repair gas mixing equipment. Verify gas cylinder contents and connections. Recalibrate the headspace gas analyzer. Retrain operators on proper machine setup and monitoring procedures. 3. INVESTIGATION: Determine the exact time and duration of the deviation. Review production records, gas analyzer calibration logs, and previous monitoring results. Identify all affected product lots. Implement a plan to prevent recurrence.

ACTION RESPONSIBLE**Packaging Line Operator / QA Technician / Maintenance Manager**

Step 7 Compliance

CCP 17

Step 43: METAL DETECTION

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Isolate and hold all product produced since the last acceptable monitoring check. Adjust gas mixer settings immediately to bring CO₂ concentration within 5-10%. Re-test packages to confirm effectiveness. Evaluate affected product for disposition (e.g., re-evaluation, diversion, destruction) based on deviation severity and risk assessment. 2. PROCESS CORRECTION: Identify root cause (e.g., gas mixer malfunction, incorrect gas cylinder, operator error, analyzer calibration issue). Repair/replace faulty equipment, retrain operators on proper gas mixer operation and monitoring procedures, and update SOPs if necessary. 3. INVESTIGATION: Review production, maintenance, and training records. Determine the extent of the affected lot and assess the potential impact on product safety and quality. Implement preventive measures (e.g., scheduled maintenance for gas mixer, enhanced operator training, more frequent analyzer calibration).

ACTION RESPONSIBLE**Production Manager / QA Supervisor**

Step 7 Compliance

CCP 17

Step 44: METAL DETECTION

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Stop the packaging line immediately. Isolate and hold all product produced since the last acceptable monitoring check. Remove all defective packages. Adjust or repair the sealing mechanism (e.g., heat settings, pressure, dwell time, sealing jaws for damage or contamination). Restart the line and re-verify seal integrity before resuming full production. 2. PROCESS CORRECTION: Identify root cause of the seal failure (e.g., worn sealing jaws, incorrect temperature/pressure/dwell time settings, product trapped in seal, film defects, operator error). Repair/replace faulty parts, adjust machine settings, retrain operators on proper machine operation and visual inspection techniques. Update SOPs for packaging machine operation and seal integrity checks if necessary. 3. INVESTIGATION: Review production, maintenance, and training records. Determine the extent of the affected lot and assess the potential impact on product safety and quality. Implement preventive measures (e.g., scheduled maintenance for sealing equipment, enhanced operator training, more frequent checks of sealing parameters).

ACTION RESPONSIBLE

Production Manager / QA Supervisor

Step 7 Compliance

CCP 18

Step 45: POST-SLICING WASHING & RINSING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately quarantine and hold all product packaged using the non-food grade gas. Identify and segregate the non-conforming gas cylinder/supply line to prevent further use. Evaluate affected product for potential chemical contamination and dispose of it if deemed unsafe. 2. PROCESS CORRECTION: Reject the non-conforming gas shipment and return it to the supplier. Contact the supplier to understand the deviation and request a new Certificate of Analysis (COA) for future deliveries, ensuring it confirms food-grade compliance. Verify that all subsequent gas deliveries are accompanied by valid food-grade COAs before acceptance and use. Retrain the QA Manager/Receiving Supervisor on the critical importance of verifying COAs for MAP gases. 3. INVESTIGATION: Determine the root cause of the non-conformance (e.g., supplier error, documentation error, oversight during receiving). Review records of previous MAP gas deliveries from the same supplier to ensure no prior non-conformances. Assess the impact on product safety and quality. Verify the effectiveness of the corrective actions through follow-up monitoring and internal audits.

ACTION RESPONSIBLE

QA Manager / Receiving Supervisor

Step 7 Compliance

CCP 18

Step 46: POST-SLICING WASHING & RINSING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately quarantine and hold all product packaged using gas from the unapproved supplier. Identify and segregate the gas from the unapproved supplier to prevent further use. Evaluate affected product for potential chemical contamination and dispose of it if deemed unsafe. 2. PROCESS CORRECTION: Reject the gas shipment from the unapproved supplier. Initiate the vendor qualification process for the supplier if they wish to become approved, or cease procurement from them. Reinforce the procedure for cross-referencing supplier names with the Approved Supplier List during receiving. Retrain Receiving Supervisor/QA Technician on the Approved Supplier List procedure and the importance of adhering to it. 3. INVESTIGATION: Determine the root cause of the deviation (e.g., purchasing error, outdated Approved Supplier List, lack of training, emergency procurement bypassing procedure). Review purchasing records to identify any other instances of procurement from unapproved suppliers. Assess the impact on product safety and quality. Verify the effectiveness of the corrective actions through follow-up monitoring and audits of purchasing practices.

ACTION RESPONSIBLE

Receiving Supervisor / QA Technician

Step 7 Compliance

CCP 19 Step 47: LABELING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Hold all affected label rolls/batches and prevent their use. Quarantine any product that may have been labeled with non-conforming labels. 2. PROCESS CORRECTION: Reject the non-conforming label roll/batch from the supplier. Obtain and verify new, correct label rolls/batches. Retrain personnel on label verification procedures. 3. INVESTIGATION: Determine the root cause of the incorrect label (e.g., supplier error, incorrect order, human error). Review label receiving and verification procedures and implement preventive measures such as enhanced supplier audits or improved incoming label inspection protocols.

ACTION RESPONSIBLE**Quality Control Technician / Line Supervisor**

Step 7 Compliance

CCP 19 Step 48: LABELING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Stop the production line immediately. Identify and segregate all product units that have been incorrectly labeled or are suspected of being incorrectly labeled. Hold all product from the affected lot/pallet until verification is complete. If mislabeled product has left the labeling station, initiate recall procedures if necessary. 2. PROCESS CORRECTION: Correctly label all segregated product units if feasible and safe, otherwise dispose of the product. Verify that the correct label roll is loaded onto the labeling machine. Adjust or repair the labeling machine if it is applying labels incorrectly. Retrain the Line Operator/QC Technician on proper label application and verification procedures. 3. INVESTIGATION: Determine the root cause of the mislabeling (e.g., incorrect label roll loaded, machine malfunction, human error during inspection, inadequate training, vision system malfunction). Review the effectiveness of the monitoring procedure and implement preventive measures such as improved changeover procedures, enhanced machine maintenance, or additional verification steps.

ACTION RESPONSIBLE**Line Operator / Quality Control Technician**

Step 7 Compliance

CCP 20 Step 49: VISUAL INSPECTION & SORTING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: Segregate and place all affected product on 'HOLD' status. Conduct a risk assessment based on the duration and peak temperature of the deviation. If the critical limit was exceeded, product must be deemed unsafe and disposed of or destroyed. PROCESS CORRECTION: Immediately identify and rectify the cause of the temperature excursion (e.g., close door, repair refrigeration unit, adjust thermostat). Verify the cold storage unit returns to and maintains the critical limit of $\leq 4^{\circ}\text{C}$. INVESTIGATION: Conduct a root cause analysis (e.g., equipment malfunction, human error, power outage). Review temperature monitoring records and implement preventive measures (e.g., scheduled maintenance, staff retraining, alarm system review) to prevent recurrence.

ACTION RESPONSIBLE**Cold Storage Operator / Quality Control Technician / QA Supervisor**

Step 7 Compliance

CCP 20

Step 50: VISUAL INSPECTION & SORTING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: Segregate and place all affected product on 'HOLD' status. Product exposed to temperatures above 4°C for more than 2 hours or exceeding 7°C at any time must be deemed unsafe and disposed of or destroyed. PROCESS CORRECTION: Immediately identify and rectify the cause of the temperature excursion (e.g., repair refrigeration unit, close door, adjust thermostat). Verify the cold storage unit returns to and maintains the critical limit of $\leq 4^{\circ}\text{C}$. INVESTIGATION: Conduct a thorough root cause analysis of the deviation (e.g., equipment failure, power supply interruption, human error). Review continuous temperature monitoring data and alarm logs. Implement robust preventive measures (e.g., preventative maintenance schedule review, staff training on alarm response, equipment upgrade) to prevent recurrence.

ACTION RESPONSIBLE

Cold Storage Operator / Quality Control Supervisor / Plant Manager

Step 7 Compliance

CCP 21

Step 51: DISPATCH

Log: Non-Conformance & Corrective Action Report / Dispatch Log

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately hold and segregate all affected product lots. Do not load or dispatch product if its internal temperature or the transport vehicle's cargo area temperature exceeds 4°C. Any product found above 4°C must be rejected and disposed of, as it is considered potentially unsafe due to the risk of pathogenic bacterial growth. Notify QA/HACCP Team Leader. Ensure the transport vehicle's refrigeration unit is immediately checked and brought back to the critical limit, or an alternative, compliant vehicle is sourced. 2. PROCESS CORRECTION: Identify and rectify the cause of the temperature deviation (e.g., refrigeration unit malfunction, slow loading procedures, pre-cooling failure). Repair or adjust the refrigeration unit. Retrain dispatch and loading personnel on proper temperature monitoring, rapid loading techniques, and pre-cooling requirements for transport vehicles. Update SOPs if necessary to prevent recurrence. 3. INVESTIGATION: Conduct a thorough investigation into the root cause of the temperature excursion (e.g., equipment failure, human error, environmental factors). Review maintenance logs for the transport vehicle and training records for personnel. Analyze historical temperature data. Determine if other products or batches were affected. Implement long-term preventive measures such as scheduled maintenance, improved equipment, enhanced training, or revised loading protocols.

ACTION RESPONSIBLE

Dispatch Supervisor / QA Manager

Step 7 Compliance

CCP 22

Step 52: VISUAL INSPECTION & SORTING Log: Non-Conformance & Corrective Action Report / Vehicle Inspection Log

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately hold and segregate any product that has been loaded or is intended for loading into the contaminated transport vehicle. Do not load product into a vehicle that does not meet the critical limit. Any product that has come into contact with contaminated surfaces or is suspected of contamination must be rejected and disposed of. Notify QA/HACCP Team Leader. Instruct the driver to thoroughly clean and sanitize the vehicle according to food safety standards, or provide an alternative, compliant vehicle. 2. PROCESS CORRECTION: Ensure the transport vehicle is thoroughly cleaned, sanitized, and re-inspected before it can be used for food transport. If the contamination source is a vehicle malfunction (e.g., leaking fluid), the vehicle must be repaired before use. Retrain dispatch and loading personnel on the critical importance of pre-loading vehicle inspection, including visual and olfactory assessment, and the specific criteria for cleanliness and freedom from odors/residues. Reinforce transport vehicle hygiene policies with the transport company/driver. 3. INVESTIGATION: Investigate the root cause of the contamination (e.g., inadequate cleaning by the transport company, vehicle malfunction, previous load issue). Review the transport company's cleaning protocols and records. Communicate with the transport company to ensure their vehicles consistently meet food safety standards. Determine if this is an isolated incident or a recurring issue with a specific vehicle or transport provider. Implement preventive measures such as requiring proof of cleaning or conducting more frequent audits of transport vehicles.

ACTION RESPONSIBLE**Dispatch Supervisor / QA Manager****Step 7 Compliance**

CCP 24

Step 53: VISUAL INSPECTION & SORTING

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: If tears, holes, improper installation, or incorrect mesh size are detected in screens/filters, immediately stop the line. Isolate and hold all product processed since the last acceptable check of the screens/filters. Remove and replace or repair the faulty screen/filter. Conduct a thorough inspection of the product processed through the faulty screen/filter for potential foreign material contamination. If contamination is found or suspected, divert or dispose of the affected product. 2. PROCESS CORRECTION: Identify the cause of the screen/filter damage or improper installation (e.g., wear and tear, incorrect handling, maintenance error). Implement improved maintenance procedures for screens/filters, including regular inspection schedules and proper handling protocols. Retrain maintenance and production personnel on proper screen/filter installation and inspection. Verify that all screens/filters meet specified mesh size requirements. 3. INVESTIGATION: Investigate the root cause of the screen/filter failure or non-conformance. Review maintenance records, equipment specifications, and training logs. Assess the integrity of other screens/filters in the system. Implement long-term preventive measures to prevent recurrence of screen/filter issues.

ACTION RESPONSIBLE**Quality Control Technician / Maintenance Supervisor****Step 7 Compliance**

CCP 26 Step 54: WEIGHING & PORTIONING

Log: Foreign Material Incident Report / Non-Conformance Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Stop the line immediately. Isolate and hold all product processed since the last acceptable check. Remove any visible foreign material from the product or equipment. Clean and sanitize affected equipment. If foreign material is found in product, dispose of affected product or re-sort if feasible and safe, ensuring no further contamination. 2. PROCESS CORRECTION: Retrain personnel on GMPs, foreign material control procedures, and proper equipment inspection. Repair or replace damaged equipment parts (e.g., cracked plastic, rusted metal, loose components). Implement enhanced pre-operational checks for equipment integrity. 3. INVESTIGATION: Determine the source of the foreign material (e.g., personnel, equipment, raw material). Review maintenance records for equipment. Review personnel hygiene and GMP adherence. Assess the effectiveness of previous steps in removing plant debris. Verify the effectiveness of the corrective actions through subsequent monitoring.

ACTION RESPONSIBLE**Production Supervisor / Quality Control Technician****Step 7 Compliance****CCP 28 Step 55: PRIMARY PACKAGING**

Log: CCP Deviation Log / Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Hold all product packaged since the last compliant check. Segregate and evaluate the affected product for disposition (rework, re-chill, or discard) based on temperature excursion duration and magnitude. Re-establish product temperature to $\leq 4^{\circ}\text{C}$. 2. PROCESS CORRECTION: Identify and rectify the cause of the temperature deviation (e.g., adjust chiller settings, repair refrigeration unit, reduce packaging line speed, ensure proper pre-chilling). Retrain operators on temperature control procedures. 3. INVESTIGATION: Quality Assurance (QA) to investigate the root cause of the temperature deviation, including equipment malfunction, operational error, or environmental factors. Review monitoring records and maintenance logs.

ACTION RESPONSIBLE**Packaging Line Operator / Quality Control Technician / Production Manager****Step 7 Compliance****CCP 30 Step 56: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING**

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Segregate and hold all product packaged during the deviation period. Evaluate product temperature and quality. If product temperature exceeded 4°C for an unacceptable duration, divert for cooking or dispose of as waste. Clean and sanitize affected equipment and area. 2. PROCESS CORRECTION: Adjust refrigeration settings or packaging line speed to ensure product remains below 4°C . Retrain personnel on proper temperature monitoring and control procedures. Review and verify sanitation procedures if environmental contamination is suspected. 3. INVESTIGATION: Determine the root cause of the temperature deviation (e.g., equipment malfunction, line stoppage, ambient temperature, operator error). Review monitoring records and maintenance logs. If ATP monitoring indicated a sanitation failure, investigate sanitation procedures and effectiveness.

ACTION RESPONSIBLE**Quality Control Technician / Production Supervisor****Step 7 Compliance**

CCP
32**Step 57: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING**

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Hold all affected product lots. Segregate and quarantine any product potentially exposed to non-food-grade lubricants. Stop the packaging line until the issue is resolved. Initiate a recall if affected product has left the facility. 2. PROCESS CORRECTION: Remove all non-NSF H1 certified lubricants from the production area. Clean and sanitize any equipment surfaces that may have come into contact with non-food-grade lubricants. Replace with NSF H1 certified lubricants. Retrain maintenance personnel on the exclusive use of NSF H1 certified lubricants in food contact areas and proper application techniques to prevent overspray or drip. 3. INVESTIGATION: Determine the root cause of the deviation (e.g., incorrect lubricant purchase, lack of awareness, improper storage, equipment malfunction). Review purchasing records, Safety Data Sheets (SDS), and maintenance logs. Evaluate the extent of potential contamination and product safety impact. Implement preventive measures (e.g., update approved lubricant list, enhance supplier verification for maintenance supplies, revise maintenance SOPs, conduct regular audits of lubricant use).

ACTION RESPONSIBLE**Maintenance Manager / QA Manager****Step 7 Compliance**

CCP 34

Step 58: METAL DETECTION

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: Isolate and hold all product produced since the last acceptable temperature check. If vacuum leak or bubble tests reveal seal defects, or visual inspection shows obvious seal defects, consider this product potentially compromised for temperature control and pathogen growth. Evaluate affected product for temperature deviation. Product found to be above 4°C for an unacceptable duration, or with compromised seals, must be diverted for cooking, reprocessed if safe, or destroyed. PROCESS CORRECTION: Identify the root cause of the temperature deviation (e.g., refrigeration unit malfunction, line stoppage, ambient temperature issues, improper handling). Adjust or repair refrigeration equipment, optimize processing flow, or reinforce cold chain procedures. Simultaneously, identify and correct the cause of seal defects (e.g., equipment malfunction, packaging material issues, operator error). Verify that temperature control is restored and packaging equipment is producing sound seals. INVESTIGATION: Review temperature logs, equipment maintenance records, and operator training. Review packaging line settings, material specifications, and operator procedures for sealing. Determine the root cause of the temperature excursion and any related seal integrity failures. Update HACCP plan if necessary.

ACTION RESPONSIBLE**Quality Assurance Manager / Production Supervisor****Step 7 Compliance**

CCP 36

Step 59: METAL DETECTION

Log: Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

1. IMMEDIATE ACTION: Immediately cease use of the non-conforming gas supply. Identify and quarantine all product packaged using the non-conforming gas since the last verified compliant delivery. Evaluate the risk to affected product and determine disposition (hold, rework, or dispose). Notify the supplier of the non-conformance. 2. PROCESS CORRECTION: Ensure only food-grade certified gases from approved suppliers are used. Reinforce training for relevant personnel (QA, purchasing) on verifying Certificates of Analysis (CoAs) or Certificates of Conformance (CoCs) and approved supplier status. Update or implement stricter verification protocols for incoming gas deliveries. 3. INVESTIGATION: Conduct a root cause analysis to determine why non-conforming gas was received or used (e.g., supplier error, purchasing error, documentation failure). Review supplier's quality control records for the specific batch. Assess the effectiveness of the approved supplier program and internal verification procedures. Implement preventive measures to avoid recurrence.

ACTION RESPONSIBLE

Quality Assurance Manager / Purchasing Manager

Step 7 Compliance

CCP 38

Step 60: LABELING

Log: Non-Conformance Report & Corrective Action Log

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: All affected product (from the last good check to the point of discovery) must be immediately placed on hold, segregated, and identified. Production must cease until the issue is resolved. The mislabeled product must be evaluated for rework (re-labeling if safe and feasible) or destruction. PROCESS CORRECTION: Verify that the correct labels are loaded onto the labeling machine and that all mandatory information is present and legible. Retrain operators on proper label verification procedures. Adjust or repair labeling equipment if it contributed to the misapplication. INVESTIGATION: Conduct a root cause analysis to determine why the mislabeling occurred (e.g., human error, equipment malfunction, incorrect label stock issued, inadequate training). Review label changeover procedures and verification steps. Implement preventive measures to avoid recurrence, such as updated SOPs, additional training, or enhanced automated verification systems.

ACTION RESPONSIBLE

Quality Control Technician / Production Supervisor / QA Manager

Step 7 Compliance

CCP 40

Step 61: FINISHED PRODUCT COLD STORAGE

Log: Not provided

CORRECTIVE ACTIONS TO BE TAKEN

Not provided

ACTION RESPONSIBLE

Not provided

Step 7 Compliance

CCP 42

Step 62: DISPATCH

Log: Non-Conformance & Corrective Action Report / Temperature Deviation Log

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: If product temperature exceeds 4°C, immediately reject the affected lot for dispatch. If temperature excursion is detected during transit, notify Dispatch Supervisor. Segregate affected product and hold for evaluation (e.g., microbiological testing) or dispose if deemed unsafe. PROCESS CORRECTION: Identify and rectify the cause of temperature abuse (e.g., pre-cooling of vehicle, loading speed, refrigeration unit malfunction). Retrain loading personnel and drivers on critical limit adherence. Ensure transport vehicle refrigeration units are functioning correctly and pre-cooled. INVESTIGATION: Investigate the root cause of the temperature deviation (e.g., equipment failure, human error, inadequate pre-cooling, extended loading time). Review temperature monitoring records and maintenance logs. Implement preventive measures to avoid recurrence (e.g., scheduled maintenance, enhanced training, revised loading protocols).

ACTION RESPONSIBLE

Dispatch Supervisor / Quality Assurance Manager

Step 7 Compliance

CCP 43

Step 63: DISPATCH

Log: Vehicle Inspection Checklist / Non-Conformance & Corrective Action Report

CORRECTIVE ACTIONS TO BE TAKEN

IMMEDIATE ACTION: Immediately reject the transport vehicle for loading. Do not load any product into the contaminated vehicle. Segregate any product that may have already been loaded into the contaminated area and hold for assessment or disposal. PROCESS CORRECTION: Require the transport vehicle to be thoroughly cleaned, sanitized (if applicable), and re-inspected. If recurring, communicate with the carrier to address their cleaning and maintenance protocols. Ensure all cleaning agents used are food-grade and properly rinsed. INVESTIGATION: Investigate the source of the contamination (e.g., inadequate cleaning by carrier, vehicle malfunction causing leaks, previous cargo contamination). Review carrier's cleaning and maintenance logs. Evaluate carrier's compliance with food safety requirements. Implement preventive measures such as enhanced carrier qualification, pre-shipment inspection checklists, and driver training on vehicle hygiene.

ACTION RESPONSIBLE

Dispatch Supervisor / Quality Control Technician

Step 7 Compliance

Step 8: Verification Procedures

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 1 Step 1: RECEIVING RAW CABBAGE	1. Record Review: Daily/weekly review of receiving logs to ensure critical limits (supplier approval, temperature, visual inspection) were met and documented, and corrective actions were properly implemented. Review Approved Supplier List and supplier audit records. 2. Calibration: Verify calibration records for thermometers used by receiving personnel (e.g., monthly/quarterly) against a certified reference standard. 3. Audits/Testing: Conduct quarterly internal audits to observe receiving personnel's monitoring activities (e.g., proper temperature measurement, visual inspection technique). Annually audit approved suppliers or review third-party audit reports. Annually or as risk dictates, conduct microbiological testing on incoming raw cabbage from selected lots/suppliers to validate the effectiveness of supplier controls and critical limits.	Daily record review, monthly calibration check, quarterly internal audit, annual supplier audit/validation testing.	Receiving Logs/Records, Approved Supplier List, Supplier Audit Reports/Certificates, Thermometer Calibration Records, CCP Verification Summary Sheet, Internal Audit Reports, Microbiological Test Results.	Quality Control Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 2 Step 2: RECEIVING RAW CABBAGE	1. Record Review: Daily/weekly review of receiving logs, COAs, and transport vehicle inspection records to ensure critical limits (supplier approval, COA compliance, vehicle cleanliness/segregation) were met and documented, and corrective actions were properly implemented. Review Approved Supplier List and supplier audit records, specifically for chemical control programs. 2. Calibration: Not directly applicable to COA review or visual inspection; however, if internal analytical equipment were used for rapid screening, their calibration records would be verified. 3. Audits/Testing: Conduct quarterly internal audits to observe receiving personnel's COA review and transport vehicle inspection. Annually audit approved suppliers or review third-party audit reports focusing on pesticide application, heavy metal control, and allergen management programs. Annually or as risk dictates, conduct independent laboratory testing of incoming raw cabbage for pesticide residues, heavy metals, or specific allergens to validate COA accuracy and supplier controls.	Daily record review, quarterly internal audit, annual supplier audit/validation testing.	<i>Receiving Logs/Records, Approved Supplier List, Supplier Audit Reports/Certificates, Certificates of Analysis (COAs), CCP Verification Summary Sheet, Internal Audit Reports, Independent Laboratory Test Results.</i>	Quality Control Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 3 Step 3: RAW MATERIAL COLD STORAGE	1. Record Review: Weekly review of cold storage temperature logs and associated corrective action records to ensure monitoring was performed as required and critical limits were consistently met. 2. Calibration: Annual calibration of all temperature monitoring devices (e.g., thermometers, data loggers) used for the cold storage unit against a certified reference standard. 3. Audits/Testing: Quarterly internal audits to observe monitoring personnel performing temperature checks. Initial validation and re-validation (e.g., every 3-5 years or upon significant changes to equipment/process) through temperature mapping studies to confirm the cold storage unit consistently maintains the critical limit throughout the storage area under typical and peak load conditions.	Weekly record review, Annual calibration, Quarterly internal audit, Periodic validation	<i>Cold Storage Temperature Logs, Corrective Action Records, Thermometer Calibration Log, Verification Summary Sheet, Internal Audit Reports, Temperature Mapping Study Reports</i>	Quality Assurance Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 4 Step 4: RAW MATERIAL COLD STORAGE	1. Record Review: Weekly review of pre-processing time/temperature logs, sanitation logs (sanitizer concentration and contact time), and all associated corrective action records to ensure critical limits were met and deviations addressed. 2. Calibration: Annual calibration of timers used for monitoring holding time. Daily/weekly verification or calibration of sanitizer test kits/meters against a known standard or freshly prepared standard solution as per manufacturer's instructions. 3. Audits/Testing: Monthly internal audits to observe monitoring personnel performing time/temperature checks and sanitation procedures (e.g., sanitizer preparation, application, contact time). Monthly/Quarterly microbial swab testing of food contact surfaces post-sanitation to verify the effectiveness of the sanitation program. Initial validation and re-validation (e.g., every 3-5 years or upon significant changes to process, equipment, or sanitizer) of the 4-hour cumulative time limit (e.g., via challenge studies or scientific literature review) and the chosen sanitizer concentration/contact time (e.g., via microbial efficacy studies or reliance on validated industry standards).	Weekly record review, Annual/Daily/Weekly calibration, Monthly internal audit, Monthly/Quarterly microbial testing, Periodic validation	<i>Pre-processing Time/Temperature Logs, Sanitation Logs, Corrective Action Records, Timer Calibration Log, Sanitizer Test Kit Verification Log, Verification Summary Sheet, Internal Audit Reports, Food Contact Surface Swab Test Results, Validation Study Reports</i>	Quality Assurance Manager

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CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 6 Step 6: RAW MATERIAL COLD STORAGE	1. Record Review: Daily review of monitoring logs for visual foreign material checks, screen/sieve integrity, and metal detector performance checks, ensuring critical limits are consistently met and all corrective actions are properly documented. Review metal detector calibration records. 2. Calibration: Verify annual calibration records for metal detectors by a certified technician. Confirm the integrity and certification of metal detector test pieces. 3. Audits/Testing: Weekly independent metal detector test piece challenges (ferrous, non-ferrous, stainless steel) by different personnel to confirm monitoring accuracy. Monthly independent inspection of screens/sieves for integrity and mesh size. Quarterly internal audits to observe monitoring procedures and ensure compliance. Annual review of the foreign material control program's effectiveness through trend analysis of foreign material complaints, metal detector reject rates, and re-validation studies if equipment or processes change.	Daily record review / Weekly verification testing / Monthly calibration review / Quarterly internal audit / Annual validation review	<i>CCP 6 Monitoring Log (Foreign Material Visual, Screen/Sieve, Metal Detector), Corrective Action Log, Metal Detector Calibration Log, Metal Detector Test Piece Log, Internal Audit Report, HACCP Verification Summary Report</i>	Quality Control Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 7 Step 7: PRE-PROCESSING PREPARATION	**Record Review**: The Quality Assurance Manager (QAM) or designee will daily/weekly review process water microbiological test reports (E. coli, Total Coliforms) from the accredited laboratory to ensure all monitoring records are complete, accurate, and indicate compliance with critical limits. Corrective action records for any deviations and their effectiveness will be reviewed monthly. Previous verification activity records will be reviewed quarterly. **Calibration**: The QAM will annually verify that the external accredited laboratory performing microbiological testing maintains current accreditation and performs internal calibration and quality control checks on their equipment according to their validated methods. **Audits/Testing**: An annual internal audit of the water potability program will be conducted, covering sampling procedures, record-keeping, and corrective actions. The QAM will review external laboratory accreditation annually. Annual re-validation testing of the water source and treatment system will be performed by an independent accredited laboratory to confirm consistent potability.	Daily/Weekly record review; Monthly corrective action review; Annual audit, lab accreditation review, and re-validation.	Process Water Microbiological Test Reports, Corrective Action Log, Verification Log, Internal Audit Reports, External Lab Accreditation Certificates, Water System Validation Reports.	Quality Assurance Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 8 Step 8: PRE-PROCESSING PREPARATION	**Record Review**: The Quality Assurance Manager (QAM) or designee will daily/shift review monitoring records for free chlorine residual. Periodic (e.g., quarterly/annually) external laboratory reports for heavy metals, industrial contaminants, and cleaning chemical residues will be reviewed upon receipt. All monitoring records will be verified for completeness, accuracy, and compliance with critical limits. Corrective action records for any deviations and their effectiveness will be reviewed monthly. Previous verification activity records will be reviewed quarterly. **Calibration**: The free chlorine residual testing equipment (e.g., colorimeter) will be calibrated daily or weekly against known standards, and calibration records maintained. The QAM will annually verify that external accredited laboratories performing heavy metal, industrial contaminant, and cleaning chemical testing maintain current accreditation and perform internal calibration and quality control checks on their equipment. **Audits/Testing**: A semi-annual internal audit of the chemical water quality program will be conducted, covering monitoring procedures, calibration records, and corrective actions. The QAM will review external laboratory accreditation annually. Annual re-	Daily/Shift chlorine record review; Quarterly/Annual chemical record review; Daily/Weekly calibration; Monthly corrective action review; Semi-annual internal audit; Annual lab accreditation review and re-validation.	<i>Free Chlorine Monitoring Log, Heavy Metal/Industrial Contaminant Test Reports, Cleaning Chemical Residue Test Reports, Corrective Action Log, Verification Log, Calibration Log (for chlorine meter), Internal Audit Reports, External Lab Accreditation Certificates, Water System Validation Reports, CIP/Rinsing Validation Reports.</i>	Quality Assurance Manager

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CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 9 Step 9: PRE-PROCESSING PREPARATION	1. Record Review: Review daily monitoring logs for sanitizer concentration, pH, and water potability certificates/test results. Verify that all critical limits were met and that corrective actions were properly documented and implemented when deviations occurred. 2. Calibration: Verify calibration records for pH meters, chlorine test kits/strips, and any automated sanitizer dosing equipment. Ensure calibration is performed at the specified frequency (e.g., weekly/monthly) and within acceptable limits. 3. Audits/Testing: Conduct periodic (e.g., quarterly) unannounced internal audits of the washing and rinsing process to observe monitoring procedures, sanitizer preparation, and water quality checks. Conduct independent checks of sanitizer concentration using a calibrated reference method (e.g., titration) against the routine monitoring method (e.g., test strips) at a specified frequency (e.g., monthly). Periodically (e.g., semi-annually or annually) send samples of wash water and/or post-wash product for microbiological testing (e.g., generic E. coli, total coliforms) to validate the effectiveness of the sanitizer concentration and contact time. Review annual (or more frequent, as required by local regulations) third-party laboratory reports	Daily record review; Weekly/Monthly calibration verification; Monthly independent sanitizer checks; Quarterly internal audits; Semi-annual/Annual validation testing (microbiological, water potability).	CCP 9 Monitoring Log, Calibration Log, Corrective Action Log, Internal Audit Report, Water Potability Certificates, Microbiological Test Reports, Verification Summary Sheet.	Quality Assurance Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
	for incoming water potability.			

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 10 Step 10: PRE-PROCESSING PREPARATION	1. Record Review: Review daily monitoring logs for product temperatures and ATP swab readings. Verify that all critical limits were met and that corrective actions (e.g., re-cooling, re-cleaning, re-sanitizing) were properly documented and implemented when deviations occurred. Review environmental swab results for pathogens. 2. Calibration: Verify calibration records for thermometers used for product temperature monitoring and ATP luminometers. Ensure calibration is performed at the specified frequency (e.g., annually for thermometers, according to manufacturer's recommendations for ATP luminometers) and within acceptable limits. 3. Audits/Testing: Conduct periodic (e.g., quarterly) unannounced internal audits of the visual inspection and sorting process to observe monitoring procedures for product temperature and equipment cleanliness. Verify proper sanitation procedures are followed. Conduct independent checks of thermometers against a certified reference thermometer or ice bath at a specified frequency (e.g., monthly). Verify ATP luminometer performance using positive/negative controls or calibration verification swabs as per manufacturer's instructions (e.g., weekly/monthly).	Daily record review; Monthly thermometer checks; Weekly/Monthly ATP luminometer checks; Quarterly internal audits; Annual product temperature validation studies; Ongoing EMP review.	CCP 10 Monitoring Log, Calibration Log, Corrective Action Log, Internal Audit Report, Environmental Monitoring Program (EMP) Results, Verification Summary Sheet.	Quality Assurance Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
	Review results from the routine environmental monitoring program for <i>Listeria</i> spp. or <i>L. monocytogenes</i> on food contact surfaces and in the processing environment, analyzing trends and ensuring corrective actions are effective. Periodically (e.g., annually) conduct studies to confirm that the cooling system and handling procedures consistently maintain product temperature <= 4°C (40°F) throughout the sorting step.			
CCP 11 Step 11: VISUAL INSPECTION & SORTING	1. Record Review: Daily review of CCP 11 monitoring logs for lubricant type and chemical residue checks, corrective action records, and lubricant purchase documentation (e.g., COAs) to ensure compliance with critical limits. 2. Calibration: Annual calibration of ATP luminometers (if used) and verification of pH strip expiry dates and proper storage conditions. 3. Audits/Testing: Monthly internal audits to observe lubricant application practices and pre-operational sanitation checks, including independent verification testing for chemical residues on food contact surfaces (e.g., using alternative test methods or sending samples to an accredited lab quarterly).	Daily record review / Monthly audit / Quarterly verification testing	CCP 11 Monitoring Log (Lubricant Type) , CCP 11 Monitoring Log (Chemical Residue) , Corrective Action Log , Lubricant Inventory/Purchase Records , Supplier COAs/Specifications , Calibration Records (ATP Luminometer, pH meter) , Verification Audit Reports , Verification Testing Results	Quality Control Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 12 Step 12: PRE-PROCESSING PREPARATION	1. Record Review: Daily review of CCP 12 monitoring logs for foreign material detection and screen/filter integrity checks, and corrective action records. Review maintenance logs for screen/filter replacement. 2. Calibration: Not directly applicable for visual inspection or physical screens/filters; however, any measurement tools used to verify mesh size (e.g., calipers) would be calibrated annually. 3. Audits/Testing: Monthly internal audits to observe visual inspection processes and pre-operational checks of screens/filters, including verification of mesh size. Quarterly challenge tests for screens/filters (if feasible) to confirm effectiveness in removing specified foreign materials.	Daily record review / Monthly audit / Quarterly verification testing	CCP 12 Monitoring Log (Foreign Material), CCP 12 Monitoring Log (Screen/Filter Integrity), Corrective Action Log, Maintenance Records (Screens/Filters), Verification Audit Reports, Verification Testing Results	Quality Control Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 13 Step 13: SLICING/SHREDDING	Daily review of monitoring records for visual foreign material checks, personnel GMP compliance, and equipment integrity. Weekly review of corrective action logs and customer complaints related to foreign material. Calibration of any measuring devices used for equipment integrity checks (if applicable, e.g., calipers) as per manufacturer's schedule. Quarterly internal audits of the weighing and portioning area to observe GMP adherence, equipment condition, and product for foreign material, including a review of the effectiveness of the foreign material control program through trend analysis of monitoring data and complaint records.	Daily record review / Weekly corrective action review / Quarterly internal audit / Annual validation review	<i>CCP 13 Monitoring Log (Foreign Material/GMP/Equipment Integrity), Corrective Action Log, Internal Audit Report, Calibration Log, CCP Verification Summary Sheet, Customer Complaint Log</i>	Quality Assurance Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 14 Step 14: SLICING/SHREDDING	Daily review of monitoring records for product temperature, packaging material integrity, environmental hygiene (visual/ATP), and MAP gas composition. Weekly review of corrective action logs. Calibration of all thermometers, ATP meters, and MAP gas analyzers as per manufacturer's recommendations and internal schedule (e.g., daily pre-shift checks, weekly/monthly full calibrations, annual external calibrations). Monthly internal audits of the packaging area, including observation of monitoring activities and sanitation practices. Quarterly independent environmental swabbing for <i>Listeria</i> spp. on food contact surfaces in the packaging area. Annual validation testing of finished product for relevant pathogens (e.g., <i>Salmonella</i>, <i>Listeria monocytogenes</i>) and independent laboratory verification of MAP gas composition in packaged product.	Daily record review / Weekly corrective action review / Daily-Annual calibration / Monthly internal audit / Quarterly environmental swabbing / Annual product testing & MAP validation	CCP 14 Monitoring Log (Temperature/Packaging/Hygiene/MAP), Corrective Action Log, Calibration Log, Internal Audit Report, Environmental Swab Results, Finished Product Micro Test Results, MAP Gas Validation Report, CCP Verification Summary Sheet	Quality Assurance Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 15 Step 15: POST-SLICING WASHING & RINSING	1. Record Review: Daily review of Product Temperature Logs, ATP Swab Results Logs, and Corrective Action Records. Weekly review of thermometer and ATP meter calibration records. 2. Calibration: Quarterly calibration of all thermometers used for product temperature monitoring against a certified reference. Annual calibration/performance verification of ATP meters per manufacturer specifications. 3. Audits/Testing: Monthly internal audit of CCP monitoring procedures and critical limit adherence. Weekly microbiological environmental swabbing (e.g., for <i>Listeria</i> spp., APC) on food contact surfaces in the packaging area. Annually, or upon significant process change, conduct validation testing to confirm the effectiveness of temperature control and sanitation procedures in preventing pathogen proliferation during packaging.	Daily record review / Weekly environmental swabbing / Monthly internal audit / Quarterly thermometer calibration / Annual ATP meter calibration & validation testing	Product Temperature Log, ATP Swab Results Log, Corrective Action Log, Thermometer Calibration Log, ATP Meter Performance Log, Internal Audit Report, Environmental Monitoring Log, Validation Study Report	Quality Assurance Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 16 Step 16: SLICING/SHREDDING	1. Record Review: Daily review of Packaging Material COCs/LOGs, Chemical Residue Test Logs, Lubricant Inventory/Usage Logs, and Corrective Action Records. 2. Calibration: Quarterly verification of pH meter calibration (if used). Regular checks of chemical test kit expiry dates and storage conditions. 3. Audits/Testing: Monthly internal audit of CCP monitoring procedures, critical limit adherence, and physical inspection of packaging materials and lubricants in use. Annually, conduct a supplier audit for primary packaging material providers. Semi-annually, send swab samples from food contact surfaces post-sanitation to an accredited lab for specific chemical residue analysis. Periodically (e.g., biennially or risk-based), send packaging material samples to an accredited lab for independent food-grade compliance testing.	Daily record review / Monthly internal audit / Quarterly pH meter calibration / Semi-annual lab chemical residue testing / Annual supplier audit / Biennial packaging material lab testing	<i>Packaging Material COC/LOG, Approved Supplier List, Chemical Residue Test Log, Lubricant Inventory Log, Corrective Action Log, pH Meter Calibration Log, Internal Audit Report, Supplier Audit Report, Lab Analysis Report (Chemical Residue), Lab Analysis Report (Packaging Material)</i>	Quality Assurance Manager

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CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 18 Step 18: POST-SLICING WASHING & RINSING	1. **Record Review**: Review of MAP gas Certificates of Analysis (COA) or food-grade certifications upon receipt of each gas delivery. Monthly review of the Approved Supplier List to confirm the MAP gas supplier's current approved status. 2. **Calibration**: N/A for direct gas purity monitoring equipment calibration, as control is primarily through supplier certification and documentation review. If in-line gas purity sensors are used, their calibration records would be reviewed as per schedule. 3. **Audits/Testing**: Annual internal audits of the purchasing and receiving procedures for MAP gases to ensure compliance with the Approved Supplier Program and verification requirements. Periodic supplier audits (e.g., every 1-3 years) of MAP gas suppliers to verify their food safety programs. Annual independent laboratory testing of received MAP gas samples to confirm purity and food-grade status.	Upon each gas delivery (COA review), Monthly (supplier status review), Annual (internal audit, independent lab testing), Every 1-3 years (supplier audit).	<i>MAP Gas Certificates of Analysis (COA), Approved Supplier List, Supplier Qualification Records, Internal Audit Reports, Supplier Audit Reports, Independent Lab Test Results, Verification Summary Sheet.</i>	Quality Control Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 19 Step 19: LABELING	1. Record Review: Daily review of 'Label Verification Logs' and 'Pre-Production Label Setup Records' to ensure 100% verification was performed, critical limits were met, and all deviations and corrective actions were properly documented. 2. Calibration: Not applicable for direct measurement devices. However, ensure label printing equipment (if applicable for variable data like date codes) is maintained and verified for legibility and accuracy of printed information. 3. Audits/Testing: Weekly internal audits of the labeling station to observe monitoring activities, verify label stock, and randomly inspect finished product labels for accuracy and completeness against specifications. Validation testing involves periodic review of consumer complaints and internal non-conformance reports related to labeling to confirm the effectiveness of the 100% verification process in preventing mislabeled products from reaching consumers.	Daily review / Weekly audit / Annual validation review	Label Verification Log, Pre-Production Label Setup Record, Labeling Audit Report, Non-Conformance Report, Consumer Complaint Log	Quality Control Supervisor

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 20 Step 20: VISUAL INSPECTION & SORTING	1. Record Review: Daily review of 'Finished Product Cold Storage Temperature Logs' (manual and/or continuous recorder data) to ensure temperatures were maintained within critical limits, and all deviations and corrective actions were properly documented. 2. Calibration: Calibrate all temperature monitoring devices (e.g., data loggers, manual thermometers) used for monitoring the finished product cold storage unit against a certified reference thermometer annually, or as per manufacturer's recommendations. 3. Audits/Testing: Monthly internal audits of the cold storage unit to observe monitoring procedures, verify the accuracy of manual temperature checks, and confirm proper product storage practices. Validation testing involves periodic review of scientific literature, regulatory guidelines, or internal challenge studies (if applicable) to confirm that the established critical limits are effective in controlling the growth of <i>Listeria monocytogenes</i> and other psychrotrophic spoilage organisms in sliced cabbage for its intended shelf life.	Daily review / Monthly audit / Annual calibration / Biennial validation review	<i>Finished Product Cold Storage Temperature Log, Temperature Monitoring Device Calibration Log, Cold Storage Audit Report, Validation Study Report (or Scientific Literature Review)</i>	Quality Control Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 21 Step 21: VISUAL INSPECTION & SORTING	1. Record Review: Review of daily/per-shipment temperature monitoring logs (product and/or air) and associated corrective action records for completeness, accuracy, and adherence to critical limits. This includes verifying that all required fields are filled, deviations are properly documented, and corrective actions were effective. 2. Calibration: Review of calibration records for all temperature monitoring devices (e.g., thermometers, data loggers) used for CCP monitoring, ensuring they are calibrated against traceable standards at the specified frequency (e.g., annually or semi-annually) and are within acceptable accuracy limits. 3. Audits/Testing: Periodic internal audits (e.g., quarterly) to observe monitoring personnel performing temperature checks during loading, ensuring adherence to established procedures. This includes verifying proper use of equipment and accurate record-keeping. Validation testing involves independent checks of product or air temperature using a separate, calibrated reference thermometer during loading events to confirm the accuracy of routine monitoring and equipment.	Record Review: Daily/Weekly; Calibration Record Review: Monthly; Internal Audits/Validation Testing: Quarterly	<i>CCP 21 Verification Log, Temperature Monitoring Records, Corrective Action Log, Calibration Records, HACCP Verification Summary</i>	Quality Control Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 22 Step 22: VISUAL INSPECTION & SORTING	1. Record Review: Review of daily/per-shipment transport vehicle inspection logs and associated corrective action records for completeness, accuracy, and adherence to critical limits. This includes verifying that all required fields are filled, deviations are properly documented, and corrective actions were effective. 2. Calibration: Not applicable for visual/olfactory inspection. However, personnel competency records (e.g., training logs) are reviewed to ensure monitors are qualified to perform inspections. 3. Audits/Testing: Periodic internal audits (e.g., quarterly) to observe monitoring personnel performing transport vehicle inspections, ensuring adherence to established procedures for visual and olfactory checks. This includes verifying proper inspection techniques and accurate record-keeping. Validation testing involves an independent, trained individual performing a separate visual and olfactory inspection of a transport vehicle and comparing findings with the routine monitoring records to confirm the effectiveness of the monitoring procedure and the competency of the monitor.	Record Review: Weekly/Monthly; Internal Audits/Validation Testing: Quarterly	<i>CCP 22 Verification Log, Pre-shipment Vehicle Inspection Records, Corrective Action Log, Personnel Training Records, HACCP Verification Summary</i>	Quality Control Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 24 Step 23: VISUAL INSPECTION & SORTING	1. Record Review: Daily review of visual inspection and sorting logs, foreign material detection records, and corrective action reports to ensure critical limits were met and actions were effective. 2. Calibration: Verify the integrity and proper functioning of physical separation devices (e.g., screens, filters) through scheduled maintenance checks. Ensure personnel performing visual inspections are trained and their competency is periodically assessed. 3. Audits/Testing: Conduct monthly internal audits to observe visual inspection procedures and personnel adherence. Perform annual validation testing by intentionally introducing known, traceable foreign materials into a test batch to confirm the effectiveness of the sorting process.	Daily record review / Monthly internal audit / Annual validation testing	<i>Visual Inspection Logs, Sorting Records, Foreign Material Corrective Action Reports, Internal Audit Reports, Personnel Training Records, Validation Test Reports</i>	Quality Control Manager
CCP 25 Step 24: WEIGHING & PORTIONING	Weekly audit of CCP monitoring records and equipment calibration for WEIGHING & PORTIONING	Weekly	<i>WEIGHING & PORTIONING Verification Log Sheet</i>	QA Manager / HACCP Coordinator
CCP 26 Step 25: PRIMARY PACKAGING	Weekly audit of CCP monitoring records and equipment calibration for PRIMARY PACKAGING	Weekly	<i>PRIMARY PACKAGING Verification Log Sheet</i>	QA Manager / HACCP Coordinator
CCP 29 Step 26: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING	Weekly audit of CCP monitoring records and equipment calibration for MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING	Weekly	<i>MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING Verification Log Sheet</i>	QA Manager / HACCP Coordinator

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 30 Step 27: MODIFIED ATMOSPHERE PACKAGING (MAP) & SEALING	Record Review: Daily review of temperature monitoring logs, ATP monitoring logs, and corrective action records. Calibration: Weekly calibration checks of thermometers used for product temperature, and monthly calibration of ATP luminometers. Audits/Testing: Quarterly internal audits of the MAP & Sealing process, including observation of monitoring activities and verification of critical limit adherence. Annual validation testing to ensure temperature control effectiveness and sanitation efficacy (e.g., microbial swabs post-cleaning).	Daily record review, weekly thermometer calibration, monthly ATP luminometer calibration, quarterly internal audits, annual validation testing.	Temperature Monitoring Log , ATP Monitoring Log , Corrective Action Log , Thermometer Calibration Log , ATP Luminometer Calibration Log , Internal Audit Report , Validation Study Report .	Quality Assurance Manager
CCP 33 Step 28: METAL DETECTION	1. Record Review: Review daily MAP gas composition monitoring logs, Certificates of Analysis (COA) from gas suppliers, and records of any non-conforming gas batches. 2. Calibration: Verify monthly calibration records for gas analyzers used to monitor MAP gas composition and pressure gauges on gas lines/cylinders. 3. Audits/Testing: Conduct annual audits of MAP gas suppliers to ensure compliance with specifications. Perform quarterly independent laboratory testing of MAP gas samples for purity and absence of impurities.	Daily record review, monthly calibration checks, annual supplier audit, quarterly gas testing.	MAP Gas COA Records , Gas Analyzer Calibration Logs , Supplier Audit Reports , Independent Lab Test Reports , CCP Verification Log .	Quality Assurance Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 37 Step 29: FINISHED PRODUCT COLD STORAGE	Weekly audit of CCP monitoring records and equipment calibration for FINISHED PRODUCT COLD STORAGE	Weekly	FINISHED PRODUCT COLD STORAGE Verification Log Sheet	QA Manager / HACCP Coordinator
CCP 39 Step 30: DISPATCH	Weekly audit of CCP monitoring records and equipment calibration for DISPATCH	Weekly	DISPATCH Verification Log Sheet	QA Manager / HACCP Coordinator
CCP 40 Step 31: DISPATCH	Weekly audit of CCP monitoring records and equipment calibration for DISPATCH	Weekly	DISPATCH Verification Log Sheet	QA Manager / HACCP Coordinator

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 28 Step 32: PRIMARY PACKAGING	1. Record Review: Daily review of product temperature logs during packaging, packaging material inspection records, sanitation records for food contact surfaces, and O2/CO2 concentration logs (if applicable). Verify that critical limits were maintained and all corrective actions were properly documented and implemented. 2. Calibration: Quarterly calibration of all temperature measuring devices (e.g., thermometers, probes) used for monitoring product temperature. Annual calibration of gas analyzers (if used for O2/CO2 monitoring). 3. Audits/Testing: Conduct monthly internal audits to observe packaging procedures, temperature control adherence, and packaging material handling. Perform quarterly microbiological testing of finished product for key pathogens (e.g., Listeria monocytogenes) and environmental swabs of food contact surfaces to verify sanitation effectiveness. Conduct annual validation testing to confirm the packaging process consistently maintains product temperature below 4°C and prevents microbial growth.	Daily record review / Quarterly calibration / Monthly internal audit and environmental swabs / Quarterly finished product testing / Annual validation testing	Product Temperature Logs, Packaging Material Inspection Logs, Sanitation Records, Gas Concentration Logs, Corrective Action Reports, Calibration Records, Internal Audit Reports, Microbiological Test Results (Finished Product & Environmental), Validation Reports	Quality Assurance Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 32 Step 33: METAL DETECTION	Record Review: Weekly review of packaging material certifications (COAs), supplier compliance documentation, cleaning chemical usage logs, lubricant purchase records, and any related corrective actions. Calibration: Calibration of any analytical equipment used for testing packaging material compliance or chemical residue detection (e.g., FTIR, GC-MS) as per manufacturer specifications, typically annually. Audits/Testing: Annual supplier audits for packaging material manufacturers. Quarterly internal audits of packaging material handling, cleaning, and lubrication procedures. Annual validation testing of cleaning procedures to ensure no chemical residue remains. Biennial analytical testing of packaging materials for compliance (e.g., migration tests) and lubricants for food-grade certification.	Weekly record review, annual analytical equipment calibration, annual supplier audits, quarterly internal audits, annual cleaning validation, biennial material/lubricant testing.	Packaging Material COAs, Supplier Audit Reports, Cleaning/Sanitation Logs, Lubricant Certification Records, Analytical Equipment Calibration Log, Chemical Residue Test Reports, Packaging Material Migration Test Reports, Validation Study Report.	Quality Assurance Manager

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CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 36 Step 35: METAL DETECTION	1. Record Review: Review MAP gas supplier Certificates of Analysis (COA), food-grade certifications, and monitoring records for supplier approval status and gas quality documentation. 2. Calibration: Calibrate gas flow meters and pressure gauges used in MAP packaging annually or as per manufacturer's recommendation to ensure accurate gas mixture delivery. 3. Audits/Testing: Conduct annual audits of MAP gas suppliers to verify food-grade compliance and quality control programs. Perform periodic (e.g., annual) independent laboratory testing of gas samples from the supply line to confirm purity and food-grade specifications.	Weekly record review; Annual supplier audit and gas testing; Annual calibration of gas delivery equipment.	MAP Gas Supplier COAs, Food-Grade Certifications, Supplier Audit Reports, Gas Sample Test Results, Gas Delivery Equipment Calibration Records, CCP Verification Log.	Quality Assurance Manager

CCP REF	VERIFICATION ACTIVITY	FREQUENCY	RECORD / PROOF	WHO
CCP 38 Step 36: LABELING	1. Record Review: Review daily labeling monitoring records, pre-production label verification checklists, label inventory logs, and corrective action reports related to mislabeling incidents. 2. Calibration: Calibrate label printing equipment (if labels are printed in-house) and label applicators annually or as per manufacturer's recommendation to ensure legibility, correct information placement, and accurate application. 3. Audits/Testing: Conduct quarterly internal audits of the labeling process, including comparison of finished product labels against product specifications and master label files. Perform annual external audits of labeling procedures. Conduct monthly random sampling of finished products to verify label accuracy and	Daily record review; Quarterly internal audit; Annual external audit; Monthly random label checks; Annual calibration of labeling equipment.	Label Verification Checklists, Label Inventory Logs, Labeling Monitoring Records, Corrective Action Reports, Internal Audit Reports, External Audit Reports, Finished Product Label Sample Records, Labeling Equipment Calibration Records, CCP Verification Log.	Quality Control Supervisor

APPROVED BY (DIGITAL RECORD):

Victoria Jideani

Role: Admin

SIGNATURE IMAGE / APPROVAL MARK:

Verified via system login

ADDITIONAL HACCP TEAM SIGNATURE:

DATE OF FINAL APPROVAL:

September 20th, 2026